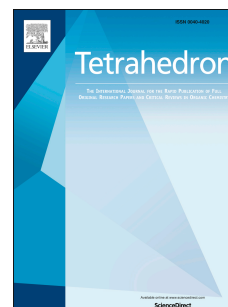


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Enantioselective titanium-promoted 1,2-additions of carbon nucleophiles to carbonyl compounds

Hélène Pellissier*

Aix-Marseille Université,

Centrale Marseille, CNRS, iSm2 UMR 7313,

13397, Marseille, France

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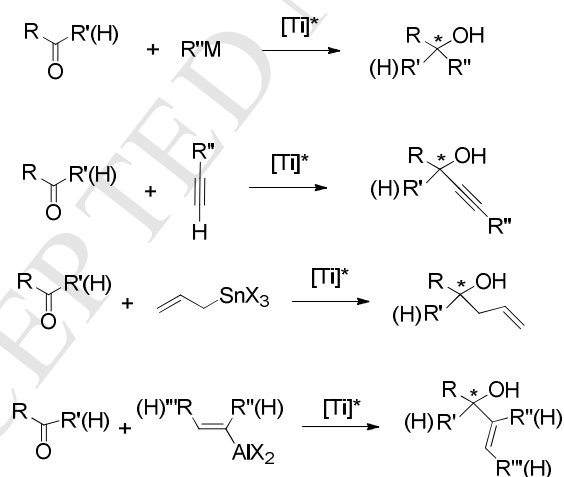
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*Corresponding author. Tel.: +33-4-91-28-27-65;

e-mail: h.pellissier@univ-amu.fr

Graphical Abstract

The goal of this review is to collect the major developments in enantioselective titanium-promoted 1,2-alkylation, 1,2-arylation, 1,2-alkynylation, 1,2-allylation, and 1,2-vinylation reactions of carbonyl compounds reported since the beginning of 2008, well illustrating the power of these inexpensive, nontoxic, and highly abundant catalysts to provide remarkably diverse chemical reactivity.

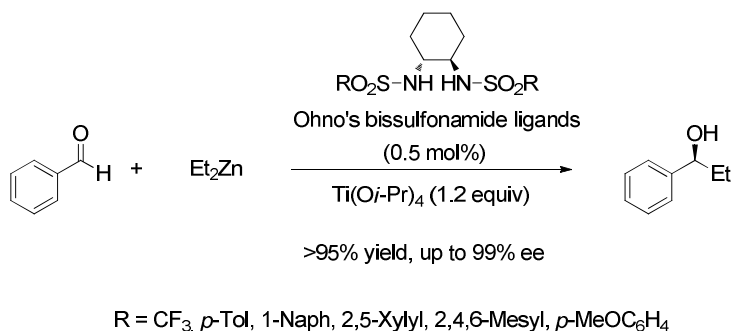


Keywords:

Asymmetric catalysis; Chirality; Titanium; 1,2-Additions; Carbon nucleophiles; Carbonyl compounds

1. Introduction

The catalysis of organic reactions by metal complexes constitutes one of the most useful and powerful tools in organic chemistry.¹ Although asymmetric synthesis is sometimes viewed as a subdiscipline of organic chemistry, actually this topical field transcends any narrow classification and pervades essentially all chemistry. Of the methods available for preparing chiral compounds, catalytic asymmetric synthesis has attracted most attention. In particular, asymmetric transition-metal catalysis has emerged as a powerful tool to perform reactions in a highly enantioselective fashion over the past few decades. Efforts to develop new asymmetric transformations focused preponderantly on the use of few metals, such as titanium, nickel, copper, ruthenium, rhodium, palladium, iridium, and more recently gold. However, by the very fact of the lower costs of titanium catalysts in comparison with other transition metals, and their nontoxicity which has permitted their use for medical purposes (prostheses), enantioselective titanium-mediated transformations have received a continuous ever-growing attention during the last decades that leads to exiting and fruitful researches.^{1j,2} This interest might also be related to the fact that titanium complexes are of high abundance, exhibit a remarkably diverse chemical reactivity, and constitute ones of the most useful Lewis acids in asymmetric catalysis. This usefulness is particularly highlighted in the area of enantioselective 1,2-alkylation, 1,2-arylation, 1,2-alkynylation, 1,2-allylation, and 1,2-vinylation reactions of carbonyl compounds. These methodologies have a strategically synthetic advantage to form a new C–C bond, a new functionality (alcohol) with concomitant creation of a stereogenic centre in a single transformation. Since the first enantioselective titanium-promoted addition of diethylzinc to benzaldehyde reported in 1989 by Ohno and Yoshioka which used chiral *trans*-1,2-bis(trifluoromethanesulfonylamino)cyclohexane as ligand (Scheme 1),³ enantioselective titanium-promoted additions of organometallic reagents to prochiral aldehydes and ketones have been studied extensively.



Scheme 1. First Ti-promoted enantioselective addition of diethylzinc to benzaldehyde reported by Ohno and Yoshioka in 1989

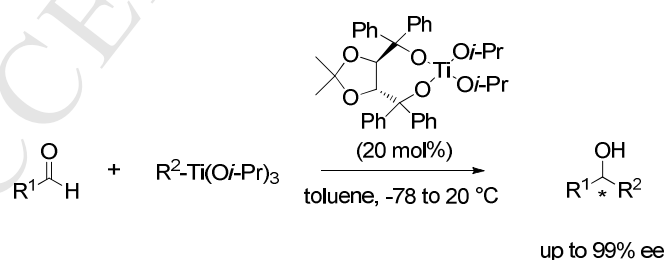
For example, important progress has been made recently in the design and development of chiral titanium Lewis acids for asymmetric catalysis of additions of organozinc reagents to carbonyl compounds to reach various chiral functionalised alcohols under relatively mild conditions on the basis of the extraordinary ability of chiral titanium catalysts to control stereochemistry which can be attributed to their rich coordination chemistry and facile modification of titanium Lewis acid centre by structurally modular ligands.^{2a,2d,4} In this context, good results have been recently reported dealing with enantioselective titanium-promoted dialkylzinc additions to more challenging aliphatic aldehydes than commonly used aromatic ones. In addition, a range of challenging functionalised alkylzinc reagents could be highly enantioselectively added to aldehydes. Concerning the alkylation and arylation of carbonyl compounds by organometallic reagents other than organozinc reagents, impressive advances have been made in the last few years by using chiral titanium catalysts. For example, the first highly efficient enantioselective titanium-promoted alkylations of aldehydes with organolithium reagents have been recently developed. Moreover, the direct additions of highly reactive alkyl and aryl Grignard reagents to all types of aldehydes at room temperature were recently demonstrated to give general excellent enantioselectivities when induced by chiral titanium catalysts. Importantly, the first direct titanium-promoted asymmetric additions of alkyl- and aryltitanium reagents to various aldehydes including aliphatic ones performed at room temperature were successfully developed. Another important advance was the first titanium-promoted enantioselective direct addition of alkylboranes including functionalised ones to aldehydes including aliphatic ones. Furthermore, in the context of enantioselective titanium-promoted additions to ketones, the first highly efficient enantioselective additions of (2-furyl)- and (2-thienyl)aluminum reagents to ketones have been described. For all these types of nucleophilic reagents, remarkable enantioselectivities were reached for

alkylation/arylation reactions. In another context, the enantioselective addition of organometallic alkynyl derivatives to carbonyl compounds is today the most expedient route toward chiral propargylic alcohols which constitute strategic building blocks for the enantioselective synthesis of a range of complex important molecules. In the last few years, impressive advances have been made in this area particularly in the variety of alkynes used to be added to aldehydes. Besides excellent results afforded with phenylacetylene, remarkable enantioselectivities were observed for a range of other terminal (functionalised) alkynes, such as *para*-tolylacetylene, trimethylsilylacetylene, ethynylcyclohexene, 4-phenyl-1-butyne, 5-chloro-1-pentyne, 1-hexyne, 1-heptyne, 1-octyne, various alkynoates, as well as 1,3-diynes and 1,3-enynes. In the context of enantioselective alkynylations of ketones, the first successful use of aryltrifluoromethyl ketones was described. Importantly, several supported chiral ligands have been recently successfully applied to the catalysis of almost all types of 1,2-additions, such as enantioselective dialkylzinc additions to ketones, enantioselective alkynylations of aldehydes, and enantioselective allylations of ketones. Although most of the novel methods collected in this review require superstoichiometric amounts of titanium sources (along with catalytic amounts of chiral ligands), they remain highly useful regarding the advantages of titanium elements, such as low cost, abundance, and low toxicity. The goal of this review is to provide a comprehensive overview of the major developments in enantioselective titanium-promoted 1,2-alkylation, 1,2-arylation, 1,2-alkynylation, 1,2-allylation, and 1,2-vinylation reactions of carbonyl compounds reported since the beginning of 2008, since this general field was previously reviewed this year by Yu and co-workers in a book chapter dealing with titanium Lewis acids.^{2a} In addition, little parts of this field were included in several reviews not especially based on titanium catalysis.^{4a,5,6,7,8,9} For readers convenience, the review has been divided into three parts. The first part deals with enantioselective titanium-promoted alkylation and arylation reactions of carbonyl compounds, the second part includes enantioselective titanium-promoted alkynylation reactions of carbonyl compounds, while the third part collects enantioselective titanium-promoted allylation and vinylation reactions of carbonyl compounds.

2. Titanium-promoted alkylation and arylation reactions

The formation of a carbon–carbon bond via nucleophilic addition of an organometallic reagent to a carbonyl substrate constitutes one of the most elementary transformations in

organic synthesis and has been studied extensively during the last several decades.⁷ The dawn of organometallic chemistry dates back to 1849 with Frankland's early work on organozinc compounds.¹⁰ By the turn of the 20th century, the routine use of organozinc reagents in organic synthesis was largely supplanted by main-group organometallics thanks to the rapid growth of Grignard chemistry,¹¹ and the development of practical routes to organolithium compounds.¹² Actually, the genesis of enantioselective addition to carbonyl compounds dates to 1940 with a report by Betti and Lucchi on the reaction of methylmagnesium iodide with benzaldehyde in the presence of *N,N*-dimethylbornylamine as solvent to give enantioenriched 1-phenylethanol.¹³ However, it was demonstrated later that the slight optical rotation observed apparently originated from an optically active by-product generated from the *N,N*-dimethylbornylamine solvent. In the 1950s, Wright et al. reported what appears to be the first successful enantioselective addition of Grignard reagents to carbonyl compounds, using chiral ethers as cosolvents, providing low enantioselectivities of 17% ee.¹⁴ Later in 1989, Ohno and Yoshioka reported the first enantioselective titanium-promoted addition of diethylzinc to benzaldehyde which allowed enantioselectivities of up to 99% ee to be achieved by using chiral *trans*-1,2-bis(trifluoromethanesulfonylamino)cyclohexane as ligand (Scheme 1).³ In 1994, Seebach and Weber described the first enantioselective truly catalytic alkyl and aryl additions to aldehydes employing a highly reactive $\text{RTi}(\text{O}i\text{-Pr})_3$ reagent which provided enantioselectivities of up to 99% ee upon catalysis with a titanium TADDOLate complex (Scheme 2).¹⁵ After these two remarkable pioneering works (Schemes 1-2), chemists have shown a continuous interest in developing highly enantioselective catalysts for the asymmetric alkyl and aryl transfer to aldehydes.



Scheme 2. First Ti-catalysed enantioselective alkyl and aryl additions to aldehydes reported by Seebach and Weber in 1994

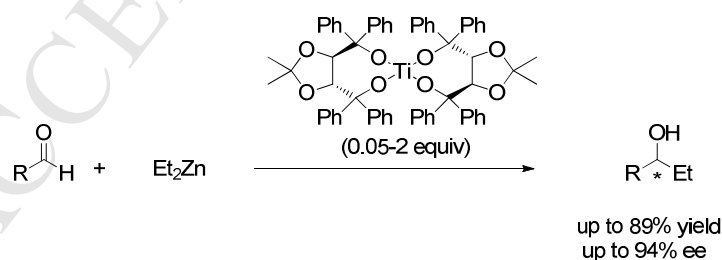
The nucleophilic 1,2-addition reactions of organometallic reagents, such as organozinc, aluminum, magnesium, titanium, and lithium species in addition to boron reagents, to carbonyl compounds can be mediated by titanium complexes, through transmetalation of

organometallic reagents or by enhancing the electrophilicity of the carbonyl compounds via titanium coordination. Alkyltitanium complexes can be obtained from metal carbanions via titanation. Introduction of chirality at the titanium centre or on the ligand (or a combination of both) enables the possibility of asymmetric induction in the carbonyl addition reaction.

2.1. Additions of dialkylzinc reagents

2.1.1. Aldehydes as electrophiles

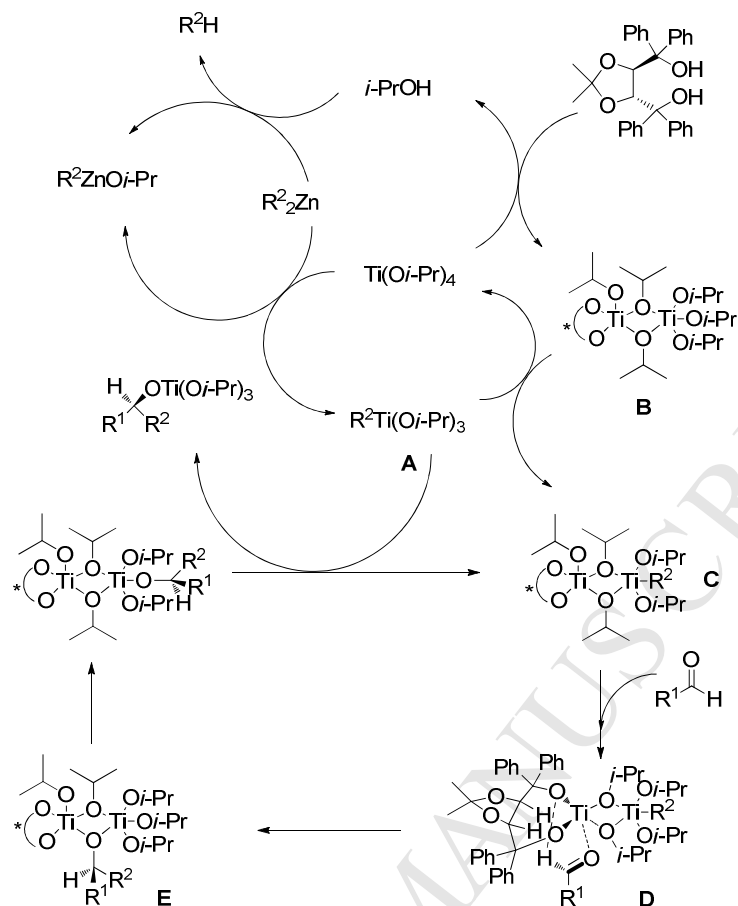
2.1.1.1. Using BINOL-derived ligands. In 1983, Oguni and Omi reported the first reaction of diethylzinc with benzaldehyde performed in the presence of a catalytic amount of (*S*)-leucinol achieved with a moderate enantioselectivity (49% ee).¹⁶ At the same time, Reetz et al. reported stereoselective reactions of titanium reagents with chiral alkoxycarbonyl compounds.¹⁷ Later in 1986, Noyori and co-workers discovered (-)-DAIB as the first highly enantioselective ligand for the dialkylzinc addition to aldehydes, providing enantioselectivities of up to 95% ee.¹⁸ Concerning titanium as promoter, it was in 1989 that Yoshioka and Ohno³ reported the first titanium-promoted enantioselective addition of dialkylzinc reagents¹⁹ to aldehydes,^{5b-d,7g} using chiral *trans*-1,2-bis(trifluoromethanesulfonylamino)cyclohexane as ligand of Ti(Oi-Pr)₄ (Scheme 1). The titanium catalyst was prepared in situ in the presence of the diorganozinc. Later in 1991, Seebach and Schmidt demonstrated that TADDOL-derived titanium complexes also functioned as efficient asymmetric catalysts (Scheme 3).²⁰



Scheme 3. Ti-promoted asymmetric addition of diethylzinc to aldehydes reported by Seebach and Schmidt in 1991

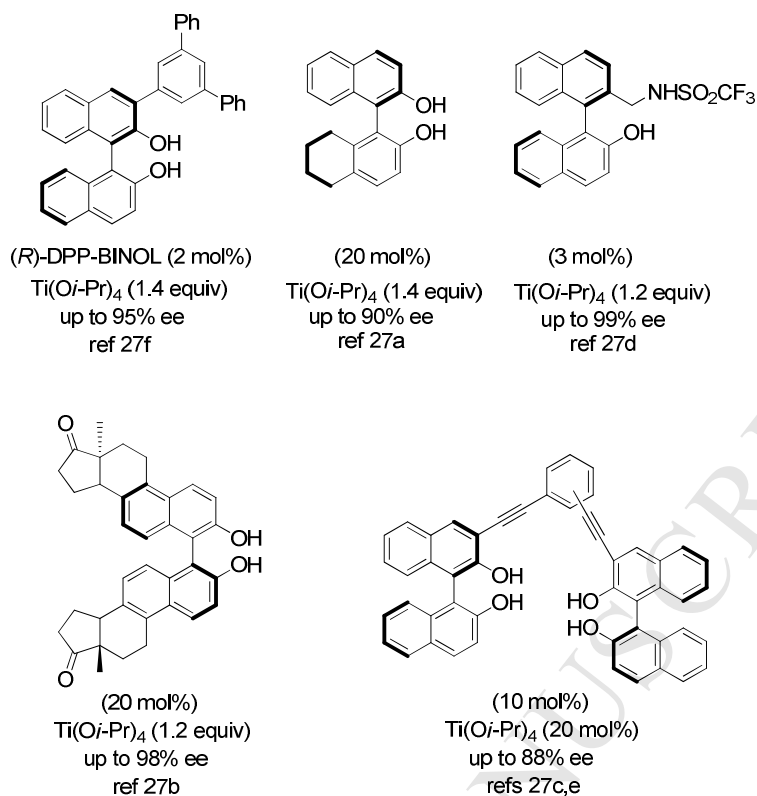
Ever since, a number of chiral titanium complexes have been developed and high enantioselectivities have been reached.^{5,21} Although the exact mechanism of the enantioselective addition of dialkylzinc reagents (R^2_2Zn) to aldehydes (R^1CHO) performed in

the presence of an excess of $\text{Ti}(\text{Oi-Pr})_4$ and substoichiometric amounts of chiral ligands is still not well-known, studies on reactions induced by TADDOL ligands and reported in 1990s by Seebach et al. have allowed the mechanism depicted in Scheme 4 to be proposed.^{15,22} It begins with the alkyl exchange between zinc reagent R^2_2Zn and $\text{Ti}(\text{Oi-Pr})_4$ to generate new alkyl titanium complex **A**, which was detected by NMR studies.^{7e,23} The role of titanium is not limited to the preparation of this complex but also to that of bimetallic complex **B** bearing only one chiral ligand.^{3a,24} This μ -oxo complex was assumed to be formed by two isopropoxide groups in the bridge, according to the symmetry of NMR spectra. The alkyl exchange between complex **B** and either alkyltitanium intermediate **A** or the starting dialkylzinc reagent provides new complex **C**. In this complex, the coordination of the aldehyde takes place, and although there are two possible coordinating atoms, the titanium atom coordinated to the chiral ligand is more active owing to a faster ligand exchange, which is due to the bulkiness of the ligand compared to isopropoxide groups. The catalytically active species seems to be the bimetallic complex **D**. In this complex as in all TADDOL derivatives, the two phenyl substituents are situated in different conformational positions. The phenyl groups placed in a pseudoaxial position are responsible for the enantioselectivity of the addition while the pseudoequatorial phenyl groups are necessary for a fast exchange between the aldehyde and the isopropoxide group or between the final chiral bulky alcohol and isopropoxide. The fact that aliphatic, aromatic, and α,β -unsaturated aldehydes afforded the same level of enantioselectivity, as well as topological reaction sense, seems to indicate that there is not π -stacking or charge transfer interactions between the aldehyde and the phenyl group on the TADDOL ligand, corroborating that only van der Waals interactions between pseudoaxial phenyl groups and the aldehyde chain control the stereochemical outcome of the addition. Furthermore, a hydrogen bond between the oxygen atom of the ligand and the hydrogen atom of the carbonyl moiety can favor this complexation process.²⁵ The final fast exchange of ligands liberates the chiral alcohol product and regenerates the starting bimetallic complex **C**. Although Scheme 4 depicts a general mechanism for the enantioselective addition of dialkylzinc to aldehydes in the presence of $\text{Ti}(\text{Oi-Pr})_4$ and any other chiral ligand, depending on the ligand and the reaction conditions used, other factors and reaction pathways must be taken into account.²⁶



Scheme 4. Proposed catalytic cycle for enantioselective titanium-catalysed dialkylzinc addition to aldehydes using a TADDOL ligand

Until 2008, a number of titanium chiral ligands have been successfully applied to induce chirality in addition of dialkylzinc reagents to aldehydes. Among them, a range of BINOL derivatives have been developed by several groups, allowing moderate to excellent enantioselectivities of up to 99% ee to be achieved by using 2-20 mol% of ligands.²⁷ In most cases, stoichiometric or superstoichiometric amounts of titanium were employed. Some of the best results are summarised in Scheme 5.

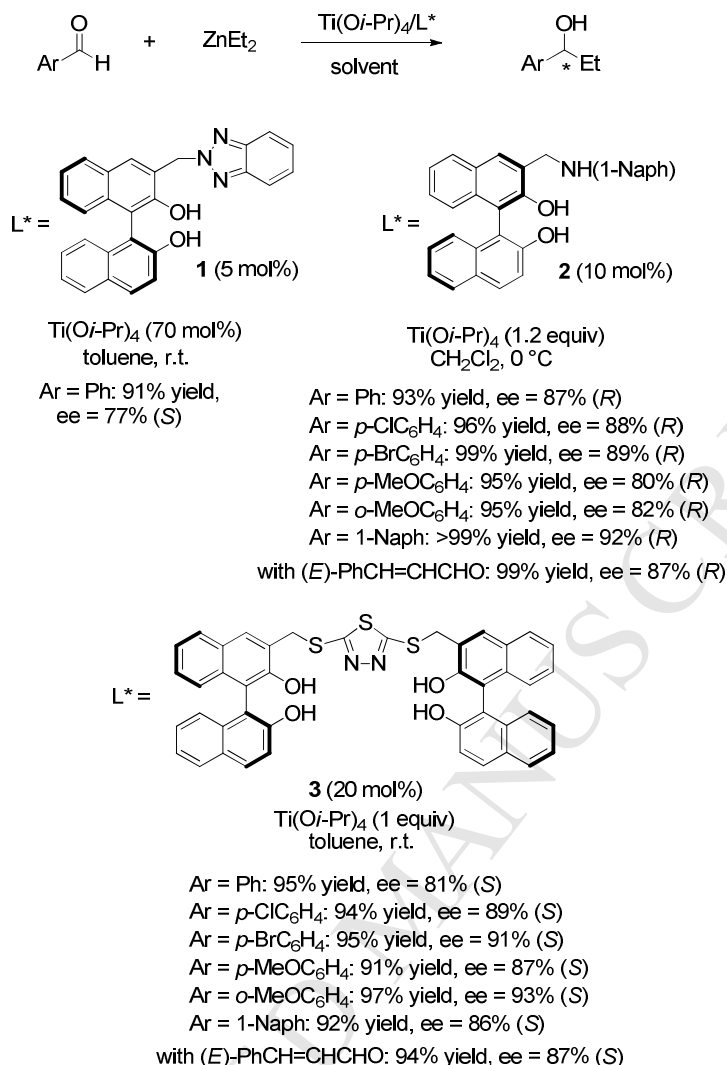


Scheme 5. Chiral BINOL-derived ligands in early titanium-promoted dialkylzinc additions to aldehydes

Inspired by these pioneering works, a variety of novel BINOL-derived chiral ligands have been designed by different groups in the last seven years to be investigated in these reactions. In a recent example, Li and co-workers reported the synthesis of a range of novel 3-substituted chiral BINOL ligands to be applied to enantioselective diethylzinc addition to aromatic aldehydes.²⁸ Therefore, three 3-aminomethyl-substituted BINOL ligands, such as (S)-3-(1*H*-imidazol-1-yl)methyl-1,1'-binaphthol, (S)-3-(1*H*-1,2,3-benzotriazol-1-yl)methyl-1,1'-binaphthol, and (S)-3-(2*H*-1,2,3-benzotriazol-2-yl)methyl-1,1'-binaphthol, were easily synthesised from (S)-2,2'-dimethoxymethyl-1,1'-binaphthol in four steps and further investigated as chiral ligands of $\text{Ti}(\text{O}i\text{-Pr})_4$ in addition of diethylzinc to benzaldehyde. The best enantioselectivity of 77% ee combined with 91% yield was achieved by using (S)-3-(2*H*-1,2,3-benzotriazol-2-yl)methyl-1,1'-binaphthol **1** at 5 mol% of catalyst loading, as shown in Scheme 6. It is important to highlight that this work represented a rare example of using only substoichiometric amounts of $\text{Ti}(\text{O}i\text{-Pr})_4$ (70 mol%) in enantioselective titanium-catalysed alkylation of aldehydes in spite of a limited scope.

Better enantioselectivities were reached by the same authors employing other chiral 3-substituted aminomethyl BINOL derivatives in the reaction of diethylzinc with a range of aromatic aldehydes.²⁹ As shown in Scheme 6, enantioselectivities of up to 92% ee were obtained by inducing the reaction with chiral 3-arylaminomethylBINOLs, such as (*R*)-3-(naphthalene-1-ylamino)methyl-1,1'-binaphthol **2**. This ligand was selected among three novel chiral 3-substituted BINOL Schiff bases which provided moderate enantioselectivities ($\leq 77\%$ ee) and their reductive 3-arylaminomethyl-BINOL derivatives. Using the most efficient ligand **2**, the best result was achieved for sterically hindered 1-naphthalenecarbaldehyde with an almost quantitative yield in combination with an enantioselectivity of 92% ee. Using this ligand, the authors found that the presence of an electron-withdrawing substituent on the substrate benzaldehyde increased the enantioselectivity of the reaction (88 and 89% ee, respectively, with $R = p\text{-ClC}_6\text{H}_5$ and $p\text{-BrC}_6\text{H}_5$) while the presence of electron-donating groups decreased the enantioselectivity (80 and 82% ee, respectively, with $R = p\text{-MeOC}_6\text{H}_5$ and $o\text{-MeOC}_6\text{H}_5$). In addition, the scope of the process could be applied to an α,β -unsaturated aldehyde, such as cinnamaldehyde, which provided a good enantioselectivity of 87% ee.

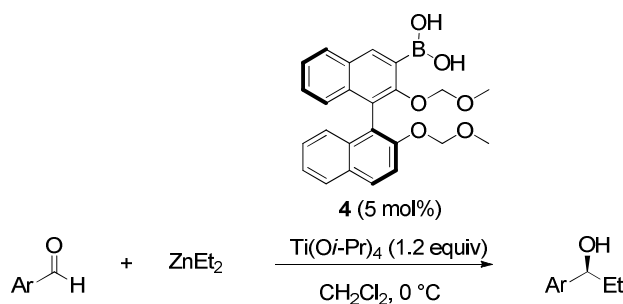
Moreover, these authors investigated novel BINOL-based ligands bounded with both sulfur-contained heterocycle, such as thiazole or thiadiazole, and thioether block in which the sulfur could serve as a talent anchor.³⁰ The active chiral titanium catalyst in the enantioselective addition of diethylzinc to benzaldehyde was in situ generated from $\text{Ti}(\text{O}i\text{-Pr})_4$ and the chiral ligand. Among four ligands tested, ligand (*S,S*)-2,5-bis(2,2'-dihydroxy-1,1'-binaphthalene-3-yl)-1,3,4-thiadiazole **3** was found to be the most efficient, providing the corresponding (*S*)-secondary alcohol in enantioselectivity of 81% ee. The scope of the process was extended to other aromatic aldehydes to give the corresponding chiral alcohols in enantioselectivities of up to 93% ee in combination with general excellent yields ranging from 91 to 97%, as shown in Scheme 6. The best enantioselectivity of 93% ee was reached with *ortho*-methoxybenzaldehyde. In addition, the scope of the process could be applied to an α,β -unsaturated aldehyde, such as cinnamaldehyde, which provided a good enantioselectivity of 87% ee.



Scheme 6. 3-Substituted chiral BINOL ligands in addition of diethylzinc to aromatic (and α,β -unsaturated) aldehydes

In 2010, the same authors described a novel 3-substituted chiral ligand derived from (*S*)-BINOL such as (*S*)-3-dihydroxyborane-2,2'-bis(methoxymethoxy)-1,1'-binaphthyl **4**.³¹ This ligand was easily synthesised in 82% overall yield starting from commercially available (*S*)-BINOL through a six-step sequence. The authors further studied its catalytic activity to induce diethylzinc addition to benzaldehyde among a range of a series of (*S*)-BINOL-derived ligands substituted at the 3 position with some five-membered nitrogen-containing aromatic heterocycles. They found that ligand **4** employed at 5 mol% of catalyst loading exhibited the best catalytic efficiency, providing an enantioselectivity of 83% ee combined with an almost quantitative yield (Scheme 7). The scope of the procedure was extended to a range of aromatic and heteroaromatic aldehydes with diverse electronic and steric properties, which

gave enantioselectivities of up to 91% ee in combination with very high to quantitative yields. As shown in Scheme 7, both the position and electronic nature of the substituents on the phenyl ring could affect enantioselectivities dramatically. On the other hand, very good yields were surprisingly gained in all cases of substrates studied. Various benzaldehyde derivatives, bearing no matter electron-withdrawing or electron-donating groups substituted at the *para* position, were smoothly converted to the corresponding alcohols with similar enantioselectivities than benzaldehyde. On the other hand, the presence of *ortho*-substituents, regardless of electronic-rich or electronic-poor groups, diminished the enantioselectivity probably because these substituents could weaken the coordination of the aldehyde to the chiral catalyst and thus reduce the effect of its chiral environment. While 1-naphthaldehyde gave a good result (91% ee), the lowest enantioselectivity (48% ee) was gained in the case of 2-furaldehyde. In addition, the scope of the process could be applied to an α,β -unsaturated aldehyde, such as cinnamaldehyde, which quantitatively provided the corresponding product in a moderate enantioselectivity of 80% ee.



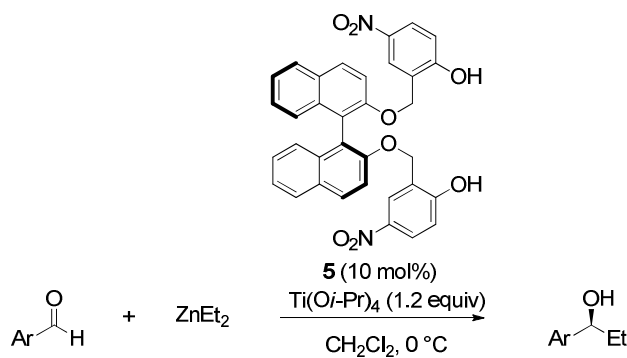
Ar	Yield (%)	ee (%)
Ph	>99	83
<i>p</i> -FC ₆ H ₄	>99	82
<i>p</i> -BrC ₆ H ₄	98	84
<i>p</i> -MeOC ₆ H ₄	97	80
<i>p</i> -CF ₃ C ₆ H ₄	78	85
<i>p</i> -(NMe ₂)C ₆ H ₄	90	82
<i>p</i> -((EtO) ₂ CH)C ₆ H ₄	98	84
<i>m</i> -MeOC ₆ H ₄	93	88
<i>m</i> -Tol	83	78
<i>p</i> -Tol	92	83
<i>o</i> -MeOC ₆ H ₄	>99	76
<i>o</i> -ClC ₆ H ₄	93	70
2,4-Cl ₂ C ₆ H ₃	>99	74
3,4-Cl ₂ C ₆ H ₃	86	75
1-Naph	>99	91
2-furyl	98	48
5-piperonyl	>99	82

with (*E*)-PhCH=CHCHO: >99% yield, ee = 80%

Scheme 7. Another 3-substituted chiral BINOL ligand in addition of diethylzinc to aromatic (and α,β -unsaturated) aldehydes

In addition to chiral 3-substituted BINOL ligands, Judeh and Gou have described the synthesis and applications of novel chiral 2,2'-disubstituted BINOL ligands to the same reactions.³² As shown in Scheme 8, the use of ligand **5** bearing two free OH groups and synthesised in one step from (*S*)-BINOL allowed a range of chiral secondary aromatic alcohols to be achieved in good to high yields and enantioselectivities of up to 89% ee. Among a range of aromatic aldehydes having electron-donating and electron-withdrawing groups investigated, it was found that poor electron-donating substituents, such as methyl and phenyl, led to a slight decrease in the enantiomeric excesses of the products, whereas the

para-ethyl group resulted in an increase in the enantioselectivity (83% ee) in comparison with the results obtained with benzaldehyde. On the other hand, electron-withdrawing groups (such as F, Cl, Br and I) showed variations in the yields, but no major differences in the enantiomeric excesses except for the strongly electron-withdrawing CF₃ group which led to a lower enantioselectivity (64% ee). Moreover, reactions of 1- and 2-naphthaldehydes resulted in excellent yields (92 and 95%, respectively) and moderate enantioselectivities (72 and 75% ee, respectively). In this study, the authors have demonstrated that ligands not bearing OH groups were unable to promote the reaction, indicating that the presence of a free OH on the ligand skeleton was indispensable for the catalytic activity.

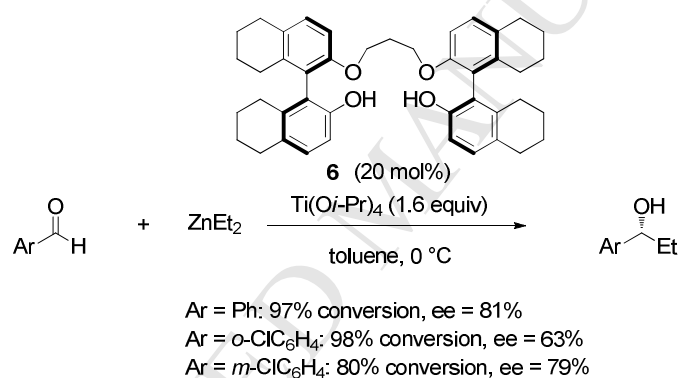


Ar	Yield (%)	ee (%)
Ph	90	77
<i>o</i> -Tol	84	73
<i>m</i> -Tol	88	67
<i>p</i> -Tol	92	68
<i>p</i> -EtC ₆ H ₄	87	83
<i>p</i> -PhC ₆ H ₄	89	67
<i>o</i> -MeOC ₆ H ₄	77	78
<i>m</i> -MeOC ₆ H ₄	73	61
<i>p</i> -MeOC ₆ H ₄	80	79
<i>m</i> -BnOC ₆ H ₄	71	83
<i>p</i> -FC ₆ H ₄	70	79
<i>p</i> -ClC ₆ H ₄	93	84
<i>p</i> -BrC ₆ H ₄	88	69
<i>p</i> -IC ₆ H ₄	81	89
<i>p</i> -CF ₃ C ₆ H ₄	76	64
<i>o</i> -IC ₆ H ₄	76	81
<i>m</i> -IC ₆ H ₄	84	82
1-Naph	92	72
2-Naph	95	75

Scheme 8. 2,2'-Disubstituted chiral BINOL ligand in addition of diethylzinc to aromatic aldehydes

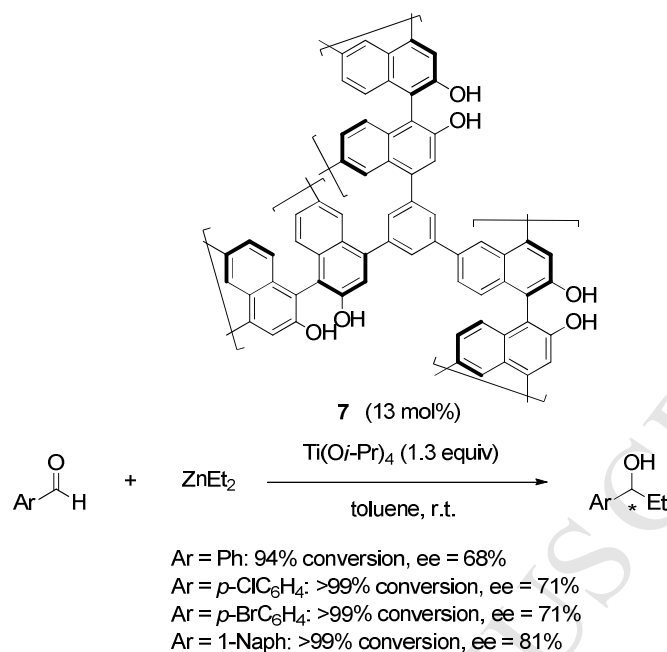
In 2010, Pereira et al. reported the preparation of novel chiral BINOL-derived ligands consisting of two BINOL or H₈-BINOL fragments joined by diverse linkages through the oxygen at the 2'-position of the aryl fragments.³³ These ligands were further investigated as promoters in the addition of diethylzinc to benzaldehyde in the presence of Ti(Oi-Pr)₄. It was shown that the performance of these catalysts was very sensitive to the nature of the ether linkage. The ligand with a propylene link provided a better enantioselectivity (70% ee) than

those with two or four carbon atoms joining the BINOL fragments. Furthermore, using the propylene link, but replacing (*R*)-BINOL by (*R*)-H₈-BINOL in ligand **6**, a significant improvement in the enantioselectivity of the reaction was achieved (81% ee), as shown in Scheme 9. The scope of this methodology was extended to several other aromatic aldehydes which provided the corresponding chiral alcohols in enantioselectivities of up to 79% ee (Scheme 9).³⁴ A significant influence of the aldehyde structure on the enantioselectivity was observed. For example, the enantiomeric excess obtained for the alkylation of 2-chlorobenzaldehyde was significantly lower (63% ee) than those obtained with 3-chlorobenzaldehyde (79% ee) or benzaldehyde (81% ee). Thus, the best results were obtained with benzaldehydes not substituted on the *ortho* position. A drawback of this process was its narrow scope.



Scheme 9. Chiral *bis*-H₈-BINOL-2,2'-propylether ligand in addition of diethylzinc to aromatic aldehydes

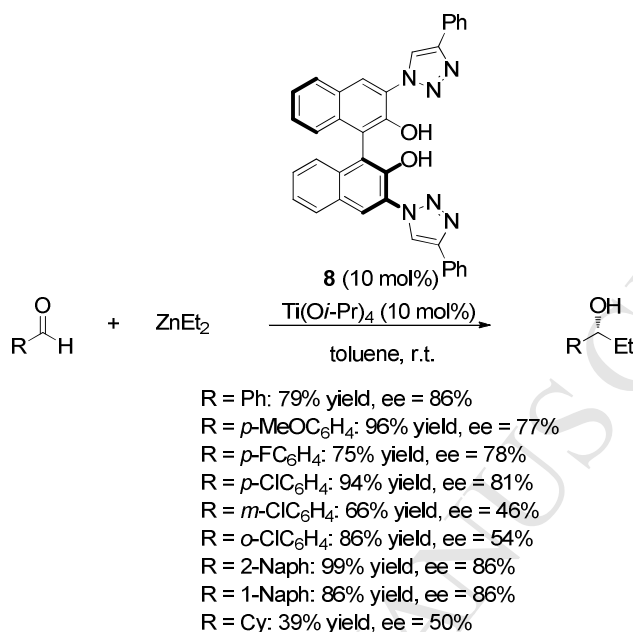
In 2011, a series of chiral cross-linked titanium polymers based on the 1,1'-binaphthyl building blocks were synthesised by Lin et al. via cobalt-catalysed trimerisation reaction of terminal alkyne groups.³⁵ These highly porous cross-linked polymers containing chiral dihydroxy functionalities were treated with Ti(Oi-Pr)₄ to generate chiral Lewis acid catalysts for asymmetric addition of diethylzinc to aromatic aldehydes. Along with excellent conversions, the observed enantioselectivities were moderate to good, since the most efficient ligand **7** provided enantioselectivities of 68 to 81% ee, as shown in Scheme 10. However, it must be noted that this polymer presented the advantage to be readily recycled and reused for up to ten times without loss of conversion and enantioselectivity.



Scheme 10. Chiral cross-linked polymer ligand in addition of diethylzinc to aromatic aldehydes

In 2012, a new class of easily tunable chiral 1,2,3-triazole-BINOL ligands was developed by Mancheno and Beckendorf and their activity in asymmetric Lewis acid catalysis explored for the first time in the diethylzinc addition to aldehydes.³⁶ It was shown that ligands with mono- and bis-methylene-bridged triazoles led to results similar to the parent BINOL (85% yield, 54-58% ee vs 84% yield, 53% ee). Conversely, ligands having triazole units directly linked to the binaphthol backbone showed more interesting and promising results. Among them, ligand **8** showed an interesting catalytic behavior, which suggested the non-innocent participation of the triazole units in both the formation and reactivity of the active titanium catalyst. Good enantioselectivities of up to 86% ee were obtained by both the right selection of the substitution pattern at the triazole ring (phenyl at 4-position) and the fine tuning of the reaction conditions (10 mol% of catalyst loading in toluene at room temperature). As shown in Scheme 11, 2- and 1-naphthaldehydes, as well as both electron-withdrawing and electron-donating *para*-substituted benzaldehydes reacted well, giving the corresponding alcohols in comparable good enantioselectivities. On the other hand, *meta*- and *ortho*-substituted benzaldehydes, as well as aliphatic cyclohexyl carboxaldehyde, afforded the corresponding

alcohols in significantly lower ee values (46-54% ee). It must be highlighted, however, that this process presented the advantage to employ only 10 mol% of $\text{Ti}(\text{Oi-Pr})_4$.

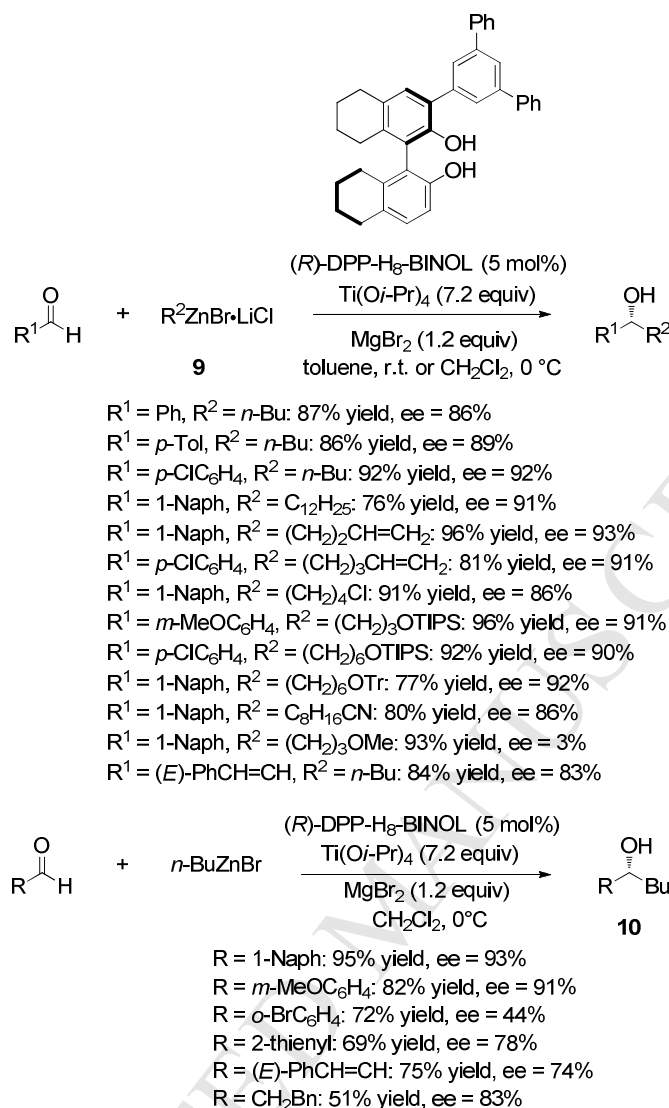


Scheme 11. Chiral 1,2,3-triazole-BINOL ligand in addition of diethylzinc to aldehydes

Always in the area of BINOL-derived ligands, Harada et al. have studied the enantioselective titanium-promoted alkylation of aldehydes by using more challenging functionalised alkylzinc bromides for the first time.³⁷ It was shown that the reactivity of these organozinc halide reagents was enhanced by mixing them with $\text{Ti}(\text{Oi-Pr})_4$ and MgBr_2 . In the presence of chiral ligand (*R*)-DPP-H₈-BINOL, a variety of functionalised alkylzinc reagents prepared from readily available bromide precursors underwent enantioselective addition to aromatic and α,β -unsaturated aldehydes to give the corresponding functionalised chiral alcohols in good to high enantioselectivities (Scheme 12, first equation). For example, enantioselectivities of up to 93% ee were achieved in combination with general high yields by using a range of variously functionalised zinc reagents **9** prepared from the corresponding bromide precursors by treatment with zinc dust in the presence of LiCl. Remarkably, silyloxy-, and alkoxy-substituted alkylzinc reagents afforded the corresponding mono-protected diols in high enantioselectivities (90-92% ee). However, an exception was observed for the reaction of 3-methoxypropyl zinc reagent for which a non-enantioselective reaction was observed (ee

= 3%), probably due to a background racemic reaction promoted by an intermolecular coordination of the zinc atom of the reagent by the neighboring methoxy group. On the other hand, a zinc reagent bearing a remote cyano group led to the corresponding product in 86% ee.

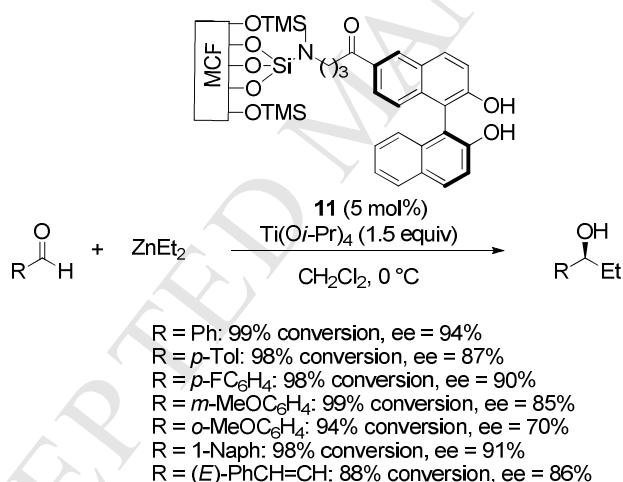
The same reaction conditions were applied to the enantioselective addition of *n*-BuZnBr to various aromatic, heteroaromatic, and α,β -unsaturated aldehydes which provided the corresponding chiral alcohols **10** in generally lower enantioselectivities except for naphthaldehyde (93% ee), as shown in Scheme 12 (second equation). For *para*- and *meta*-substituted benzaldehyde derivatives, both high yields and enantioselectivities (89-92% ee) were reached while a moderate enantioselectivity (44% ee) was observed in the case of *ortho*-bromobenzaldehyde. Moreover, the authors showed that the reaction tolerated an aliphatic aldehyde (R = CH₂Bn) but the reaction required four days at 5 °C to afford the corresponding product in relatively high enantioselectivity of 83% ee. It is worth mentioning that this work belongs to the rare examples of enantioselective additions of functionalised alkylzinc reagents to a variety of aldehydes reported so far, and moreover it presents the advantage to use only 5 mol% of ligand loading and offers excellent results.



Scheme 12. Chiral 3,5-diphenylphenyl-H₈-BINOL ligand in additions of (functionalised) organozinc bromide reagents to aldehydes

In the last few years, two important results for the enantioselective ethylation of aromatic aldehydes (and α,β -unsaturated aldehydes) have to be highlighted related to the fact that they concerned the successful use of chiral supported ligands (Schemes 13-14). The first example was reported by Abdi et al. who described good to high enantioselectivities of up to 94% ee for the reaction of various aldehydes with diethylzinc induced by silica-supported chiral BINOL ligand **11** used at 5 mol% of catalyst loading in the presence of 1.5 equivalent of Ti(Oi-Pr)₄.³⁸ This heterogenised ligand was covalently anchored on two different relatively large pore sized mesoporous silicas (SBA-15) (7.5 nm) and mesocellular foams (MCF) (14 nm) by covalent grafting method using *N*-methyl-3-aminopropyltriethoxysilane as reactive

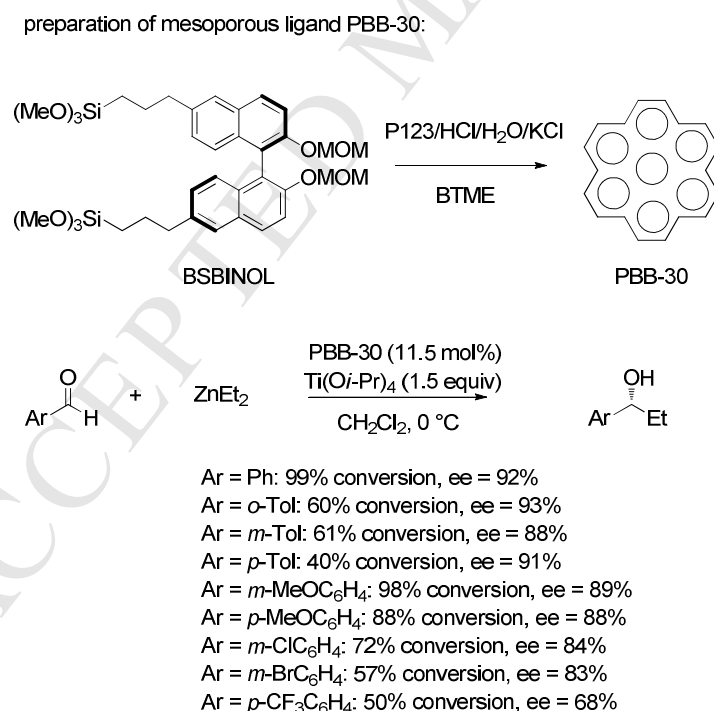
surface modifier. As shown in Scheme 13, the reaction of small as well as bulkier aldehydes afforded the corresponding chiral secondary alcohols with excellent conversions of 94 to 99% and good to very high enantioselectivities of up to 94% ee in the case of aromatic aldehydes while cinnamaldehyde provided an 88% conversion combined with an enantioselectivity of 86% ee. It must be noted that the substituents on benzaldehyde derivatives had some influence on the reactivity and enantioselectivity of the reaction, since *para*-substituted aldehydes showed better reactivity with respect to conversion and enantioselectivity than *ortho*-substituted benzaldehyde (Scheme 13). This was probably due to the strong steric effect of the *ortho*-substituent which could deteriorate the coordination of the substrate to the chiral catalyst thus lowering the reactivity. The MCF-supported BINOL catalyst could be reused in several catalytic runs without significant drop of the enantioselectivity. The pore size of silica supports and capping of free silanol groups with TMS groups on the silica surface were found to be important toward achieving high enantioselectivities.



Scheme 13. Silica-supported chiral BINOL ligand in addition of diethylzinc to aromatic and α,β -unsaturated aldehydes

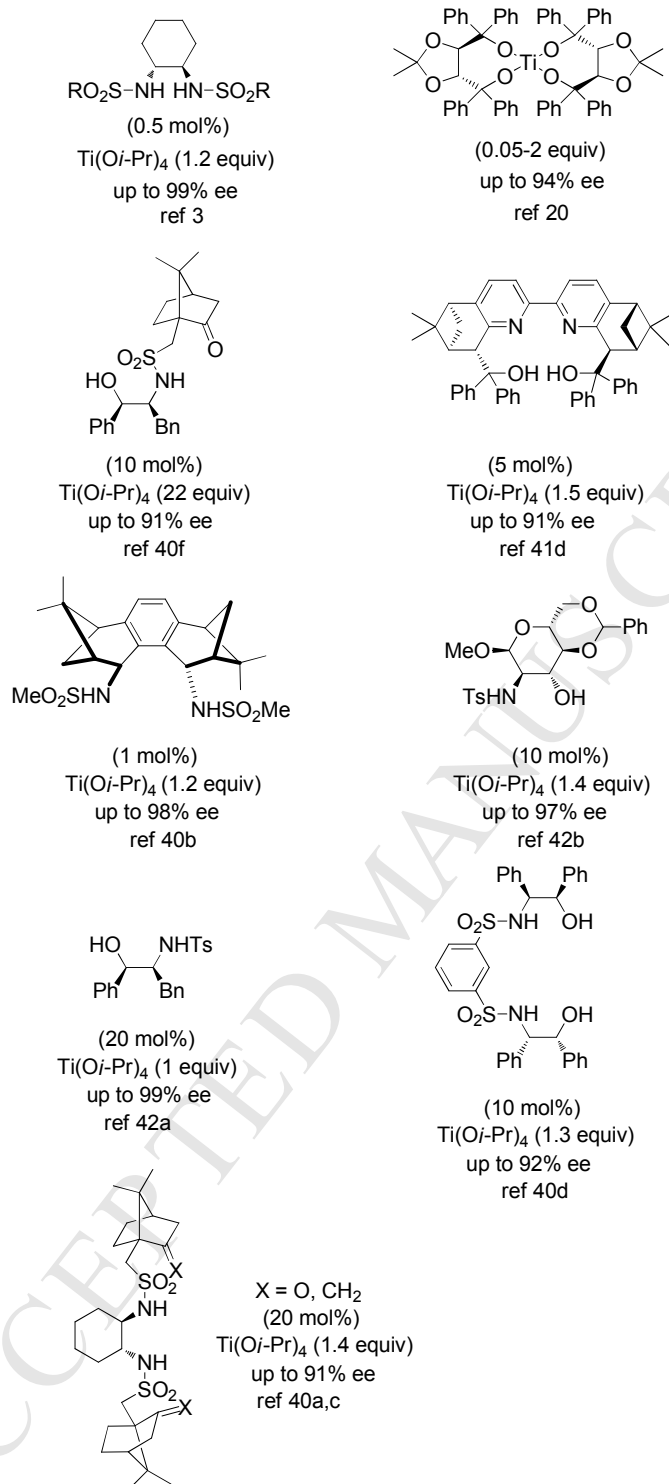
Chiral periodic mesoporous organosilicas with chiral ligands in the framework are novel chiral porous solids which have demonstrated application potential in asymmetric catalysis, constituting a challenge in the field of heterogeneous asymmetric catalysis. In 2010, Yang et al. reported the synthesis of (*R*)-BINOL-functionalised mesoporous organosilicas PPB-30, based on the cocondensation of 1,2-bis(trimethoxysilyl)ethane (BTME) and (*R*)-2,2'-di(methoxymethyl)oxy-6,6'-di(1-propyltrimethoxysilyl)-1,1'-binaphthyl (BSBINOL) as a

chiral silane precursor in an acidic medium using the P123 surfactant as the template (Scheme 14).³⁹ When applied to the same reactions as above, this chiral heterogeneous ligand exhibited higher enantioselectivity but lower catalytic activity than its homogeneous counterpart in CH_2Cl_2 as solvent. As shown in Scheme 14, a range of chiral aromatic secondary alcohols was obtained in good to high enantioselectivities of up to 93% ee. The results indicated that the size of the substrates and the electronic and steric properties of their substituents had a remarkable effect on the catalytic activity and enantioselectivity of the promotor. For example, substrates bearing electron-donating groups provided better enantioselectivities than those bearing electron-withdrawing groups (Br , Cl , CF_3). Moreover, it was shown that steric hindrance played an important role in the enantioselectivity values for the substrates with electron-withdrawing groups. In addition to present the advantage to be more enantioselective than its homogeneous counterpart, this catalyst was shown recyclable since 88% conversion combined with 89% ee were obtained in the second run of the ethylation of benzaldehyde instead of 99% conversion and 92% ee for the first run.



Scheme 14. (*R*)-BINOL-functionalised mesoporous organosilica ligand in addition of diethylzinc to aromatic aldehydes

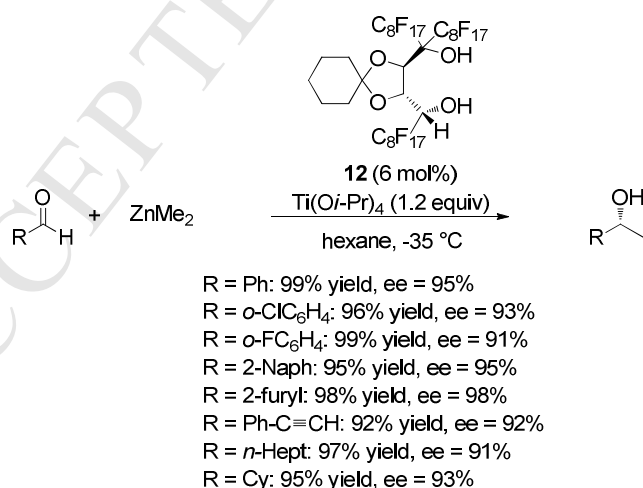
2.1.1.2. *Using other ligands.* In addition to BINOL-derived ligands, a range of other types of chiral ligands has been successfully investigated in enantioselective titanium-promoted addition of zinc reagents to aldehydes. From 1989 to 2008, remarkable results were reported by several groups using TADDOL ligands initiated by the pioneering work of Seebach and Schmidt in 1991,²⁰ chiral sulfonamide ligands⁴⁰ initiated by the pioneering work of Ohno and Yoshioka in 1989,³ chiral diol or triol ligands,⁴¹ and chiral amino alcohols⁴² among other ligands, which allowed in some cases enantioselectivities of up to 99% ee to be reached by using 0.5-20 mol% of ligand in combination with superstoichiometric amounts of $\text{Ti}(\text{Oi-Pr})_4$. Some of the best early results are collected in Scheme 15.



Scheme 15. Other chiral ligands in early titanium-promoted dialkylzinc additions to aldehydes

As a more recent and highly efficient example inspired by the pioneering work reported by Seebach and Schmidt on TADDOL-derived catalysts (Scheme 15, ref 20, or Scheme 3), Ando

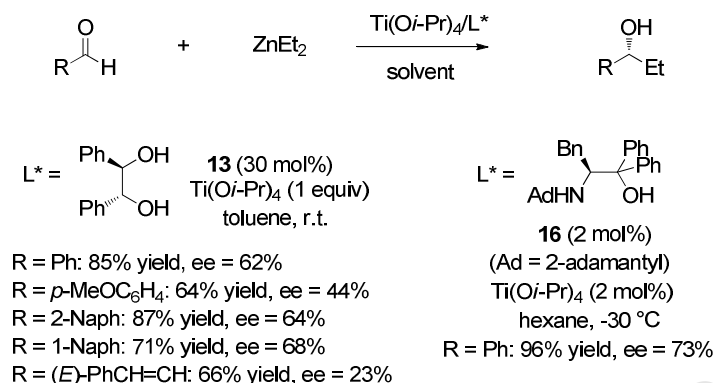
et al. have developed novel recyclable fluorous chiral ligands designed as the first fluorinated analogues of TADDOL.⁴³ Unlike TADDOL,⁴⁴ which has four aromatic substituents, these ligands have only three perfluoroalkyl substituents. Applied to induce chirality in addition of dimethylzinc to aromatic as well as aliphatic aldehydes, these novel ligands provided excellent homogeneous results. Among them, diol **12** seemed to be the most effective since very high enantioselectivities of up to 98% ee were achieved along with general almost quantitative yields for various aromatic as well as aliphatic aldehydes, as shown in Scheme 16. Moreover, the authors have studied the recyclability of ligand **12** taking advantage of its low solubility in cold toluene. Upon cooling a solution of the crude reaction product in toluene, the ligand precipitated and was separated from the products by filtration. By this simple method, the authors could recycle ligand **12** up to four times without decreasing the enantioselectivity. It is worth mentioning that these results are remarkable since the asymmetric addition of dimethylzinc to aldehydes is known to be very slow and mostly gives low enantioselectivities. Moreover, the methodology presented the advantage to be compatible with aliphatic aldehydes which is not yet common. It is probably one of the best general methods for adding enantioselectively dimethylzinc to all types of aldehydes promoted by a chiral titanium complex, which ensued from the pioneering methodology reported by Seebach and Schmidt using TADDOL-derived titanium complexes.²⁰



Scheme 16. Chiral TADDOL-based fluorinated ligand in addition of dimethylzinc to aldehydes

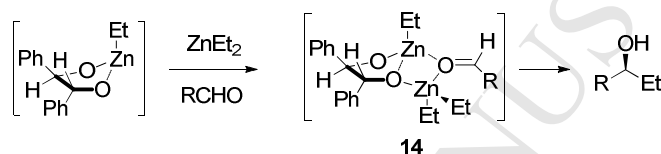
In 2008, Hitchcock and Dean investigated (*R,R*)-hydrobenzoin **13** as chiral ligand in asymmetric addition of diethylzinc to aromatic and α,β -unsaturated aldehydes in the presence or absence of $\text{Ti}(\text{Oi-Pr})_4$.⁴⁵ The enantioselectivity of the process involving no titanium catalyst was as high as 85% ee in the case of 2-naphthaldehyde favoring the formation of the (*S*)-enantiomer of the corresponding alcohol. Surprisingly, the enantioselectivities of the reactions of aromatic aldehydes when performed in the presence of $\text{Ti}(\text{Oi-Pr})_4$ were of up to 68% ee, favoring the formation of the (*R*)-enantiomers. The formation of the opposite enantiomers was attributed to the different transition states **14** and **15** mediated by either zinc or titanium (Scheme 17). As shown in Scheme 17, the use of one equivalent of $\text{Ti}(\text{Oi-Pr})_4$ allowed moderate to good enantioselectivities to be achieved (44-68% ee) for various aromatic aldehydes, whereas a low enantioselectivity of 23% ee was obtained in the case of cinnamaldehyde as substrate.

Later in 2012, Johnson et al. studied sterically encumbered chiral L-amino alcohols with secondary amines and tertiary alcohols as ligands in addition of diethylzinc to benzaldehyde.⁴⁶ The ligands were substituted at the amino nitrogen with isopropyl, cyclohexyl, or adamantyl groups, and at the position α to the alcohol with hydrogen, methyl, *n*-butyl, or phenyl groups. The catalyst which gave the highest enantioselectivities was ligand **16** exhibiting the most steric hindrance, containing adamantyl substituent on the nitrogen and phenyl groups α to the oxygen. Performing the reaction in the presence of 2 mol% of ligand **16** without $\text{Ti}(\text{Oi-Pr})_4$ provided the corresponding (*R*)-1-phenylpropanol with an enantioselectivity of 58% ee. On the other hand, when the reaction was induced by a catalytic amount (2 mol%) of $\text{Ti}(\text{Oi-Pr})_4$, it afforded the same product with the similar configuration in 96% yield and a better enantioselectivity of 73% ee, as shown in Scheme 17. Actually, in almost all cases, the addition catalysed by the titanium complexes exhibited higher enantioselectivity than that of the amino alcohol ligand alone. A steric argument could be employed to rationalise these results. The dimeric titanium complex shown in **17** had significant steric bulk near the ligand nitrogen due to the *N*-adamantyl substituent. Therefore, when benzaldehyde bound to the titanium centre, it did so in avoiding the steric environment due to the chiral backbone substituent (Bn) and orientating the phenyl group away from the *N*-alkyl substituent. The addition of the ethyl group from the other titanium centres or incoming zinc reagent was then directed to the *Re* face, as shown in Scheme 17. It must be highlighted that the reaction catalysed by ligand **16** constituted one rare example of alkylation of aldehydes by zinc reagents involving only a catalytic amount (2 mol%) of titanium.

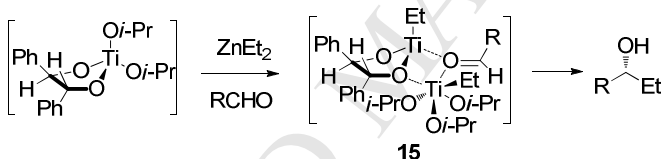


proposed mechanism for enantioselective pathways using ligand **13**:

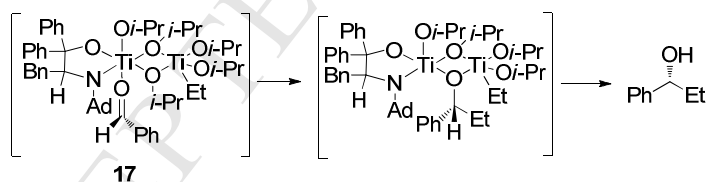
without titanium:



with titanium:



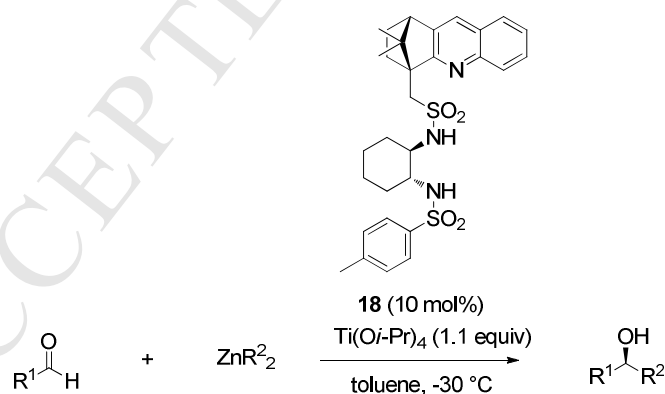
proposed mechanism for enantioselective pathways using ligand **16**:



Scheme 17. Chiral 1,2-diol and 1,2-amino alcohol ligands in addition of diethylzinc to aromatic and α,β -unsaturated aldehydes

Inspired by the first enantioselective titanium-promoted addition of diethylzinc to benzaldehyde reported in 1989 by Ohno and Yoshioka which used chiral *trans*-1,2-bis(trifluoromethanesulfonylamino)cyclohexane as ligand (Scheme 1),³ several groups have recently investigated various ligands of this type to promote enantioselective titanium-promoted additions of organozinc reagents to aldehydes. For example, excellent general results were reported by Cozzi, Ramon, and Yus in 2008 for enantioselective titanium-

promoted additions of diethylzinc to a variety of aldehydes spanning from aromatic to aliphatic ones.⁴⁷ These reactions were induced by a novel class of chiral camphorsulfonamide-based quinoline ligands employed at 10 mol% of catalyst loading in combination with 1.1 equivalent of Ti(Oi-Pr)₄ in toluene at -30 °C. Among a range of C₂- and C₁-symmetric ligands of this type, ligand **18** bearing only one camphor-quinoline unit gave the best results in terms of yield and enantioselectivity (Scheme 18). The yields were found similar using different sources of nucleophiles, such as diethylzinc and dimethylzinc, albeit the enantioselectivity using dimethylzinc was lower (80% ee vs 92% ee for addition of diethylzinc to benzaldehyde). The electronic character of the arenecarbaldehyde derivatives seemed not to have any important impact on the results, since *para*-substituted benzaldehydes with either electron-withdrawing or electron-donating groups gave results similar to those found with benzaldehyde. A possible difference was found for the cyano derivative providing the lowest results (56% yield, 75% ee) in the series by competing with the oxygen atom of the carbonyl group for complexation with the Lewis acid centre. The best result (96% ee) of the study was reached when the highly hindered naphthaldehyde was used as the electrophile. Surprisingly, the reactions of α,β -unsaturated aldehydes afforded lower enantioselectivities of 50-63% ee. However, the reaction using an aliphatic aldehyde, such as benzylacetaldehyde, gave a very high level of enantioselectivity of 93% ee.



R¹ = Ph, R² = Me: 94% yield, ee = 80%

R¹ = Ph, R² = Et: 95% yield, ee = 92%

R¹ = *p*-ClC₆H₄, R² = Et: 89% yield, ee = 90%

R¹ = *p*-NCC₆H₄, R² = Et: 56% yield, ee = 75%

R¹ = *p*-CF₃C₆H₄, R² = Et: 95% yield, ee = 87%

R¹ = *p*-MeOC₆H₄, R² = Et: 95% yield, ee = 89%

R¹ = 1-Naph, R² = Et: 95% yield, ee = 96%

R¹ = (*E*)-PhCH=CH, R² = Et: 65% yield, ee = 63%

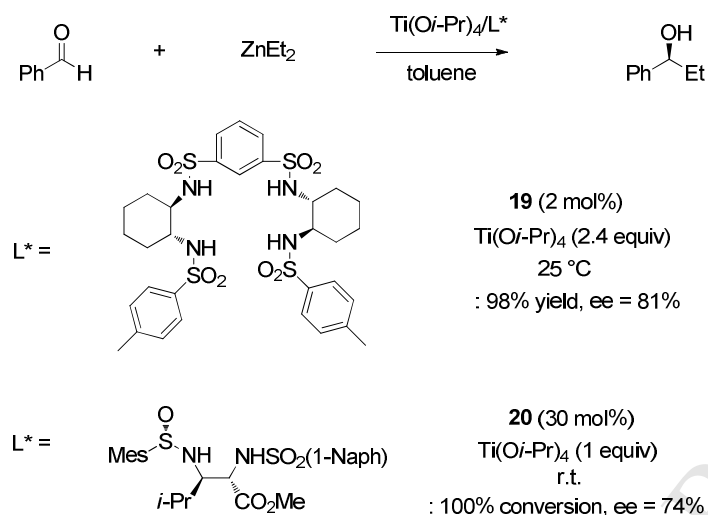
R¹ = PhC≡C, R² = Et: 82% yield, ee = 50%

R¹ = BnCH₂, R² = Et: 86% yield, ee = 93%

Scheme 18. Chiral camphorsulfonamide-based quinoline ligand in addition of dialkylzinc to aldehydes

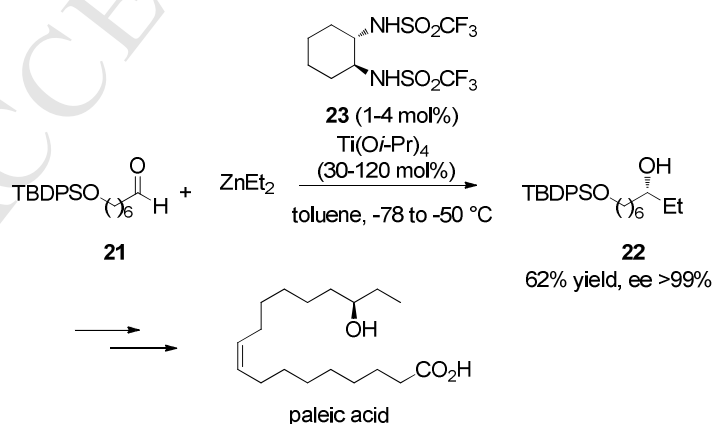
The enantioselective titanium-promoted addition of diethylzinc to benzaldehyde was also performed in the presence of chiral tetrakis(sulfonamides) as ligands by de Parrodi and Somanathan, in 2010.⁴⁸ Indeed, these authors reported the synthesis of a series of novel C_2 -symmetric tetrakis(sulfonamides) with the aim of surrounding the Lewis acid titanium metal centre with four chiral nitrogen atoms in a *cisoid* conformation, hoping that the additional chirality could enhance the enantioselectivity of the addition reaction in comparison with the corresponding more simple bis(sulfonamide) ligands. Among various tetrakis(sulfonamide) ligands investigated, ligand **19** provided the best enantioselectivities of up to 81% ee, along with excellent yields of up to 98% (Scheme 19). In this study, it was found that the presence of electron-withdrawing and -donating groups on the sulfonamide benzene ring of the ligand had a modest effect on the enantioselectivity of the process.

Always in the context of sulfonamide ligands, Viso et al. have developed a family of novel chiral sulfinamido-sulfonamide ligands synthesised from sulfinimines, which were evaluated in the same reaction.⁴⁹ Interestingly, experimental evidences showed a crucial cooperation between the sulfinyl and sulfonyl functionalities to reach enantiocontrol in the alkylation process since suppressing the sulfur chiral atom by oxidation into the sulfonamide led to the corresponding bis(sulfonamide) ligands which provided racemic 1-phenylpropanol when tested under the same conditions. Among a range of various sulfinamido-sulfonamides investigated, the best enantioselectivity of 74% ee associated with a complete conversion was obtained with ligand **20** (Scheme 19).



Scheme 19. Chiral bis(sulfonamide) and sulfinamido-sulfonamide ligands in addition of diethylzinc to benzaldehyde

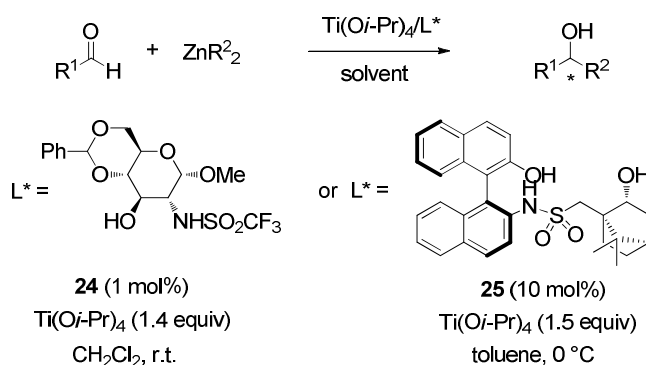
In 2010, Watanabe et al. reported a total synthesis of paleic acid, an antimicrobial agent effective against *Mannheimia* and *Pasteurella*, which was based on an enantioselective titanium-promoted alkylation of 7-hydroxyheptylaldehyde protected as a *tert*-butyldiphenylsilyl ether **21** to give the corresponding almost enantiopure alcohol **22** in 62% yield (Scheme 20).⁵⁰ The process employed a catalytic amount of Ohno's chiral ligand³ (1*S*,2*S*)-bis(trifluoromethanesulfonylamino)cyclohexane **23**. Product **22** was subsequently converted into expected paleic acid through five supplementary steps.



Scheme 20. Chiral bis(sulfonamide) ligand in addition of diethylzinc to an aliphatic functionalised aldehyde and synthesis of paleic acid

Inspired by their previously reported excellent work dealing with the use of D-glucosamine-derived sulfonamide ligands in this type of reactions,^{42b} Bauer and Smolinski later described other excellent results in the enantioselective addition of diethylzinc to aliphatic and aromatic aldehydes using other D-glucosamine-derived ligands.⁵¹ The obvious advantage of this type of ligands was its modular synthesis. Indeed, three sites of this type of ligands could be easily altered during their synthesis, thus leading to various ligands having the same chiral precursor. Among them, D-glucosamine-derived β -hydroxy *N*-trifluoromethylsulfonamide **24** was found to be the most active ligand when employing at a catalyst loading as low as 1 mol%, providing remarkable enantioselectivities of up to >99% ee especially with aromatic aldehydes while aliphatic aldehydes gave enantioselectivities of up to 88% ee (Scheme 21). It must be noted that generally high yields were achieved except in the cases of *ortho*-, *meta*-, and *para*-nitrobenzaldehydes and *para*-*N,N*-dimethylaminobenzaldehyde which gave yields ranging from 15 to 38% along with enantioselectivities of 10 to 36% ee.

Finally, a series of novel camphorsulfonylated ligands derived from L-camphor and chiral NOBIN were synthesised by Song et al. to be tested in the titanium-promoted addition of dialkylzinc reagents to aromatic and aliphatic aldehydes.⁵² The highest catalytic efficiency was obtained with mono-*N*-hydroxycamphorsulfonylated (*S*)-NOBIN **25** in toluene, which gave (*S*)-addition products with high yields of up to 98% and enantioselectivities of up to 87% ee, as shown in Scheme 21. It must be noted that better enantioselectivities were observed for the addition of diethylzinc in comparison with that of dimethylzinc which provided enantioselectivities ranging from 11 to 44% ee. In spite of the fact that ligand **25** possesses a NOBIN unit and consequently could also be part of section 2.1.1.1, it was decided to situate its utilisation in this section dealing with ligands other than BINOL-derived ones since it is also derived from L-camphor and bears a sulfonamide function.



with $\text{L}^* = \mathbf{24}$:

with $\text{R}^2 = \text{Et}$:

$\text{R}^1 = \text{Ph}$: 83% yield, ee = 93% (*R*)

$\text{R}^1 = o\text{-MeOC}_6\text{H}_4$: 83% yield, ee = 75% (*R*)

$\text{R}^1 = m\text{-MeOC}_6\text{H}_4$: 74% yield, ee >99% (*R*)

$\text{R}^1 = p\text{-MeOC}_6\text{H}_4$: 98% yield, ee = 77% (*R*)

$\text{R}^1 = o\text{-FC}_6\text{H}_4$: 84% yield, ee = 94% (*R*)

$\text{R}^1 = p\text{-FC}_6\text{H}_4$: 73% yield, ee = 78% (*R*)

$\text{R}^1 = o\text{-ClC}_6\text{H}_4$: 94% yield, ee = 98% (*R*)

$\text{R}^1 = p\text{-ClC}_6\text{H}_4$: 78% yield, ee = 88% (*R*)

$\text{R}^1 = p\text{-BrC}_6\text{H}_4$: 74% yield, ee = 81% (*R*)

$\text{R}^1 = o\text{-NO}_2\text{C}_6\text{H}_4$: 15% yield, ee = 36% (*R*)

$\text{R}^1 = m\text{-NO}_2\text{C}_6\text{H}_4$: 23% yield, ee = 20% (*R*)

$\text{R}^1 = p\text{-NO}_2\text{C}_6\text{H}_4$: 38% yield, ee = 16% (*R*)

$\text{R}^1 = p\text{-(NMe}_2\text{)C}_6\text{H}_4$: 22% yield, ee = 10% (*R*)

$\text{R}^1 = (E)\text{-PhCH=CH}$: 82% yield, ee = 87% (*R*)

$\text{R}^1 = \text{Cy}$: 97% yield, ee = 85% (*R*)

$\text{R}^1 = n\text{-Hex}$: 84% yield, ee = 88% (*R*)

with $\text{L}^* = \mathbf{25}$:

with $\text{R}^2 = \text{Et}$:

$\text{R}^1 = \text{Ph}$: 95% yield, ee = 87% (*S*)

$\text{R}^1 = p\text{-ClC}_6\text{H}_4$: 98% yield, ee = 85% (*S*)

$\text{R}^1 = p\text{-MeOC}_6\text{H}_4$: 96% yield, ee = 83% (*S*)

$\text{R}^1 = p\text{-CF}_3\text{C}_6\text{H}_4$: 95% yield, ee = 84% (*S*)

$\text{R}^1 = o\text{-Tol}$: 96% yield, ee = 22% (*S*)

$\text{R}^1 = o\text{-MeOC}_6\text{H}_4$: 97% yield, ee = 64% (*S*)

$\text{R}^1 = 2\text{-Naph}$: 93% yield, ee = 78% (*S*)

$\text{R}^1 = 1\text{-Naph}$: 90% yield, ee = 77% (*S*)

$\text{R}^1 = (E)\text{-PhCH=CH}$: 75% yield, ee = 55% (*S*)

$\text{R}^1 = \text{Cy}$: 90% yield, ee = 84% (*S*)

with $\text{R}^2 = \text{Me}$:

$\text{R}^1 = \text{Ph}$: 95% yield, ee = 44% (*S*)

$\text{R}^1 = o\text{-Tol}$: 98% yield, ee = 11% (*S*)

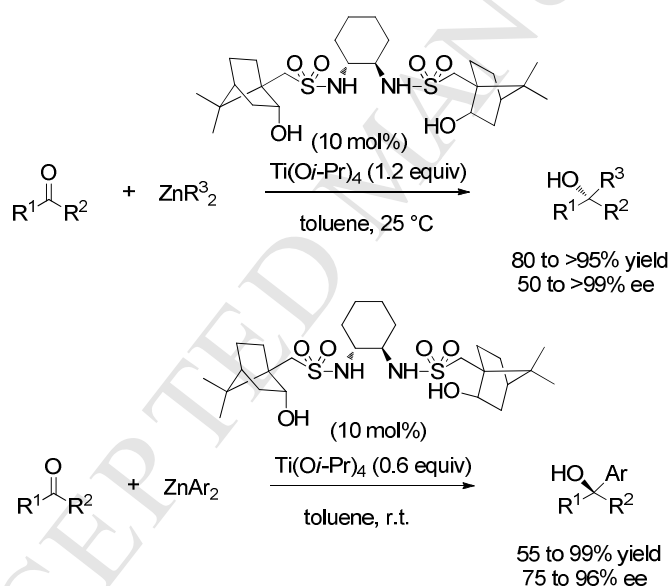
$\text{R}^1 = p\text{-CF}_3\text{C}_6\text{H}_4$: 97% yield, ee = 29% (*S*)

$\text{R}^1 = 1\text{-Naph}$: 97% yield, ee = 36% (*S*)

Scheme 21. Chiral ligands derived from D-glucosamine and L-camphor in addition of dialkylzincs to aldehydes

2.1.2. Ketones as electrophiles. In comparison with aldehydes, the catalytic asymmetric addition of alkyl group to ketones is a more challenging task for synthetic chemists owing to their low electrophilicity, the reduced propensity of ketone carbonyl to coordinate with Lewis

acids, and the difficult discrimination of the both faces of the double bond. Early in 2000s, Yus and Ramon described examples of enantioselective dialkylzinc additions to ketones using a superstoichiometric amount of $\text{Ti}(\text{Oi-Pr})_4$ combined with a catalytic amount of camphorsulfonamide derivatives as chiral ligands (Scheme 22, first equation).⁵³ In this work, the corresponding tertiary alcohols were achieved in enantioselectivities of up to >99% ee with a variety of ketones. It must be noted that efficient chiral tertiary alcohols synthesis constitutes currently one of the most rapidly advancing fields in organic chemistry, since these compounds are versatile building blocks for the synthesis of natural products and pharmaceuticals. In 2003, Walsh and Garcia used the same catalyst derived from dihydroxy bis(sulfonamide) ligand and a substoichiometric amount of $\text{Ti}(\text{Oi-Pr})_4$ with diarylzinc as the nucleophiles, to afford the corresponding chiral tertiary alcohols in good to excellent enantioselectivities of up to 96% ee (Scheme 22, second equation).⁵⁴

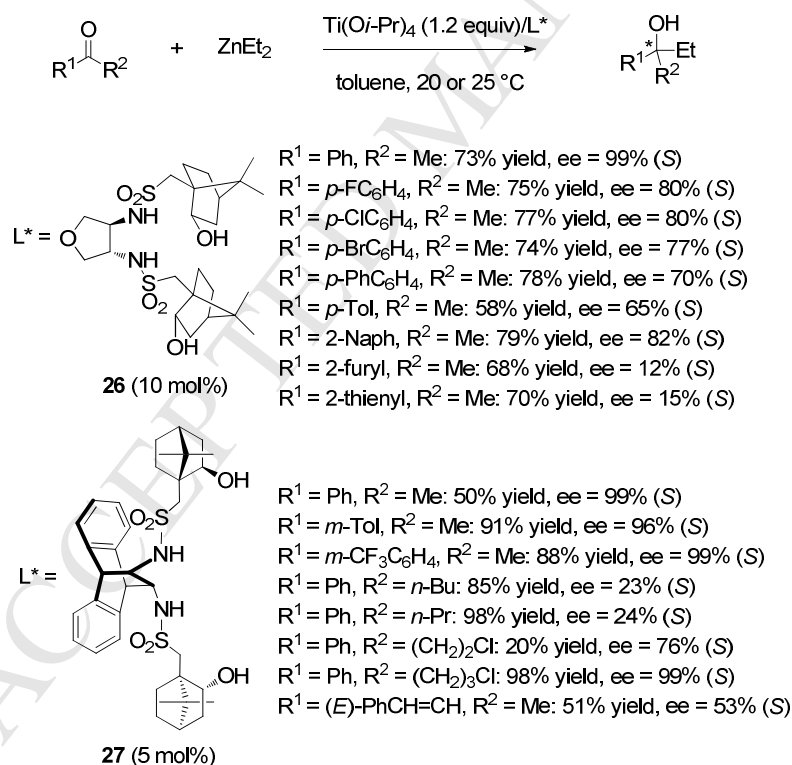


Scheme 22. Chiral camphorsulfonamide ligand in additions of dialkyl and diarylzincs to ketones reported by Ramon and Yus, and Walsh and Garcia in 2000s

Inspired by these pioneering works, Wang et al. more recently designed novel chiral ligand **26** derived from L-tartaric acid to be employed as promoter in the titanium-promoted addition of diethylzinc to ketones.⁵⁵ As shown in Scheme 23, acetophenone provided the best enantioselectivity of 99% ee in combination with a good yield (73%). Various substituted acetophenones were investigated under similar conditions, and the authors found that the presence of substituents at the *ortho*- and *meta*-positions of acetophenone were incompatible

to the reaction, since no desired products were formed. These results were ascribed to steric repulsion of the substituent and the ethyl group. Moreover, ketones containing heteroaromatic groups, such as 2-acetyl furan and 2-acetyl thiophene, led to the corresponding products with low enantioselectivities of 12 and 15% ee, respectively, along with moderate yields (68 and 70%, respectively). These results could stem from the binding of the heteroatom in the substrate with the titanium centre. On the other hand, 2-acetyl naphthalene having more steric hindrance than other ketones gave a high enantioselectivity of 82% ee.

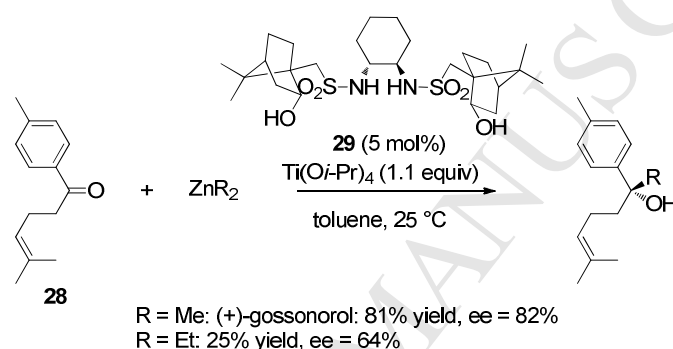
In another context, de Parrodi et al. developed later novel chiral ligand **27** derived from L-camphor and based on a C_2 -symmetric 11,12-diamino-9,10-dihydro-9,10-ethanoanthracene backbone.⁵⁶ This ligand exhibited a large N-C-C-N dihedral angle and larger bite angle than *trans*-1,2-diaminocyclohexane. When applied at 5 mol% of catalyst loading to the titanium-promoted addition of diethylzinc to a variety of aryl alkyl ketones, it afforded low to excellent enantioselectivities of up to 99% ee, as summarised in Scheme 23.



Scheme 23. Chiral ligands derived from L-tartaric acid and L-camphor in addition of diethylzinc to ketones

Inspired by their previous works based on the use of chiral isoborneolsulfonamide ligand (Scheme 22, ref 53), Ramon et al. later reported an enantioselective total synthesis of

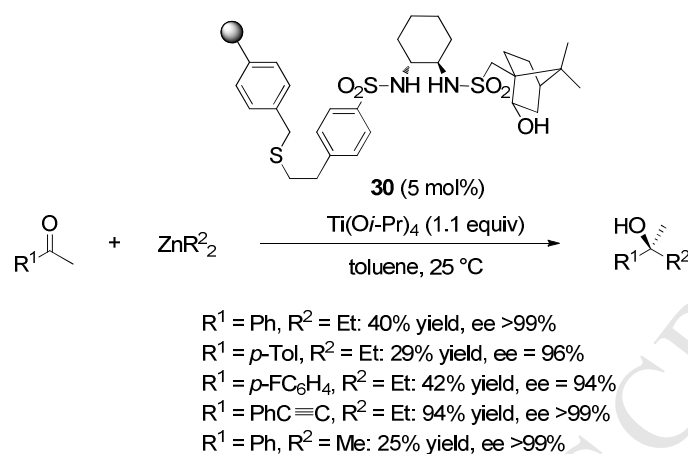
biologically active (+)-gossonorol the key step of which was the titanium-promoted addition of dimethylzinc to 5-methyl-1-(2-methylphenyl)hex-4-en-1-one **28** to give the corresponding chiral tertiary alcohol (+)-gossonorol.⁵⁷ This process was performed with 5 mol% of chiral isoborneolsulfonamide ligand **29** in the presence of 1.1 equivalent of $\text{Ti}(\text{O}i\text{-Pr})_4$, providing the key alcohol in 81% yield and enantioselectivity of 82% ee, as shown in Scheme 24. It must be noted that the synthesis of (+)-gossonorol was accomplished in 60% yield through a three-step process from commercially available reagents. When applying the same conditions to the addition of diethylzinc, the corresponding tertiary alcohol was achieved in both lower yield (25%) and enantioselectivity (64% ee).



Scheme 24. Chiral isoborneolsulfonamide ligand in addition of dialkylzinc to 5-methyl-1-(4-methylphenyl)hex-4-en-1-one and total synthesis of (+)-gossonorol

Finally, Ramon and Yus investigated the titanium-promoted additions of diethyl-, dimethyl-, and diphenylzincs to various methyl ketones in the presence of novel grafted isoborneolsulfonamide polymer **30** employed at 5 mol% of catalyst loading along with 1.1 equivalent of $\text{Ti}(\text{O}i\text{-Pr})_4$.⁵⁸ Whereas the highest and remarkable enantioselectivities of up to >99% ee were achieved in the ethylation process (Scheme 25), the highest chemical yields of up to 98% were obtained in the phenylation process albeit associated to moderate enantioselectivities (28-66% ee). Concerning the addition of diethylzinc to acetophenone derivatives, the authors found that both electron-donating and electron-withdrawing groups had a small negative impact on the enantioselectivities. The best results were actually obtained for the ethylation of an α,β -unsaturated aldehyde, which provided a remarkable enantioselectivity (>99% ee) combined with a high yield (94%). It must be noted that this heterogeneous ligand could be re-used at least three times without any significant loss of activity. It has to be highlighted that this novel remarkable methodology, which presents the

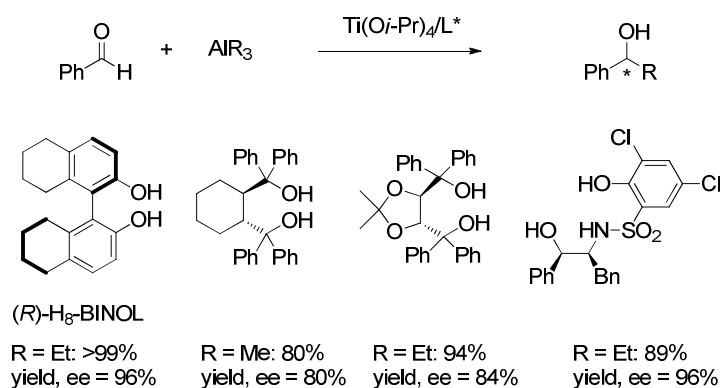
advantage to be compatible with the heterogeneous enantioselective arylation and alkylation of simple ketones, constituted an important progress in the context of polymeric catalysis.



Scheme 25. Chiral grafted isoborneolsulfonamide polymer ligand in addition of dialkylzinc to methyl ketones

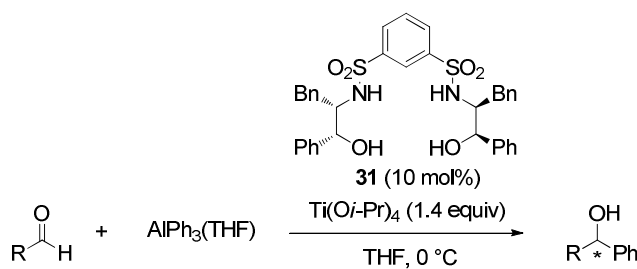
2.2. Additions of organoaluminum reagents

2.2.1. Aldehydes as electrophiles. Unfortunately, it must be recognised that few organozinc reagents are commercially available and their preparation is not always straightforward. To circumvent such limitations, attention has been focused on the use of other organometallic reagents. For example, trialkylaluminum reagents are readily available and constitute valuable alkylating reagents for the enantioselective addition to aldehydes and ketones. Additional advantages of organoaluminum compounds include low toxicities and considerable stabilities. In most cases, chiral aluminates are first generated by reaction of chiral ligands with trialkylaluminum reagents, enabling the in situ formation of chiral titanium catalysts through transmetallation. The first example of asymmetric addition of AlEt_3 to aldehydes promoted by chiral titanium complexes derived from BINOL was developed by Chan et al., in 1997.⁵⁹ The best enantioselectivity of 96% ee was reached by using 20 mol% of chiral $\text{H}_8\text{-BINOL}$ (Scheme 26). This work was followed by several other reports by Gau and Carreira's groups among others dealing with titanium-promoted additions of other alkylaluminum reagents using a variety of chiral titanium ligands providing good to high enantioselectivities of up to 96% ee (Scheme 26).⁶⁰



Scheme 26. Various chiral ligands in early additions of AlR_3 to benzaldehyde.

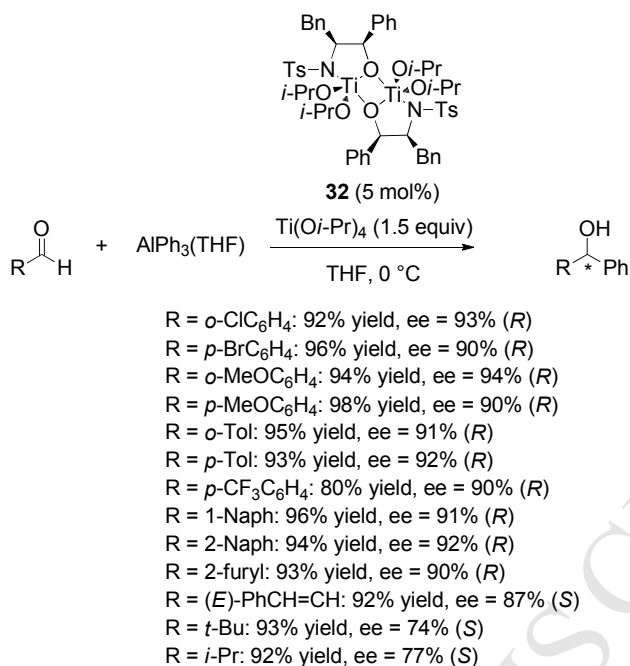
In the last few years, important progress has been made in the area of enantioselective titanium-promoted additions of aluminum reagents to aldehydes using several novel types of chiral ligands. Indeed, several excellent works have been independently reported by the groups of Gau and Yus in particular, allowing both the arylation and alkylation of all types of aldehydes to be achieved in remarkable enantioselectivities. For example, Gau and co-workers developed remarkable enantioselective additions of $\text{AlPh}_3(\text{THF})$ to both aliphatic and aromatic aldehydes in the presence of $\text{Ti}(\text{O}i\text{-Pr})_4$ and a catalytic amount of chiral disulfonamide ligand **31** in THF.⁶¹ As shown in Scheme 27, the corresponding chiral secondary alcohols were produced with enantioselectivities of $\geq 94\%$ ee except for two substrates, such as *n*-butanal and *trans*-cinnamaldehyde, which provided enantioselectivities of 87% and 85% ee, respectively. Furthermore, the products were obtained in general excellent yields of up to 98%. It has to be highlighted that this novel process is remarkable by its generality, providing excellent enantioselectivities and yields for the phenylation of all types of aldehydes. It must be noted that chiral ligand **31** gave slightly better enantioselectivities than a closely related ligand bearing four phenyl substituents which was previously employed by Ramon in 2002 (up to 92% ee, Scheme 15, ref 40d).



R = *o*-MeOC₆H₄: 95% yield, ee = 96% (*R*)
 R = *p*-MeOC₆H₄: 98% yield, ee = 95% (*R*)
 R = *o*-Tol: 97% yield, ee = 98% (*R*)
 R = *p*-Tol: 97% yield, ee = 98% (*R*)
 R = *o*-ClC₆H₄: 95% yield, ee = 95% (*R*)
 R = *m*-ClC₆H₄: 97% yield, ee = 97% (*R*)
 R = *p*-ClC₆H₄: 96% yield, ee = 96% (*R*)
 R = *p*-CF₃C₆H₄: 97% yield, ee = 96% (*R*)
 R = *o*-BrC₆H₄: 97% yield, ee = 95% (*R*)
 R = 1-Naph: 96% yield, ee = 95% (*R*)
 R = 2-Naph: 97% yield, ee = 96% (*R*)
 R = 1-thiophene: 97% yield, ee = 95% (*R*)
 R = (*E*)-PhCH=CH: 93% yield, ee = 85% (*R*)
 R = *n*-Bu: 90% yield, ee = 87% (*R*)
 R = *i*-Pr: 90% yield, ee = 97% (*R*)
 R = *t*-Bu: 95% yield, ee = 99% (*R*)
 R = Cy: 85% yield, ee = 95% (*R*)

Scheme 27. Chiral disulfonamide ligand in addition of AlPh₃(THF) to aldehydes

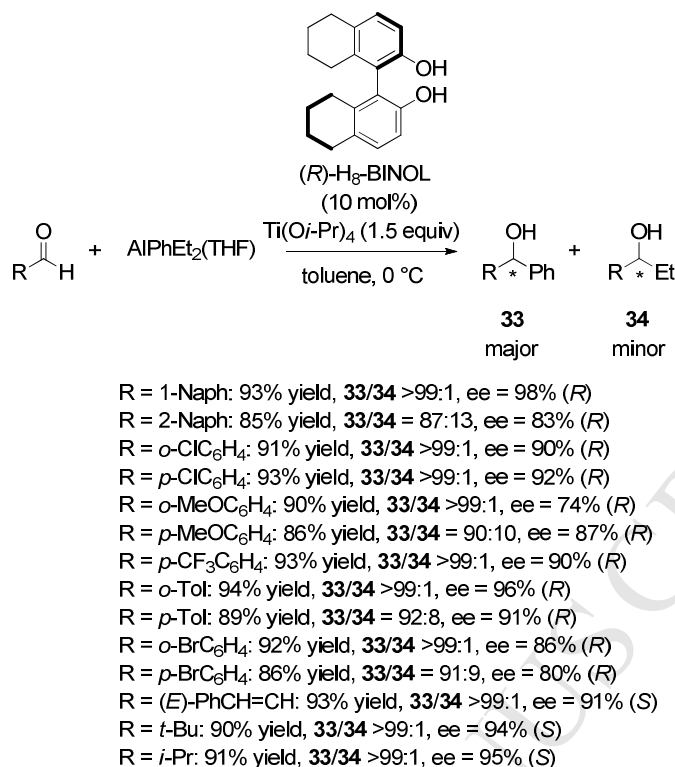
In order to further explore arylaluminum reagents, the same authors reported later the synthesis of other arylaluminum reagents containing various adducts of Et₂O, OPPh₃, DMAP, in addition to THF, and their asymmetric titanium-promoted aryl additions to aldehydes employing a catalyst loading of 5 mol% of titanium complex **32** bearing chiral *N*-sulfonylated amino alcohols as a catalyst precursor.⁶² It was demonstrated that the adduct ligand had a strong influence on the reactivity and enantioselectivity of the arylation reactions. Indeed, the phenylaluminum reagents with OPPh₃ or DMAP were unreactive toward aldehydes, and AlPh₃(THF) was found to be superior to AlPh₃(OEt₂) or AlPhEt₂(THF). In the presence of 1.5 equivalent of Ti(Oi-Pr)₄ and 5 mol% of complex **32**, the asymmetric additions of AlPh₃(THF) to aldehydes afforded the corresponding chiral secondary alcohols in high yields and enantioselectivities of up to 94% ee, as shown in Scheme 28. The best results were achieved in the cases of aromatic aldehydes. Moreover, the reaction of Al(*p*-Tol)₃(THF) with benzaldehyde provided the corresponding product in both high yield (91%) and enantioselectivity (91% ee).



with R = Ph and Al(*p*-Tol)₃(THF): 91% yield, ee = 91% (*S*)

Scheme 28. Chiral *N*-sulfonylated amino alcohol titanium complex in addition of AlPh₃(THF) to aldehydes

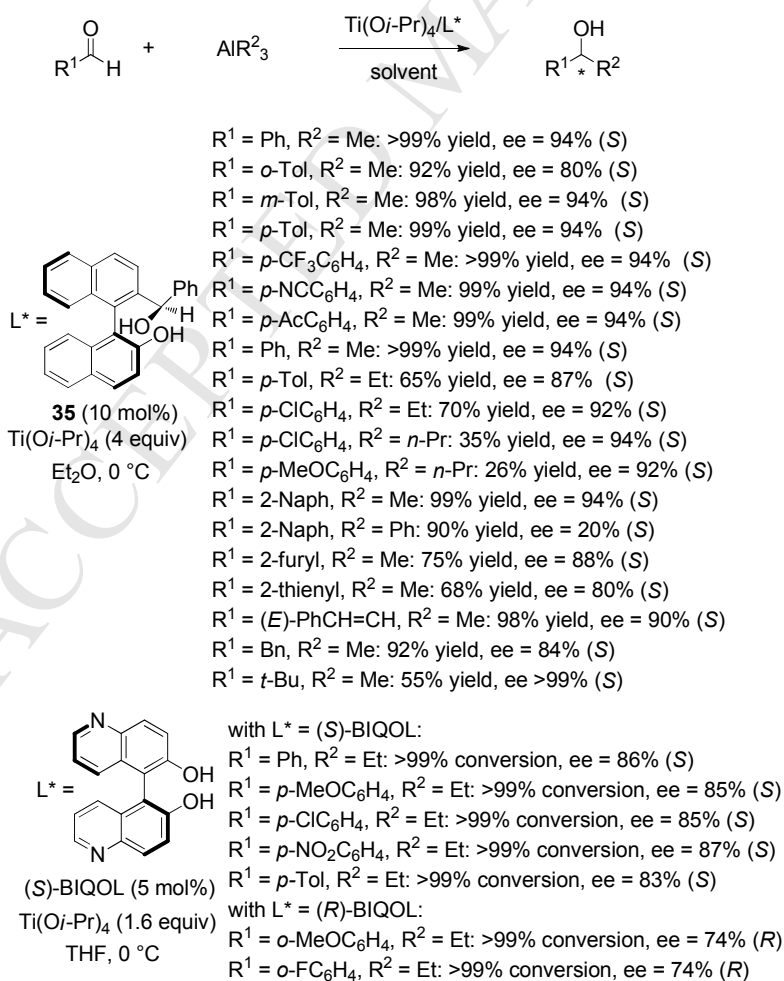
In addition, these authors have developed an easy preparation of AlPhEt₂(THF) which was also added to aromatic and aliphatic aldehydes in the presence of 10 mol% of a titanium complex of (*R*)-H₈-BINOL and 1.5 equivalent of Ti(Oi-Pr)₄ in toluene.⁶³ As shown in Scheme 29, the process afforded the corresponding chiral secondary aryl alcohols **33** as exclusive products in high yields and excellent enantioselectivities of up to 98% ee except for 2-naphthaldehyde, 4-methoxybenzaldehyde, 4-methylbenzaldehyde, and 4-bromobenzaldehyde for which the minor ethylation products **34** were obtained with yields of 8 to 13%. Remarkably, the process afforded excellent yields and enantioselectivities of up to 95% ee for aliphatic aldehydes.



Scheme 29. (*R*)-H₈-BINOL ligand in addition of AlPhEt₂(THF) to aldehydes

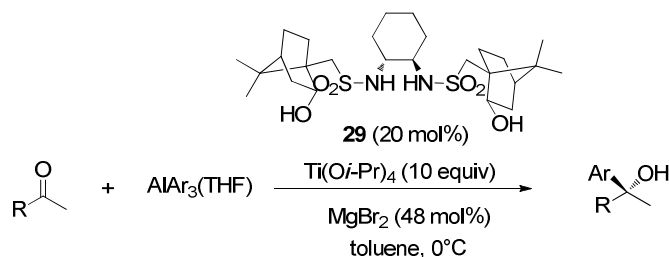
While the precedent works depicted in Schemes 27-29 dealt with the enantioselective arylation of aldehydes, Yus et al. developed remarkable asymmetric additions of alkylaluminum reagents to a range of aromatic as well as aliphatic aldehydes by using a combination of an excess of Ti(O*i*-Pr)₄ with a catalytic amount of chiral readily available BINOL-derived ligand **35** in diethylether as solvent.⁶⁴ As shown in Scheme 30, the asymmetric methylation, ethylation, and propargylation of a wide variety of aldehydes proceeded with good yields and high enantioselectivities of up to 94% ee. In particular, the system proved to be remarkably efficient for a variety of aromatic substrates, providing enantioselectivities ranging from 80 to 94% ee combined with yields of 87-99%. Heteroaromatic substrates also gave high enantioselectivities of up to 88% ee albeit with slightly lower yields (68-75%). The substrate generality was furthermore examined for aliphatic aldehydes which provided good yield (92%) and moderate enantioselectivity (84% ee) in the case of phenylacetaldehyde, while the bulky pivaldehyde gave the highest enantioselectivity (>99% ee) of the series. As a practical feature of the process, it must be mentioned that all the reactions were finished in less than one hour without the formation of by-products and that the ligand could be easily recovered.

Very recently, Chen et al. reinvestigated the asymmetric ethylation of aromatic aldehydes by using 5 mol% of (*S*)- or (*R*)-BIQOL as chiral ligand in the presence of 1.6 equivalent of $\text{Ti}(\text{O}i\text{-Pr})_4$ in THF.⁶⁵ The reactions provided the corresponding aromatic alcohols in remarkable complete conversions and good enantioselectivities of up to 87% ee (Scheme 30). A number of substituents at the aromatic ring of the aldehyde were well tolerated, including *para*- and *ortho*-methoxy, *para*-chlorine/methyl/nitro, and *ortho*-fluorine groups, leading to the corresponding alcohols in quantitative yields and with enantioselectivities of 74-87% ee. It is worth mentioning that the addition to aromatic aldehydes containing *ortho*-methoxy or *ortho*-fluorine group led to a lower enantiomeric excess (74% ee) due to the *ortho* steric effect, which could weaken the coordination strength between the aldehyde and the chiral catalyst, while *para*-substituents including electron-withdrawing or electron-donating groups of the aromatic aldehydes had less effect on the enantioselectivity of the reaction. Also noteworthy was that BIQOL induced the addition with an enantioselectivity higher than that induced by BINOL.



Scheme 30. Chiral BINOL-derived ligand and chiral BIQOL ligand in addition of AlR_3 to aldehydes

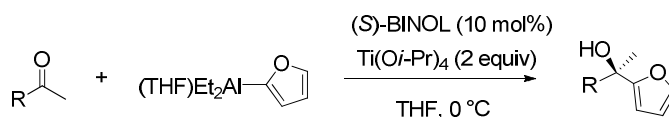
2.2.2. Ketones as electrophiles. In the last few years, important progress has also been made in the area of enantioselective titanium-promoted additions of aluminum reagents to methyl ketones. Indeed, excellent works have been described by the group of Gau, allowing both the arylation and heteroarylation of methyl ketones to be achieved in remarkable enantioselectivities. For example, these authors reported enantioselectivities of up to 97% ee in asymmetric additions of $\text{AlAr}_3(\text{THF})$ to ketones catalysed by a titanium catalyst in situ generated from $\text{Ti}(\text{Oi-Pr})_4$ and 20 mol% of *trans*-1,2-bis(hydroxycamphorsulfonylamino)cyclohexane **29** in the presence of MgBr_2 as an additive.⁶⁶ Several important features were demonstrated in this study. First, a novel aspect of the inorganic salt MgBr_2 as a key additive to promote the aryl addition of $\text{AlAr}_3(\text{THF})$ to ketones was demonstrated. Second, the catalytic system worked very well for aromatic ketones bearing either an electron-withdrawing or an electron-donating substituent on the aromatic group to afford the corresponding chiral tertiary alcohols in enantioselectivities of $\geq 90\%$ ee except for 2'-methoxyacetophenone, as shown in Scheme 31. Third, longer reaction times were required for *ortho*-substituted aromatic ketones to furnish products in good yields. Fourth, the reactions of $\text{PhTi}(\text{Oi-Pr})_3$ in additions to 2'-acetophenone catalysed by the same catalyst provided the product in low yield and low enantioselectivity, suggesting that $\text{AlAr}_3(\text{THF})$ addition reactions could not proceed via aryltitanium species. It must be noted that phenyl additions to aliphatic methyl ketones and 1-acetyl-1-cyclohexene were also examined in addition to aryl methyl ketones. The resulting tertiary chiral alcohols were obtained in good to excellent yields and good enantioselectivities of 75 to 83% ee, except for the alcohol derived from linear 2-hexanone which gave 52% ee. It has to be highlighted that this nice methodology presents the advantage to have a broad scope in terms of aldehydes as well as arylaluminum reagents, allowing general very high enantioselectivities and yields to be achieved.



R = 2-Naph, Ar = Ph: 99% yield, ee = 92%
 R = 1-Naph, Ar = Ph: 80% yield, ee = 93%
 R = *o*-Tol, Ar = Ph: 85% yield, ee = 96%
 R = *o*-ClC₆H₄, Ar = Ph: 95% yield, ee = 97%
 R = *p*-ClC₆H₄, Ar = Ph: 98% yield, ee = 92%
 R = *o*-BrC₆H₄, Ar = Ph: 87% yield, ee = 98%
 R = *o*-MeOC₆H₄, Ar = Ph: 99% yield, ee = 18%
 R = *m*-CF₃C₆H₄, Ar = Ph: 98% yield, ee = 97%
 R = *p*-CF₃C₆H₄, Ar = Ph: 97% yield, ee = 93%
 R = *p*-NO₂C₆H₄, Ar = Ph: 99% yield, ee = 92%
 R = 2-Naph, Ar = *p*-Tol: 97% yield, ee = 90%
 R = Ph, Ar = *p*-TMSOC₆H₄: 83% yield, ee = 81%
 R = Ph, Ar = 2-Naph: 95% yield, ee = 91%
 R = *n*-Bu, Ar = Ph: 97% yield, ee = 52%
 R = *i*-Pr, Ar = Ph: 82% yield, ee = 83%
 R = Cy, Ar = Ph: 87% yield, ee = 81%

Scheme 31. Chiral 1,2-bis(hydroxycamphorsulfonylamino)cyclohexane ligand in addition of AlAr₃(THF) to methyl ketones

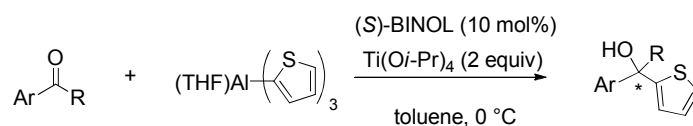
Later, the same authors studied the first asymmetric addition of a (2-furyl)aluminum reagent to aromatic methyl ketones having either an electron-donating or an electron-withdrawing substituent on the aromatic group, and to one α,β -unsaturated methyl ketone catalysed by a titanium catalyst of (*S*)-BINOL to afford the corresponding chiral tertiary 2-furyl alcohols in good to excellent enantioselectivities of 87 to 93% ee (Scheme 32).⁶⁷ Although the furylaluminum reagent employed was prepared as a mixture of three species of formulas (2-furyl)_xAlEt_{3-x}(THF) ($x = 0, 1, \text{ or } 2$), the addition reactions remarkably gave only chiral furyl alcohols with no observations of the corresponding ethylation products. In addition to its remarkable unprecedented results, this novel methodology presents the advantage to have opened up a new and easy route for the synthesis of highly reactive and extremely flexible furyl chiral alcohols which constitute key intermediates to bioactive compounds.



R = Ph: 90% yield, ee = 93%
 R = *o*-Tol: 74% yield, ee = 90%
 R = *p*-Tol: 78% yield, ee = 93%
 R = *p*-MeOC₆H₄: 74% yield, ee = 92%
 R = *o*-ClC₆H₄: 70% yield, ee = 92%
 R = *p*-ClC₆H₄: 86% yield, ee = 91%
 R = *o*-BrC₆H₄: 40% yield, ee = 93%
 R = *p*-BrC₆H₄: 92% yield, ee = 90%
 R = *p*-CF₃C₆H₄: 90% yield, ee = 91%
 R = *p*-NO₂C₆H₄: 88% yield, ee = 92%
 R = 1-Naph: 28% yield, ee = 87%
 R = 2-Naph: 94% yield, ee = 90%
 R = (*E*)-BrC=CHPh: 88% yield, ee = 88%

Scheme 32. (S)-BINOL as ligand in addition of (2-furyl)AlEt₂(THF) to methyl ketones

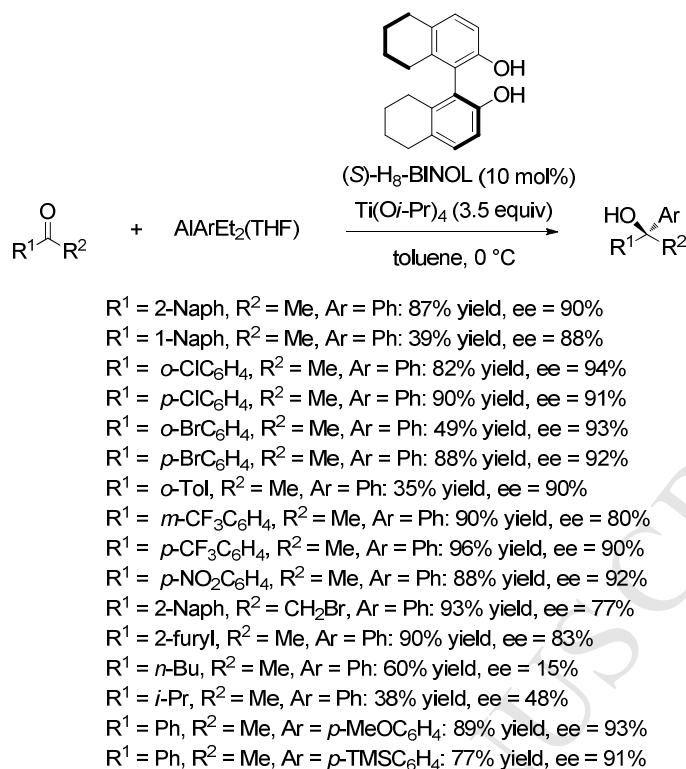
As an extension of the precedent methodology, these authors applied the same catalyst system to the first asymmetric thienylaluminum addition to a variety of ketones.⁶⁸ As shown in Scheme 33, the additions of Al(2-thienyl)₃(THF) to aromatic alkyl ketones having either an electron-donating or an electron-withdrawing substituent on the aromatic ring and to 1-acetylcyclohexene provided the corresponding chiral tertiary 2-thienyl alcohols in excellent enantioselectivities of up to 97% ee. In contrast, the additions of 2-thienyl to dialkyl ketones produced the corresponding alcohols in low enantioselectivities of 8 to 17% ee. In spite of the limitation of its scope to aromatic alkyl ketones, the importance of this remarkable unprecedented methodology is related to the fact that tertiary thienyl alcohols are well-known for their biological activities as well as key substructures in bioactive compounds and pharmaceuticals. For example, this methodology was applied to a concise synthesis of (S)-tiemonium iodide in three steps.



Ar = Ph, R = Me: 94% yield, ee = 93%
 Ar = *o*-Tol, R = Me: 75% yield, ee = 91%
 Ar = *m*-Tol, R = Me: 96% yield, ee = 90%
 Ar = *p*-Tol, R = Me: 95% yield, ee = 93%
 Ar = *o*-MeOC₆H₄, R = Me: 96% yield, ee = 45%
 Ar = *p*-MeOC₆H₄, R = Me: 92% yield, ee = 92%
 Ar = *o*-ClC₆H₄, R = Me: 92% yield, ee = 97%
 Ar = *m*-ClC₆H₄, R = Me: 95% yield, ee = 90%
 Ar = *p*-ClC₆H₄, R = Me: 96% yield, ee = 95%
 Ar = *o*-BrC₆H₄, R = Me: 93% yield, ee = 96%
 Ar = *p*-CF₃C₆H₄, R = Me: 96% yield, ee = 93%
 Ar = *p*-NO₂C₆H₄, R = Me: 95% yield, ee = 94%
 Ar = 2-Naph, R = Me: 94% yield, ee = 92%
 Ar = 1-Naph, R = Me: 70% yield, ee = 93%
 Ar = 2-Naph, R = CH₂Br: 95% yield, ee = 80%
 Ar = Ph, R = (CH₂)₂Br: 90% yield, ee = 94%
 Ar = *i*-Pr, R = Me: 96% yield, ee = 17%

Scheme 33. (S)-BINOL as ligand in addition of Al(2-thienyl)₃(THF) to ketones

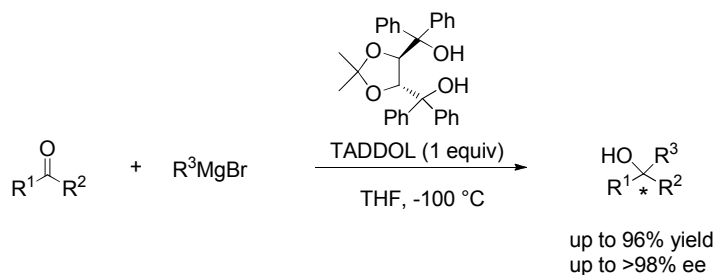
Finally, the same authors have developed an easy preparation of AlArEt₂(THF) reagents from the reaction of AlEt₂Br(THF) with one equivalent of ArMgBr in THF at 0 °C, which were added to a variety of ketones in the presence of 10 mol% of an in situ generated titanium complex of (*R*)-H₈-BINOL and 3.5 equivalents of Ti(O*i*-Pr)₄ in toluene.⁶³ The results collected in Scheme 34 show that the catalytic system worked very well in terms of stereocontrol for a wide range of aromatic ketones, regardless of the electronic nature or the steric effect of the substituents on the aryl groups, affording the corresponding chiral aryl tertiary alcohols as the sole products with high enantioselectivities of up to 94% ee. However, for aliphatic ketones, such as 3-methyl-2-butanone and 2-hexanone, the phenyl additions afforded the corresponding alcohols in low yields (38 and 60%, respectively) and poor enantioselectivities of 48 and 15% ee, respectively.



Scheme 34. (S)-H₈-BINOL as ligand in addition of AlArEt₂(THF) to ketones

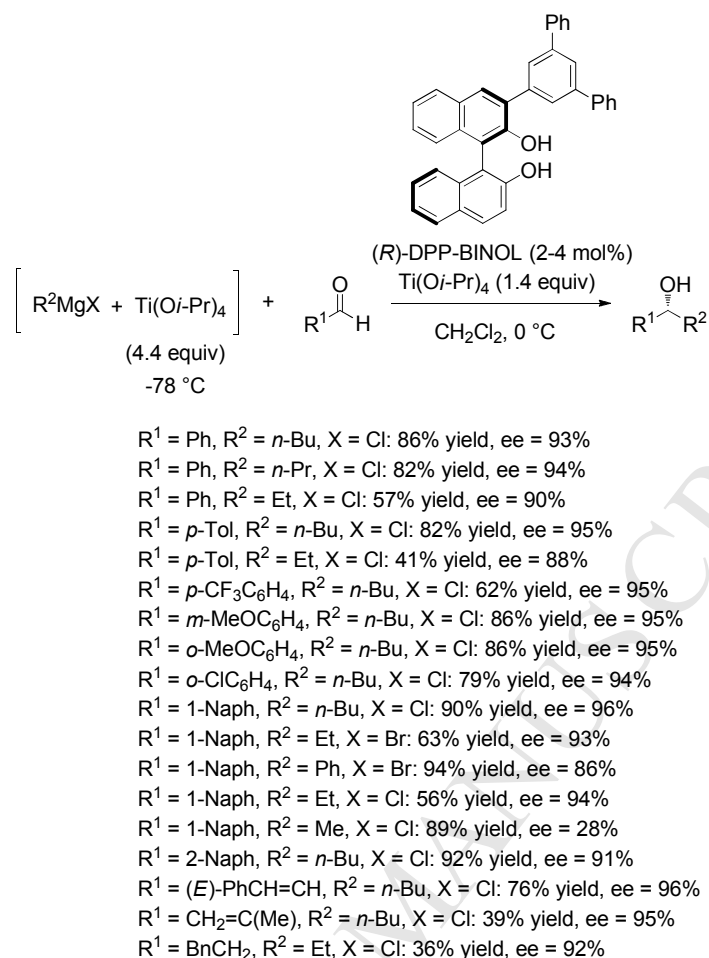
2.3. Additions of Grignard reagents

Grignard reagents are among the least expensive and most commonly used organometallic reagents in both laboratory and industry. Because of the high reactivity of these compounds, direct highly enantioselective Grignard addition to aldehydes has rarely been disclosed. The recent procedures using Grignard reagents as starting materials in addition to aldehydes often focused on transmetalation to form less reactive intermediates, such as RTi(Oi-Pr)₃, in situ generated from RMgX and Ti(Oi-Pr)₄. However, this titanium species has not been clearly determined to be RTi(Oi-Pr)₃, and titanate RTi(Oi-Pr)₄MgX could also constitute another candidate as intermediate. In 1990s, Weber and Seebach were the first authors to report the successful asymmetric addition of Grignard reagents to ketones performed in the presence of chiral TADDOL ligands, providing chiral tertiary alcohols in enantioselectivity greater than 95% ee (Scheme 35).⁶⁹



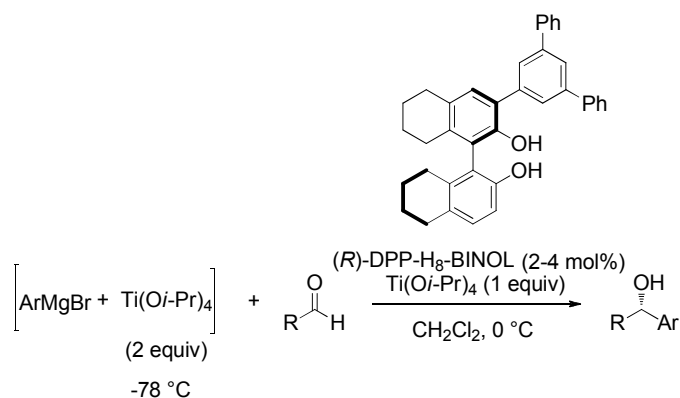
Scheme 35. First asymmetric addition of Grignard reagents to ketones reported by Weber and Seebach in 1990s

In the last few years, various enantioselective titanium-promoted additions of Grignard reagents to aldehydes have been developed by several groups all based on the use of BINOL derivatives as chiral ligands of $\text{Ti}(\text{O}i\text{-Pr})_4$. As an example, Harada et al. have reported general excellent enantioselectivities of up to 96% ee for ethylation, propylation, butylation, and even phenylation of aromatic, α,β -unsaturated, and aliphatic aldehydes starting from the corresponding Grignard reagents which were previously treated by $\text{Ti}(\text{O}i\text{-Pr})_4$ at $-78\text{ }^\circ\text{C}$ and then introduced to the reaction mixture (Scheme 36).⁷⁰ The reaction proceeded in the presence of an excess of $\text{Ti}(\text{O}i\text{-Pr})_4$ and a catalytic amount (2 mol%) of [(*R*)-3-(3,5-diphenylphenyl)-2,2'-dihydroxy-1,1'-binaphthyl] ((*R*)-DPP-BINOL), affording the corresponding chiral secondary alcohols in moderate to very high yields of up to 94%. It must be noted that chloromagnesium reagents and bromomagnesium reagents could be employed with comparable efficiency and selectivity. On the other hand, the reaction of 1-naphthaldehyde with MeMgCl resulted in a low enantioselectivity (28% ee). In contrast, a relatively high enantioselectivity (86% ee) was obtained for the phenylation of the same aldehyde. Furthermore, α,β -unsaturated aldehydes provided high enantioselectivities of up to 96% ee, while although sluggish, the reaction of aliphatic aldehydes also provided high enantioselectivities of up to 92% ee. The authors have compared the results obtained by using (*R*)-DPP-BINOL as ligand with those arisen from the use of the corresponding (*R*)-DPP- H_8 -BINOL, concluding that these two ligands had comparable efficiencies.



Scheme 36. (R)-DPP-BINOL as ligand in addition of alkyl- and phenylmagnesium halides to aldehydes

As an extension of the precedent methodology, these authors applied a related catalyst system based on (R)-DPP-H₈-BINOL to the asymmetric arylation of a wide range of aldehydes starting from the corresponding aryl Grignard reagents in combination with Ti(Oi-Pr)₄.^{70b,71} As shown in Scheme 37, the results showed high enantioselectivities and yields of up to 97% ee and 99%, respectively, for various combinations of aromatic, heteroaromatic, aliphatic, and α,β-unsaturated aldehydes with aryl bromomagnesium reagents, including those with functional groups, except for *ortho*-methoxybenzaldehyde which gave a low enantioselectivity (9% ee). Notably, an excellent enantioselectivity of 96% ee was reached for sterically hindered 2,4,6-Me₃C₆H₂MgBr.

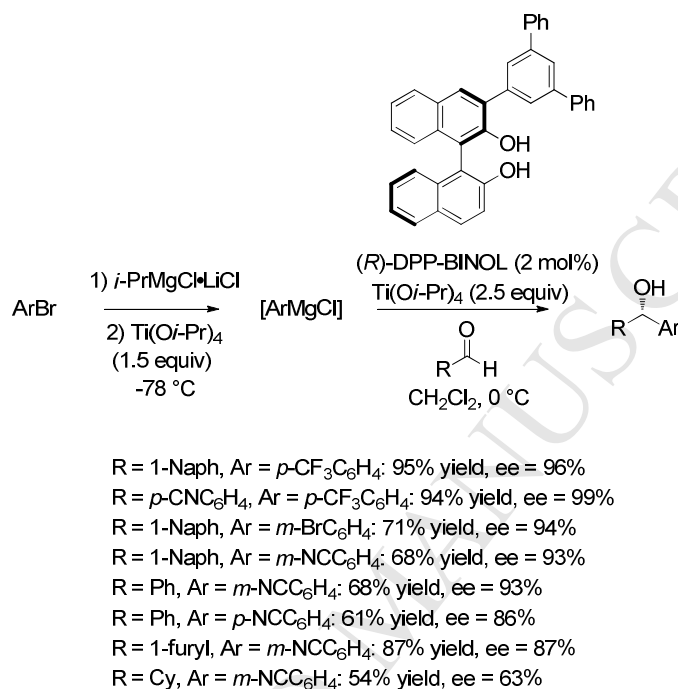


R	Ar	Yield (%)	ee (%)
Ph	<i>p</i> -Tol	99	91
Ph	<i>p</i> -ClC ₆ H ₄	95	91
Ph	<i>p</i> -FC ₆ H ₄	94	97
Ph	<i>o</i> -MeOC ₆ H ₄	66	97
Ph	2,4,6-Me ₃ C ₆ H ₂	89	96
<i>p</i> -Tol	Ph	93	90
<i>p</i> -FC ₆ H ₄	Ph	97	95
<i>p</i> -ClC ₆ H ₄	Ph	96	94
<i>p</i> -CNC ₆ H ₄	Ph	96	92
<i>p</i> -PhC ₆ H ₄	Ph	99	91
<i>m</i> -MeOC ₆ H ₄	Ph	90	95
1-Naph	Ph	97	95
2-Naph	Ph	96	91
2-furyl	Ph	83	89
2-thienyl	Ph	82	90
4-pyridyl	Ph	85	82
CH ₂ =C(Me)	Ph	87	97
<i>n</i> -Bu	Ph	88	88
Cy	Ph	86	80

Scheme 37. (R)-DPP-H₈-BINOL as ligand in addition of arylmagnesium bromides to aldehydes

In 2011, the same authors reported an efficient novel method using (R)-DPP-BINOL as titanium ligand for the enantioselective arylation of aromatic aldehydes albeit starting from aryl bromides.⁷² Indeed, in this case, functionalised aryl Grignard reagents were prepared in situ by bromine–magnesium exchange of the corresponding aryl bromide with *i*-PrMgCl (Scheme 38). This novel methodology was based on the fact that aryl bromides constitute preferable precursors for the preparation of functionalised Grignard reagents in light of their stability, good availability, and low price in comparison to the corresponding iodides. The method was applicable to aryl bromides bearing CF₃, Br, and CN groups, affording a range of

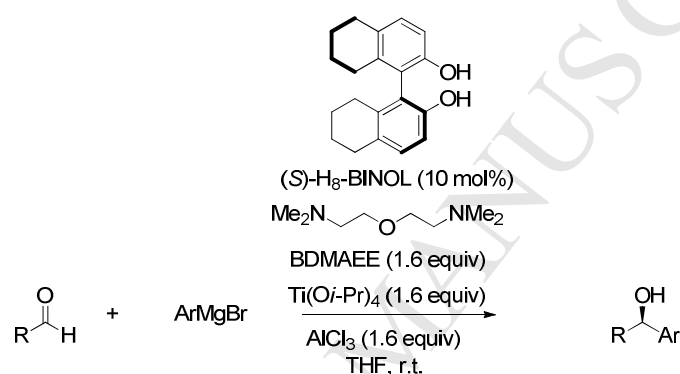
chiral functionalised aryl secondary alcohols of synthetic importance in good to high yields and enantioselectivities of up to 99% ee. Unfortunately, the reaction of an aliphatic aldehyde (R = Cy) resulted in the formation of the corresponding product in only moderate yield (54%) and enantioselectivity of 63% ee.



Scheme 38. (R)-DPP-BINOL as ligand in addition of in situ generated arylmagnesium chlorides from aryl bromides to aldehydes

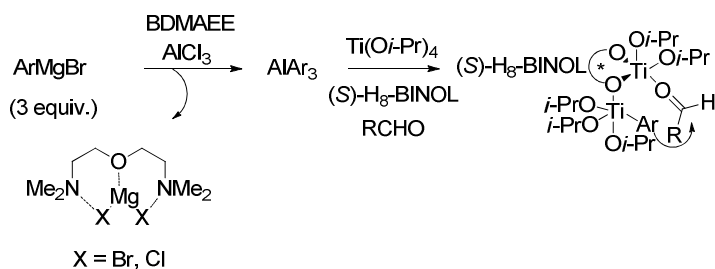
A drawback of the method reported by Harada et al. (Schemes 36-38) was the need to add the Grignard reagent to Ti(Oi-Pr)₄ at -78 °C and then to introduce the resulting mixture into the reaction for two hours at 0 °C. Although very useful on a laboratory scale, large-scale reactions at very low temperatures are impractical. A significant operational improvement was reported by Da and co-workers, consisting in converting the Grignard reagents (3 equivalents) into less reactive triarylaluminum intermediates in situ by treatment with AlCl₃.⁷³ In this novel direct process, MgBr₂ and MgBr(Oi-Pr) were formed, and could promote as Lewis acids the background reaction to form the racemic product and lower the enantioselectivity of the reaction. In this context, 2,2'-oxy-bis(*N,N*-dimethylethanamine) (BDMAEE) was used as an additive to chelate the in situ generated Lewis acids MgBr₂ and MgBr(Oi-Pr) and to suppress their activity so that the asymmetric additions promoted by 10 mol% of (S)-H₈-BINOL as chiral ligand of Ti(Oi-Pr)₄ were remarkably highly

enantioselective at room temperature for a variety of aromatic as well as aliphatic aldehydes with enantioselectivities of up to 99% ee (Scheme 39). Moreover, the chiral alcohols were obtained in general excellent yields of up to 97%. The authors have proposed the mechanism depicted in Scheme 39 to explain the role of AlCl_3 in the reaction. It consisted in accepting the aryl group from the Grignard reagent to generate AlAr_3 , which ultimately transferred the aryl to the aldehyde. BDMAEE was believed to sequester the magnesium salts to prevent them from promoting the racemic background process. It has to be noted that this unprecedented direct methodology is remarkable by its broad scope, affording excellent enantioselectivities and yields for all types of aldehydes and various aryl Grignard reagents.



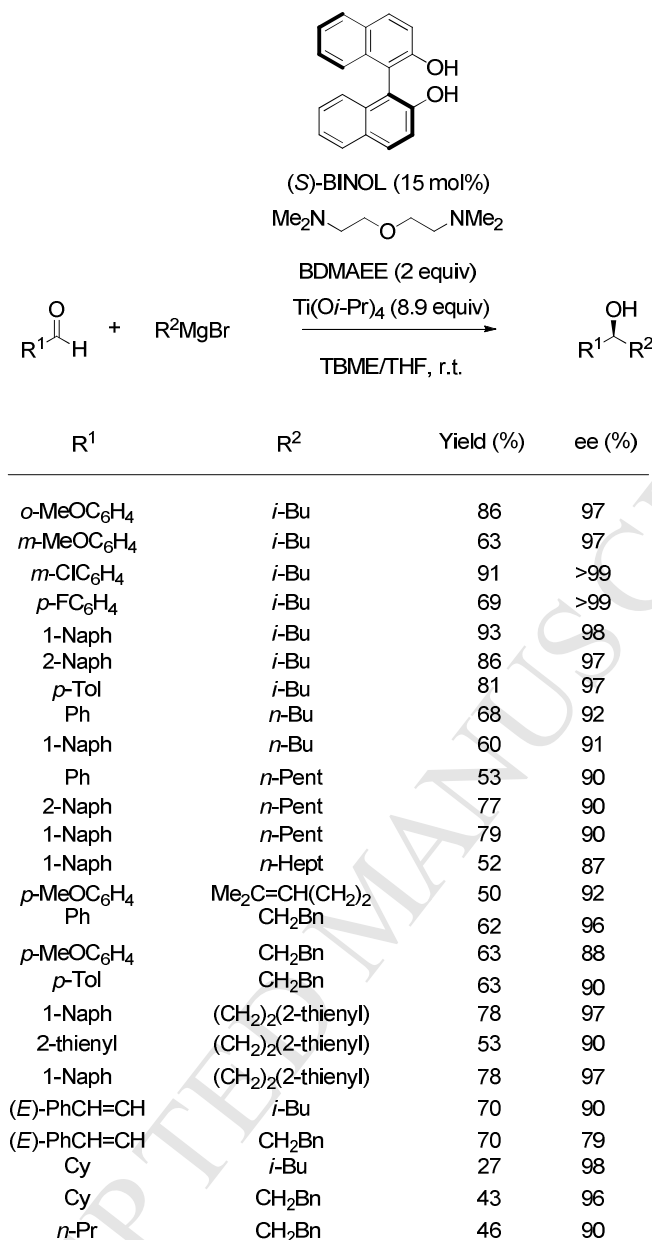
$\text{R} = o\text{-ClC}_6\text{H}_4$, $\text{Ar} = \text{Ph}$: 94% yield, ee = 94%
 $\text{R} = p\text{-FC}_6\text{H}_4$, $\text{Ar} = \text{Ph}$: 94% yield, ee = 97%
 $\text{R} = p\text{-ClC}_6\text{H}_4$, $\text{Ar} = \text{Ph}$: 92% yield, ee = 96%
 $\text{R} = p\text{-BrC}_6\text{H}_4$, $\text{Ar} = \text{Ph}$: 95% yield, ee = 95%
 $\text{R} = m\text{-MeOC}_6\text{H}_4$, $\text{Ar} = \text{Ph}$: 93% yield, ee = 97%
 $\text{R} = p\text{-Tol}$, $\text{Ar} = \text{Ph}$: 94% yield, ee = 97%
 $\text{R} = p\text{-CF}_3\text{C}_6\text{H}_4$, $\text{Ar} = \text{Ph}$: 95% yield, ee = 98%
 $\text{R} = 1\text{-Naph}$, $\text{Ar} = \text{Ph}$: 95% yield, ee = 99%
 $\text{R} = 2\text{-Naph}$, $\text{Ar} = \text{Ph}$: 96% yield, ee = 95%
 $\text{R} = 1\text{-Naph}$, $\text{Ar} = 2\text{-Naph}$: 92% yield, ee = 95%
 $\text{R} = p\text{-Tol}$, $\text{Ar} = 1\text{-Naph}$: 96% yield, ee = 96%
 $\text{R} = p\text{-FC}_6\text{H}_4$, $\text{Ar} = 1\text{-Naph}$: 94% yield, ee = 99%
 $\text{R} = 1\text{-Naph}$, $\text{Ar} = m\text{-Tol}$: 93% yield, ee = 95%
 $\text{R} = 1\text{-Naph}$, $\text{Ar} = \text{benzofuran-5-yl}$: 83% yield, ee = 87%
 $\text{R} = p\text{-ClC}_6\text{H}_4$, $\text{Ar} = 2\text{-thienyl}$: 95% yield, ee = 97%
 $\text{R} = \text{Cy}$, $\text{Ar} = \text{Ph}$: 91% yield, ee = 95%
 $\text{R} = \text{Cy}$, $\text{Ar} = 2\text{-Naph}$: 90% yield, ee = 98%
 $\text{R} = n\text{-Non}$, $\text{Ar} = \text{Ph}$: 91% yield, ee = 94%

proposed mechanism:



Scheme 39. (S)-H₈-BINOL as ligand in direct addition of arylmagnesium bromides to aldehydes with BDMAEE and AlCl₃

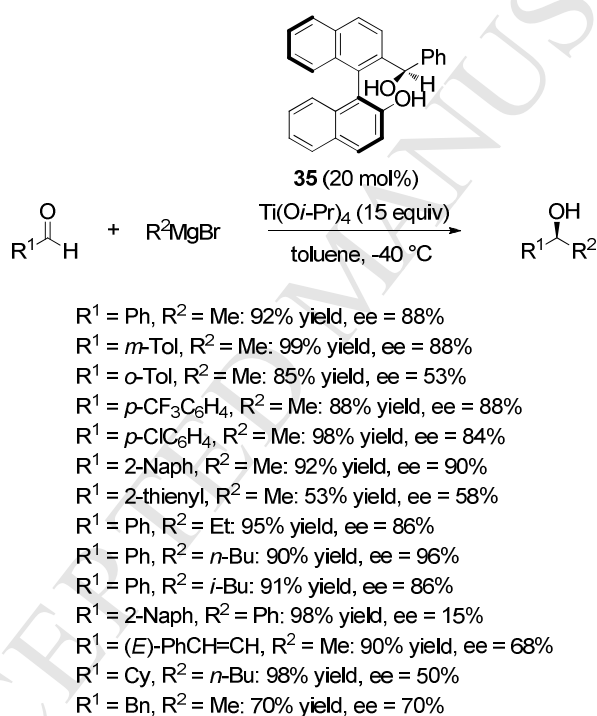
Another methodology developed by the same authors to deactivate alkyl Grignard reagents in the presence of Ti(Oi-Pr)₄ was to perform their enantioselective direct additions to aldehydes in the presence of 2,2'-oxy-bis(*N,N*-dimethylethanamine) (BDMAEE).⁷⁴ As in the precedent method, BDMAEE was supposed to chelate the in situ generated salts MgBr₂ from equilibrium of RMgBr and MgBr(Oi-Pr) from transmetalation of RMgBr with Ti(Oi-Pr)₄. When this process was promoted by 15 mol% of (S)-BINOL as chiral ligand of Ti(Oi-Pr)₄ employed in a large excess (8.9 equivalents), it afforded at room temperature a range of chiral secondary alcohols in remarkable enantioselectivities of up to >99% ee combined with good yields, as shown in Scheme 40. The wide scope of the reaction must be highlighted since butylation, pentylation, heptylation, arylethylation, as well as alkenylethylation could be successfully achieved with homogeneous excellent results with aromatic, aliphatic, as well as α,β -unsaturated aldehydes. It must be noted that this unprecedented methodology dealing with alkyl Grignard reagents efficiently completed that depicted in Scheme 39 dealing with aryl Grignard reagents. Again, it is remarkable by its broad scope, affording high to excellent enantioselectivities for all types of aldehydes and various alkyl Grignard reagents.



Scheme 40. (S)-BINOL as ligand in direct addition of alkylmagnesium bromides to aldehydes with BDMAEE

In 2011, Yus et al. reported the use of another efficient chiral catalyst for the direct addition of alkylmagnesium bromides to aromatic, α,β -unsaturated, and some aliphatic aldehydes in the presence of fifteen equivalents of Ti(Oi-Pr)₄.⁷⁵ Chiral ligand **35** was derived from (S)-BINOL and employed at 20 mol% of catalyst loading in toluene at -40 °C in combination with a large excess of Ti(Oi-Pr)₄ (15 equivalents), providing various chiral secondary alcohols in good yields and moderate to high enantioselectivities of up to 96% ee, as shown in Scheme 41. The highest enantioselectivities were generally obtained for

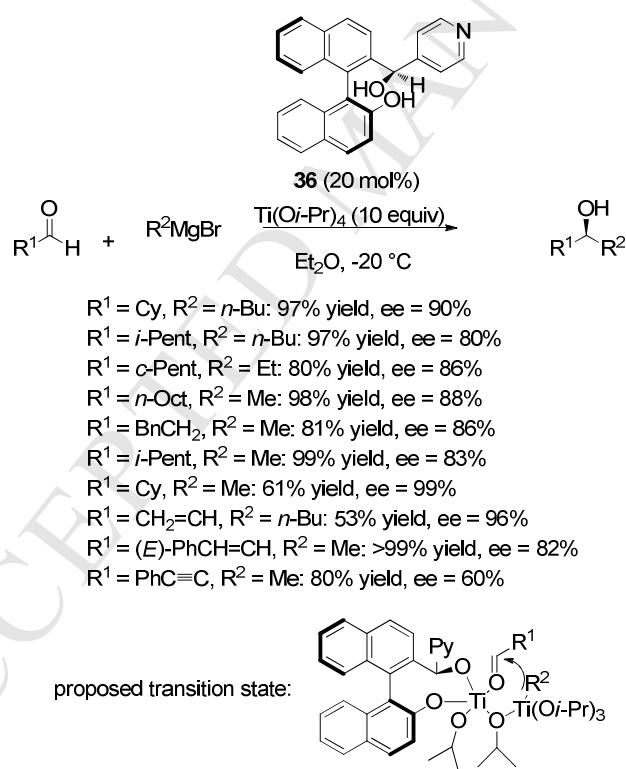
benzaldehyde and its derivatives bearing electron-poor as well as electron-rich substituents in the *meta* and *para* positions while the alkylation of *ortho*-methylbenzaldehyde proceeded with a lower enantioselectivity (53% ee), probably due to the steric hindrance close to the reactive site. On the other hand, the reaction of phenylacetaldehyde proceeded with moderate enantioselectivity (70% ee), as well as that of cinnamaldehyde which gave 68% ee. The use of 2-thiophenecarboxaldehyde prompted a decrease in the enantioselectivity to 58% ee and that of an aliphatic aldehyde, such as cyclohexanecarboxaldehyde, provided an even lower enantioselectivity (50% ee). Furthermore, it was shown that the addition of sp^2 -hybridised Grignard reagents, such as PhMgBr, to 2-naphthaldehyde proceeded in excellent yield (98%) albeit with a low enantioselectivity (15% ee).



Scheme 41. Chiral BINOL derivative as ligand in direct addition of organomagnesium bromides to aldehydes

In the same context, these authors have more recently developed a related readily available BINOL-derived chiral ligand **36** bearing a pyridine which was applied to the enantioselective direct additions of alkylmagnesium bromides to aliphatic aldehydes in the presence of ten equivalents of Ti(Oi-Pr)_4 .⁷⁶ As shown in Scheme 42, this methodology performed at $-20\text{ }^\circ\text{C}$ in diethylether allowed the synthesis of a range of chiral secondary aliphatic alcohols to be

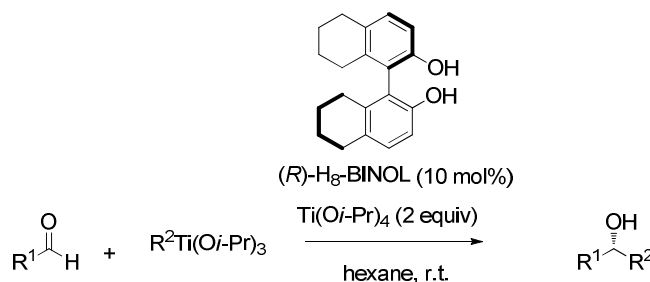
achieved in good to quantitative yields and enantioselectivities of up to 99% ee. Both linear and α -branched aliphatic substrates were found to be suitable for the reaction as well as α,β -unsaturated aldehydes, such as cinnamaldehyde (82% ee), acrolein (96% ee), and phenylpropargylic aldehyde (60% ee), demonstrating the robustness and applicability of the methodology. This novel method overcame the main problems associated with the use of aliphatic substrates, such as their multiple conformations, the absence of possible π -stacking interactions with the catalyst and/or their highly enolisable character. The authors have proposed the transition state depicted in Scheme 42 to explain the results. Even if the two methodologies depicted in Schemes 41 and 42 require low temperatures to be achieved (-40 and -20 °C, respectively), they represent considerable advances in the direct addition of alkyl Grignard reagents to aldehydes by their practical feature, broad scope, and remarkable levels of yields and enantioselectivities reached.



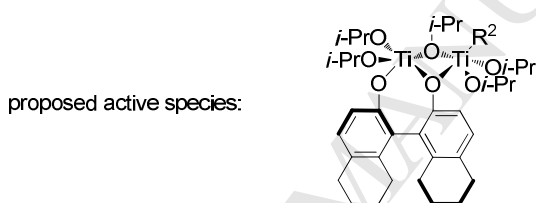
Scheme 42. Another chiral BINOL derivative as ligand in direct addition of alkylmagnesium bromides to aliphatic and α,β -unsaturated aldehydes

2.4. Additions of organotitanium reagents

Roles of excess of $\text{Ti}(\text{Oi-Pr})_4$ in titanium-promoted asymmetric additions of organometallic compounds to carbonyl compounds have been suggested as to generate a dititanium active species bearing a chiral ligand and also to facilitate the removal of the product. It has been suggested that the reactions involved the additions of organotitanium species, which were in situ generated from reactions of organometallic compounds with $\text{Ti}(\text{Oi-Pr})_4$. However, direct asymmetric additions of organotitanium reagents have been demonstrated only rarely since the publication by Seebach and Weber in 1994 describing the first catalytic asymmetric addition of $\text{RTi}(\text{Oi-Pr})_3$ to aldehydes by using a TADDOL-derived chiral titanium catalyst (Scheme 2).¹⁵ In the last few years, however, several excellent results have been independently described in this context by the groups of Gau and Harada. For example, Gau and Li have reported highly enantioselective direct additions of alkyltitanium reagents $\text{RTi}(\text{Oi-Pr})_3$ to aldehydes catalysed by an in situ generated titanium catalyst of (*R*)- H_8 -BINOL (Scheme 43).⁷⁷ This ligand was employed at 10 mol% of catalyst loading along with two equivalents of $\text{Ti}(\text{Oi-Pr})_4$ in hexane at room temperature, allowing the formation of a wide variety of chiral secondary alcohols in good to high yields and enantioselectivities of up to 94% ee, as shown in Scheme 43. Remarkably, aromatic, heteroaromatic as well as α,β -unsaturated aldehydes were compatible with this protocol and three different alkyltitanium reagents $\text{RTi}(\text{Oi-Pr})_3$ ($\text{R} = \text{Cy}$, *n*-Bu, and *i*-Bu) were successfully employed with reactivity and enantioselectivity differences in terms of steric bulkiness of the R nucleophiles. Therefore, the additions of secondary cyclohexyl to aldehydes were slower than those of primary *i*-butyl or *n*-butyl nucleophiles. For the primary alkyls, lower enantioselectivities were obtained for products from additions of the linear *n*-butyl as compared with the enantioselectivities of products arisen from additions of the branched *i*-butyl group. The authors have proposed the dititanium catalytic active species depicted in Scheme 43 containing one (*R*)- H_8 -BINOL ligand and the nucleophile.



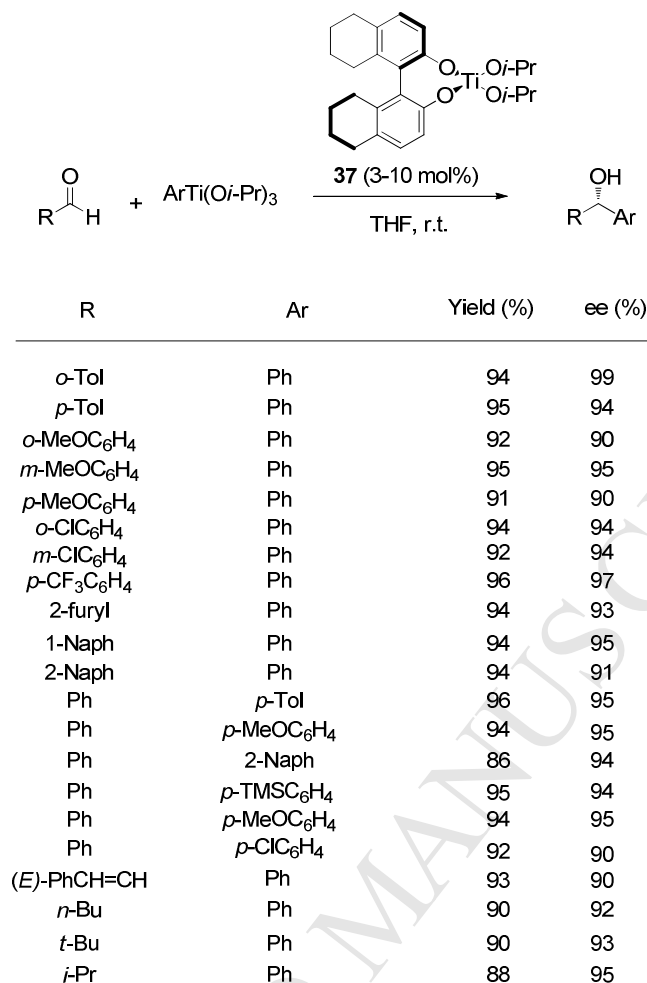
$\text{R}^1 = \text{Ph}, \text{R}^2 = \text{Cy}: 90\% \text{ yield, ee} = 92\%$
 $\text{R}^1 = p\text{-Tol}, \text{R}^2 = \text{Cy}: 83\% \text{ yield, ee} = 90\%$
 $\text{R}^1 = o\text{-FC}_6\text{H}_4, \text{R}^2 = \text{Cy}: 66\% \text{ yield, ee} = 93\%$
 $\text{R}^1 = o\text{-CF}_3\text{C}_6\text{H}_4, \text{R}^2 = \text{Cy}: 77\% \text{ yield, ee} = 94\%$
 $\text{R}^1 = 2\text{-furyl}, \text{R}^2 = \text{Cy}: 85\% \text{ yield, ee} = 90\%$
 $\text{R}^1 = \text{Ph}, \text{R}^2 = i\text{-Bu}: 78\% \text{ yield, ee} = 94\%$
 $\text{R}^1 = p\text{-Tol}, \text{R}^2 = i\text{-Bu}: 78\% \text{ yield, ee} = 92\%$
 $\text{R}^1 = m\text{-ClC}_6\text{H}_4, \text{R}^2 = i\text{-Bu}: 80\% \text{ yield, ee} = 91\%$
 $\text{R}^1 = \text{Ph}, \text{R}^2 = n\text{-Bu}: 78\% \text{ yield, ee} = 84\%$
 $\text{R}^1 = m\text{-ClC}_6\text{H}_4, \text{R}^2 = n\text{-Bu}: 80\% \text{ yield, ee} = 86\%$
 $\text{R}^1 = (E)\text{-PhCH=CH}, \text{R}^2 = n\text{-Bu}: 74\% \text{ yield, ee} = 75\%$
 $\text{R}^1 = (E)\text{-PhCH=CH}, \text{R}^2 = i\text{-Bu}: 76\% \text{ yield, ee} = 83\%$
 $\text{R}^1 = (E)\text{-PhCH=CH}, \text{R}^2 = \text{Cy}: 77\% \text{ yield, ee} = 76\%$



Scheme 43. (*R*)-H₈-BINOL as ligand in direct addition of alkyltitanium reagents to aromatic and α,β -unsaturated aldehydes

In addition, the same authors have described remarkable enantioselective direct additions of aryltitanium reagents ArTi(Oi-Pr)_3 to aromatic, α,β -unsaturated aldehydes as well as aliphatic aldehydes based on the use of a catalytic amount (3-10 mol%) of a preformed chiral titanium catalyst **37** derived from (*R*)-H₈-BINOL (Scheme 44).⁷⁸ It must be noted that the authors obtained comparable excellent results when using the corresponding in situ generated titanium catalyst of (*R*)-H₈-BINOL. Using preformed catalyst **37** presented the advantage to allow the employment of excess amounts of Ti(Oi-Pr)_4 to be avoided. Remarkably, the reactions catalysed by complex **37** proceeded instantaneously at room temperature, affording a wide range of chiral secondary alcohols with general excellent enantioselectivities always $\geq 90\%$ ee and up to 99% ee, as summarised in Scheme 44. For aromatic aldehydes bearing either an electron-donating or an electron-withdrawing substituent at *ortho*-, *meta*-, or *para*-position, PhTi(Oi-Pr)_3 addition reactions employing 3-10 mol% of ligand **37** afforded the

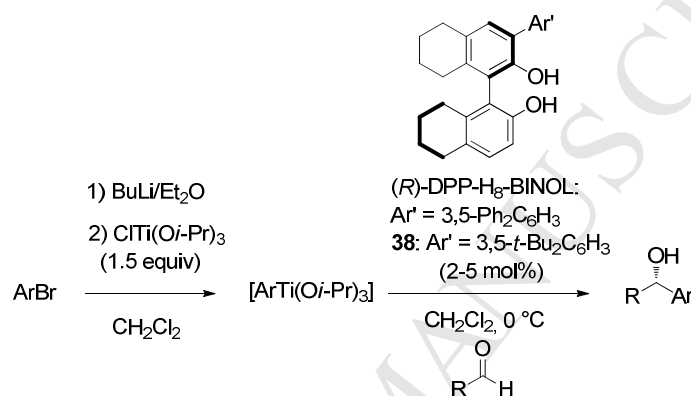
corresponding chiral secondary diarylmethanols in >90% yields and enantioselectivities of $\geq 90\%$ ee. It was worth noting that 3 mol% of ligand **37** was effective enough for the reaction of *para*-methoxybenzaldehyde (90% ee), and 5 mol% of **37** was used for 1- and 2-naphthaldehydes (giving 95 and 91% ee, respectively). The catalytic system applied equally well to the additions of $\text{PhTi}(\text{Oi-Pr})_3$ to (*E*)-cinnamaldehyde or 2-furaldehyde, affording the corresponding products in enantioselectivities of 90 and 93% ee, respectively. Regardless of the steric bulk of aliphatic aldehydes, the $\text{PhTi}(\text{Oi-Pr})_3$ addition reactions of linear pentanal or of bulkier 2-methylpropanal or 2,2-dimethylpropanal furnished the corresponding alcohols in good to excellent yields and excellent enantioselectivities of $\geq 92\%$ ee. Moreover, additions of other aryl nucleophiles $\text{ArTi}(\text{Oi-Pr})_3$ ($\text{Ar} = p\text{-tolyl}$, $p\text{-MeOC}_6\text{H}_4$, $p\text{-Cl C}_6\text{H}_4$, $p\text{-TMS C}_6\text{H}_4$, or 2-naphthyl) to benzaldehyde afforded the corresponding aryl addition products in enantioselectivities $\geq 90\%$ ee but in opposite absolute structure as compared to products from additions of the phenyl nucleophile to aryl aldehydes. In addition to the mild reaction conditions (room temperature) and the rapidity of the reaction, another great advantage of this nice process was that excess amounts of $\text{Ti}(\text{Oi-Pr})_4$ were not necessary. Furthermore, in all cases of substrates studied, the yields were very high of up to 96%. The authors have investigated other chiral ligands to induce these reactions, such as (*S*)-BINOL, TADDOL derivatives, chiral 1,2-diols, and chiral 1,2-diamines which all provided lower effectiveness in terms of stereocontrol.



Scheme 44. Preformed titanium complex of (*R*)-H₈-BINOL as catalyst in direct addition of aryltitanium reagents to aldehydes

More recently, Harada et al. reported another highly efficient and original method for the enantioselective arylation and heteroarylation of aldehydes with organotitanium reagents prepared in situ through the reaction of the corresponding aryl- and heteroaryllithium reagents with ClTi(Oi-Pr)₃.⁷⁹ Chiral titanium complexes in situ generated from (*R*)-DPP-H₈-BINOL ligand and (*R*)-3-aryl-H₈-BINOL ligand **38** exhibited an excellent catalytic activity in terms of enantioselectivity and turnover efficiency for the reaction, providing chiral diaryl-, aryl heteroaryl-, and diheteroarylmethanol derivatives in high enantioselectivities of up to 98% ee, as shown in Scheme 45. In most cases of substrates, a catalyst loading as low as 2 mol% was sufficient to reach these results. In some cases, only 0.5 mol% of ligand **38** allowed enantioselectivities of up to 90% ee to be achieved. It was found that the reaction of benzaldehyde with titanium reagents derived from *para*- and *meta*-substituted phenyl

bromides and 2-naphthyl bromide uniformly provided high yields and enantioselectivities (90-98% ee) while reactions of benzaldehyde employing aryltitanium reagents derived from *ortho*-substituted bromides resulted in lower enantioselectivities (12% ee). The scope of the reaction was also examined with aldehydes other than benzaldehyde, providing most of the time high enantioselectivities for aromatic, heteroaromatic, as well as α,β -unsaturated aldehydes whereas low to moderate enantioselectivities (20-72% ee) were obtained for aliphatic aldehydes.



with ligand (R)-DPP-H₈-BINOL (2 mol%):

R = 1-Naph, Ar = Ph: 96% yield, ee = 98%

R = Ph, Ar = *p*-MeOC₆H₄: 75% yield, ee = 94%

R = Ph, Ar = *m*-Tol: 95% yield, ee = 92%

R = Ph, Ar = *o*-MeOC₆H₄: 92% yield, ee = 12%

R = Ph, Ar = 2-Naph: 90% yield, ee = 90%

R = Ph, Ar = 3-furyl: 84% yield, ee = 95%

R = *p*-ClC₆H₄, Ar = Ph: 92% yield, ee = 95%

R = *p*-Tol, Ar = 2-thienyl: 95% yield, ee = 89%

R = 2-furyl, Ar = 2-thienyl: 96% yield, ee = 86%

R = 2-furyl, Ar = 3-furyl: 74% yield, ee = 91%

R = 2-thienyl, Ar = 3-thienyl: 94% yield, ee = 96%

R = CH₂=C(Me), Ar = *p*-ClC₆H₄: 88% yield, ee = 98%

R = *n*-Bu, Ar = *p*-ClC₆H₄: 82% yield, ee = 72%

R = Cy, Ar = *p*-ClC₆H₄: 97% yield, ee = 39%

R = Cy, Ar = 2-thienyl: 91% yield, ee = 20%

with ligand 38 (2 mol%):

R = *p*-ClC₆H₄, Ar = 2-thienyl: 93% yield, ee = 86%

R = Ph, Ar = 2-thienyl: 95% yield, ee = 95%

with ligand 38 (5 mol%):

R = 1-Naph, Ar = 2-benzothienyl: 87% yield, ee = 90%

R = Ph, Ar = 2-benzothienyl: 94% yield, ee = 87%

R = Ph, Ar = 2-furyl: 68% yield, ee = 55%

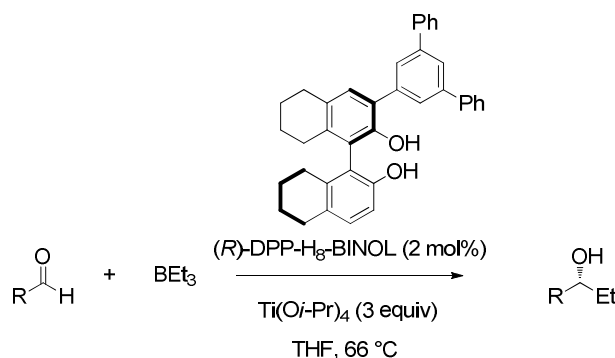
with ligand 38 (0.5 mol%):

R = Ph, Ar = 3-thienyl: 90% yield, ee = 95%

Scheme 45. (R)-3-Aryl-H₈-BINOL derivatives as ligands in addition of in situ generated (hetero)aryltitanium reagents to aldehydes

2.5. Additions of organoboron reagents

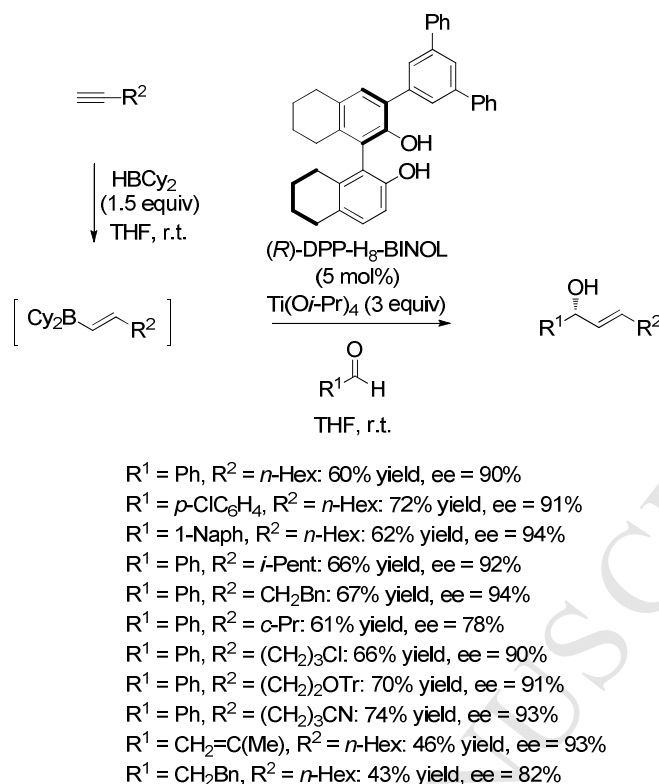
Because a variety of alkylboranes are commercially available and can be readily prepared by hydroboration of alkenes, they are promising candidates for practical alkylating reagents. Indirect use of trialkylboranes in asymmetric alkylation reactions has been reported by a boron-zinc exchange reaction.⁸⁰ On the other hand, Harada and Ukon reported, in 2008, the direct use of trialkylboranes without converting them into alkylmetallic species.⁸¹ As shown in Scheme 46, triethylborane could be directly and enantioselectively added to aromatic, aliphatic, and α,β -unsaturated aldehydes in the presence of three equivalents of $\text{Ti}(\text{O}i\text{-Pr})_4$ and a catalytic amount of (*R*)-DPP- H_8 -BINOL as chiral ligand in THF. In most cases of substrates studied, excellent enantioselectivities (92-97% ee) were obtained for the corresponding chiral secondary alcohols formed except for the reactions of *ortho*-chlorobenzaldehyde, 2-furaldehyde, and cinnamaldehyde which provided lower enantioselectivities of 84, 57, and 85% ee, respectively. Remarkably, a low catalyst loading of only 2 mol% was sufficient to reach both excellent yields and enantioselectivities. To gain insight into the mechanism of the process, the authors performed several control experiments. In the absence of the ligand, a slow alkylation reaction of 1-naphthaldehyde was observed with a mixture of triethylborane and $\text{Ti}(\text{O}i\text{-Pr})_4$ (3 equivalents). In the presence of 2 mol% of (*R*)-DPP- H_8 -BINOL, the ethylation reaction with triethylborane did not proceed at all either with a catalytic amount of $\text{Ti}(\text{O}i\text{-Pr})_4$ or without it. The requirement of $\text{Ti}(\text{O}i\text{-Pr})_4$ in more than a stoichiometric amount could suggest that the active alkylating reagent of the reaction was $\text{EtTi}(\text{O}i\text{-Pr})_3$, or its aggregate with $\text{Et}_2\text{B}(\text{O}i\text{-Pr})$, generated in equilibrium. This important work, which constituted the first direct use of trialkylboranes without converting into alkylzinc species, has significantly expanded the scope of the catalytic asymmetric alkylation of aldehydes.



R = Ph: 93% yield, ee = 96%
 R = *p*-Tol: 85% yield, ee = 96%
 R = *p*-PhC₆H₄: 87% yield, ee = 93%
 R = *p*-ClC₆H₄: 74% yield, ee = 96%
 R = *p*-FC₆H₄: 87% yield, ee = 97%
 R = *p*-CF₃C₆H₄: 73% yield, ee = 94%
 R = *m*-MeOC₆H₄: 94% yield, ee = 97%
 R = *o*-ClC₆H₄: 49% yield, ee = 84%
 R = 1-Naph: 90% yield, ee = 96%
 R = 2-Naph: 83% yield, ee = 96%
 R = 2-furyl: 83% yield, ee = 57%
 R = (*E*)-PhCH=CH: 73% yield, ee = 85%
 R = BnCH₂: 37% yield, ee = 92%

Scheme 46. (*R*)-DPP-H₈-BINOL as ligand in direct addition of BEt₃ to aldehydes

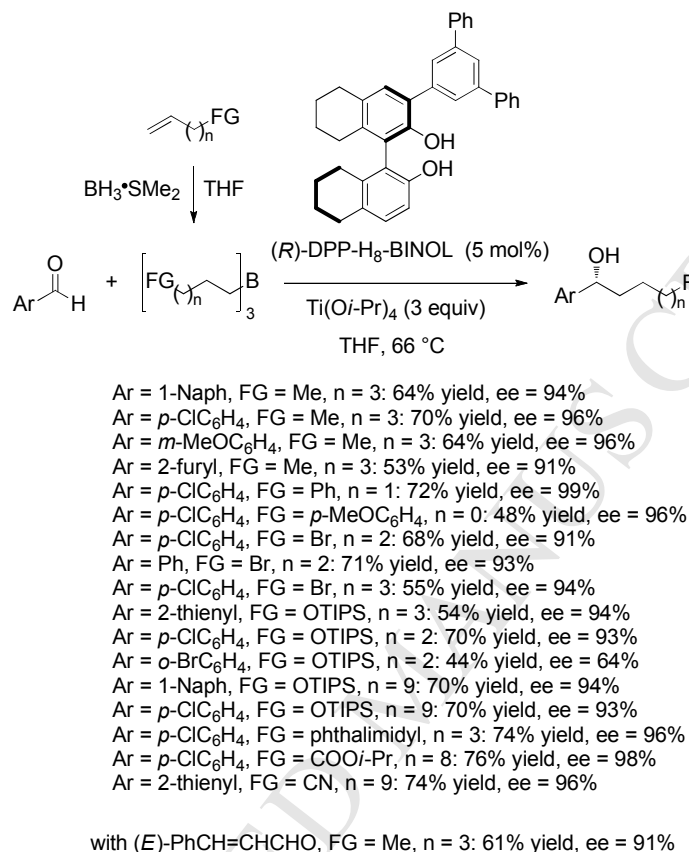
Later, the same authors reported another type of methodology for the synthesis of chiral secondary alcohols based on the use of in situ generated 1-alkenylboron reagents.⁸² As shown in Scheme 47, this one-pot procedure started from terminal alkynes and aldehydes. Hydroboration of these terminal alkynes with dicyclohexylborane and subsequent reaction of the resulting generated alkenylboron reagents with aldehydes in the presence of a catalytic amount (5 mol%) of (*R*)-DPP-H₈-BINOL as chiral ligand and an excess of Ti(Oi-Pr)₄ afforded the corresponding chiral allylic alcohols in good to high enantioselectivities of up to 94% ee. The scope of the process was found wide since a range of aromatic, aliphatic, and α,β -unsaturated aldehydes were tolerated as well as various aliphatic alkynes including those containing a chlorine atom, a protected alcohol, a nitrile, and an amide.



Scheme 47. (R) -DPP- H_8 -BINOL as ligand in addition of in situ generated 1-alkenylboron reagents to aldehydes

Despite recent significant advances in the chemistry of functionalised organometallic reagents, very few methods have been developed for the enantioselective addition of functionalised alkyl groups that would provide an efficient entry into chiral polyfunctionalised alcohols. In this context, a closely related methodology to that depicted in Scheme 47 was applied by the same authors to the synthesis of a wide range of chiral functionalised secondary alcohols, in 2013.⁸³ It was based on enantioselective additions of in situ generated functionalised alkylboron reagents to aromatic, heteroaromatic, and α,β -unsaturated aldehydes. As shown in Scheme 48, the required functionalised alkylboron reagents were in situ generated through hydroboration of the corresponding functionalised terminal olefins with $BH_3 \cdot SMe_2$. The latter subsequently underwent addition to aldehydes in the presence of 5 mol% of (R) -DPP- H_8 -BINOL and three equivalents of $Ti(Oi-Pr)_4$ to afford the corresponding alcohols in remarkable general enantioselectivities of up to 99% ee. A range of starting functionalised olefins were tolerated, since terminal alkenes bearing aromatic, aliphatic, TIPS protected alcohol, phthalimide, bromide, isopropyl ester, and cyano groups could be

successfully used in the reaction, thus demonstrating the wide generality of this novel efficient entry to chiral polyfunctionalised alcohols.

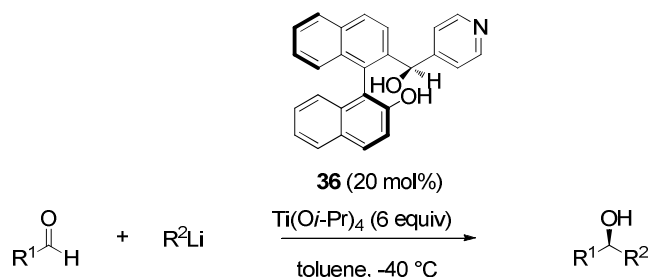


Scheme 48. (R)-DPP-H₈-BINOL as ligand in addition of functionalised alkylboron reagents to aromatic and α,β -unsaturated aldehydes

2.6. Additions of organolithium reagents

Organolithium compounds are common bench reagents found in any organic synthetic laboratory and are widely used in industry to produce numerous materials from pharmaceutical to polymers.⁸⁴ In 1969, Seebach et al. reported the first comprehensive investigation of the addition of organolithium reagents in the presence of various chiral ligands derived from diethyl tartrate.⁸⁵ It is only in 2011, that a substoichiometric enantioselective addition of methyllithium to *ortho*-tolylbenzaldehyde was reported by

Maddaluno et al.⁸⁶ On the other hand, Yus et al. soon later reported the first efficient catalytic system for the asymmetric alkylation of aldehydes with organolithium reagents performed in the presence of an excess of $\text{Ti}(\text{Oi-Pr})_4$ (Scheme 49).⁸⁷ The process involved readily available BINOL-derived chiral ligand **36** employed at a catalyst loading of 20 mol% in toluene. A variety of alkyllithium reagents could be added to aromatic and heteroaromatic aldehydes, providing the corresponding chiral aromatic secondary alcohols in good to high enantioselectivities of up to 96% ee, as shown in Scheme 49. Lower enantioselectivities (62-68% ee) were achieved in the cases of aliphatic aldehydes, α,β -unsaturated aldehydes, and also by using an aryllithium reagent, such as phenyllithium, which provided an enantioselectivity of only 17% ee by reaction with 2-naphthaldehyde. It must be noted that in this process, the potential problems associated with the high reactivity of organolithium compounds were overcome under the reaction conditions, demonstrating that this methodology was compatible with functionalised substrates. It is important to note that this work proposed the first efficient catalytic system for the asymmetric alkylation of aromatic aldehydes with alkyllithium reagents in the presence of $\text{Ti}(\text{Oi-Pr})_4$.



with $\text{R}^2 = \text{Me}$:

$\text{R}^1 = \text{Ph}$: 87% yield, ee = 90%

$\text{R}^1 = o\text{-Tol}$: 78% yield, ee = 62%

$\text{R}^1 = p\text{-Tol}$: 81% yield, ee = 88%

$\text{R}^1 = p\text{-MeOC}_6\text{H}_4$: 91% yield, ee = 89%

$\text{R}^1 = p\text{-CF}_3\text{C}_6\text{H}_4$: 82% yield, ee = 88%

$\text{R}^1 = p\text{-ClC}_6\text{H}_4$: 74% yield, ee = 84%

$\text{R}^1 = p\text{-NCC}_6\text{H}_4$: 84% yield, ee = 82%

$\text{R}^1 = 2\text{-Naph}$: 85% yield, ee = 90%

$\text{R}^1 = 2\text{-thienyl}$: 57% yield, ee = 88%

$\text{R}^1 = 2\text{-furyl}$: 56% yield, ee = 72%

$\text{R}^1 = (E)\text{-PhCH=CH}$: 84% yield, ee = 68%

$\text{R}^1 = \text{Bn}$: 23% yield, ee = 62%

with $\text{R}^2 = \text{Et}$:

$\text{R}^1 = \text{Ph}$: 78% yield, ee = 92%

$\text{R}^1 = p\text{-ClC}_6\text{H}_4$: 62% yield, ee = 92%

with $\text{R}^2 = n\text{-Bu}$:

$\text{R}^1 = \text{Ph}$: 90% yield, ee = 96%

$\text{R}^1 = p\text{-BrC}_6\text{H}_4$: 89% yield, ee = 94%

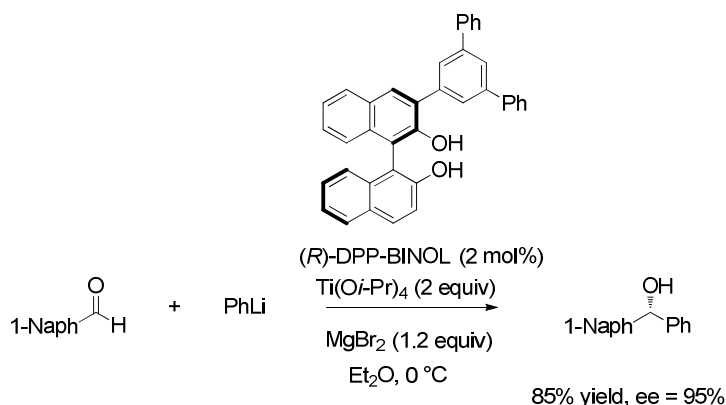
$\text{R}^1 = p\text{-MeOC}_6\text{H}_4$: 90% yield, ee = 96%

with $\text{R}^2 = \text{Ph}$:

$\text{R}^1 = 2\text{-Naph}$: 96% yield, ee = 17%

Scheme 49. (S)-BINOL derivative as ligand in addition of organolithium reagents to aldehydes

Finally, Harada and Muramatsu described the enantioselective addition of phenyllithium to 1-naphthaldehyde, providing the corresponding chiral alcohol in 85% yield and excellent enantioselectivity of 95% ee (Scheme 50).⁷¹ Actually in this process, the phenyllithium reagent could be employed after conversion into PhMgBr by treatment with MgBr_2 . It must be noted that the reaction was carried out without removing concomitantly produced LiBr , but was simply performed by mixing PhLi with MgBr_2 (1.2 equivalent) and Ti(Oi-Pr)_4 (2 equivalents). The chirality arose from using a catalytic amount (2 mol%) of (R)-DPP-BINOL as chiral ligand.

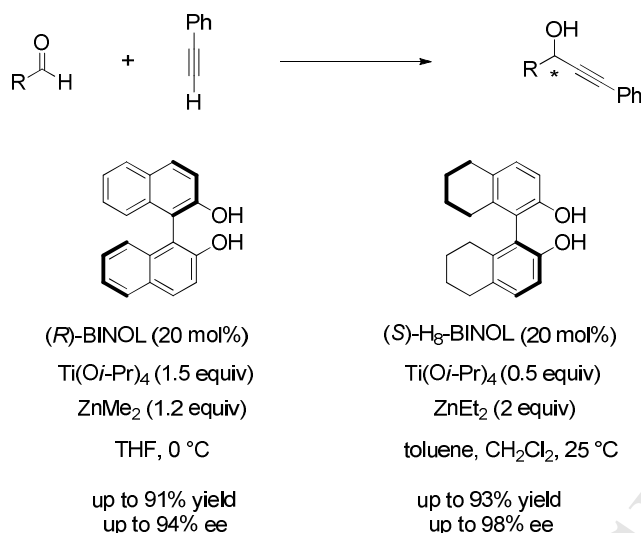


Scheme 50. (R)-DPP-BINOL as ligand in addition of PhLi to naphthylcarboxaldehyde

3. Titanium-promoted alkynylation reactions

3.1. Aldehydes as electrophiles

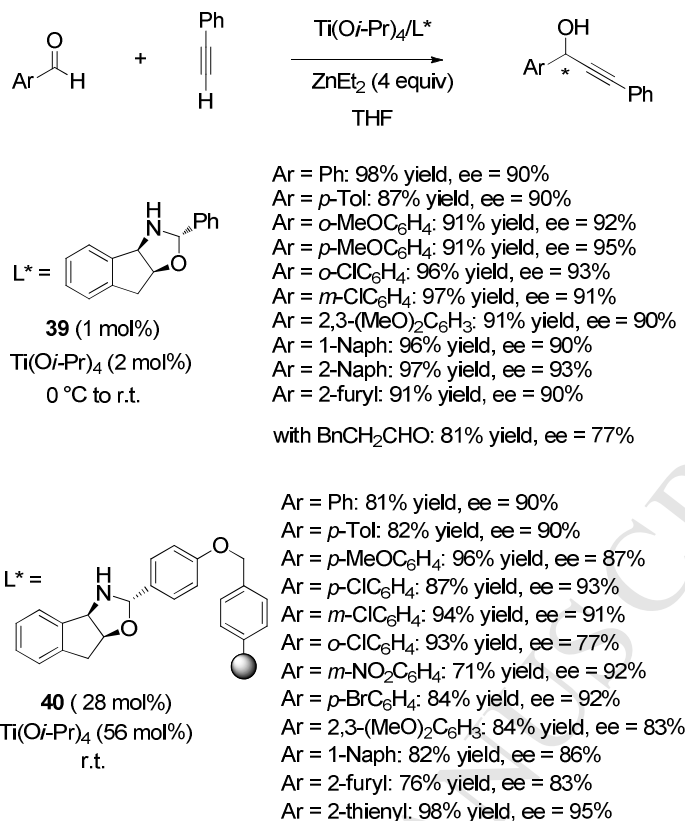
Chiral propargylic alcohols are useful building blocks for the enantioselective synthesis of a number of important chiral complex molecules. As the alkylation reaction, the alkynyl addition has a strategically synthetic advantage to form a new C–C bond with concomitant creation of a stereogenic centre in a single operation. Alkynyl-metal reagents are ideal functional carbon nucleophiles which can be prepared easily owing to the acidity of terminal alkynyl protons. Therefore, the enantioselective addition⁸ of these intermediates to carbonyl compounds constitutes an attractive alternative to the synthesis of the corresponding propargylic alcohols.⁸⁸ The first examples of enantioselective catalytic alkynylation of aldehydes using chiral titanium catalysts were independently reported by Chan and Pu, in 2002.⁸⁹ In these studies, the authors reported excellent enantioselectivities for the produced propargylic alcohols, using BINOL derivatives as chiral ligands (Scheme 51). Ever since, a number of other modified BINOL-derived ligands have been successfully used to induce these reactions but also various other types of ligands. In addition, the last few years have seen impressive advances in the variety of alkynes which could be successfully added to aldehydes with remarkable enantioselectivities.



Scheme 51. First Ti-promoted/catalysed enantioselective alkynylations of aldehydes reported by Chan and Pu in 2002

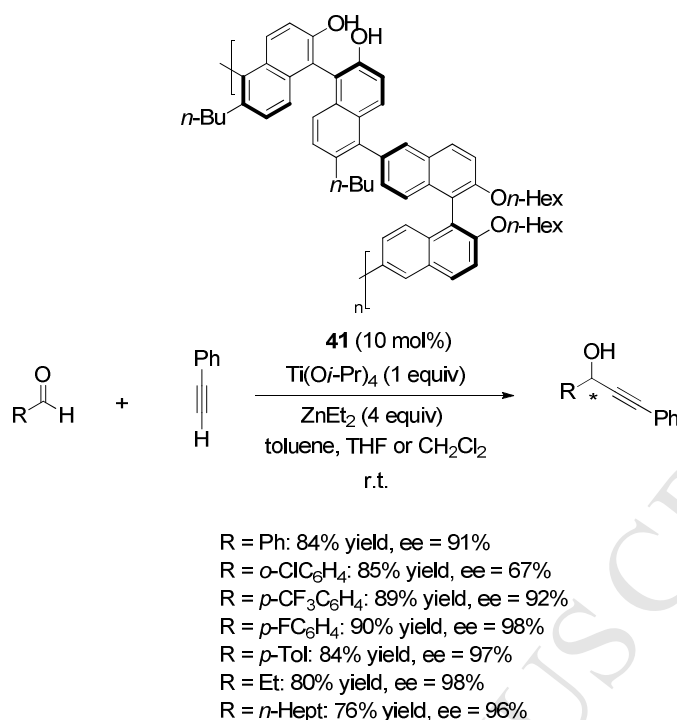
3.1.1. Additions of phenylacetylene. Among recently investigated new chiral ligands in enantioselective titanium-catalysed alkynylations of aldehydes, Mao and Zhang reported the use of a readily available and inexpensive novel chiral oxazolidine **39** as ligand in the addition of phenylacetylene to aldehydes.⁹⁰ This ligand was derived from readily available (1*R*,2*S*)-*cis*-1-amino-2-indanol. When used at 1 mol% of catalyst loading in combination with a catalytic amount (2 mol%) of $Ti(Oi-Pr)_4$ and four equivalents of diethylzinc in THF, the reaction of a variety of aromatic aldehydes afforded the corresponding chiral propargylic alcohols in both excellent yields of up to 98% and enantioselectivities of up to 95% ee, as shown in Scheme 52. It must be noted that lower enantioselectivities were obtained in the cases of aliphatic aldehydes (77% ee for $R = CH_2Bn$ for example).

In order to render this type of economical ligand recyclable, Mao et al. developed resin-supported oxazolidine ligand **40** which was found to smoothly catalyse the same reactions of aromatic aldehydes with high yields of up to 98% and enantioselectivities of up to 95% ee, as shown in Scheme 52.⁹¹ In this case, the ligand was employed at 28 mol% of catalyst loading, in combination with 56 mol% of $Ti(Oi-Pr)_4$ in the presence of four equivalents of diethylzinc. Remarkably, this novel catalytic system could be reused for five times after simple work-up. It must be noted that it was also suitable for the alkynylation of heteroaromatic aldehydes, providing enantioselectivities of up to 95% ee. The authors showed that replacing diethylzinc with dimethylzinc did not give enhanced enantioselectivities.



Scheme 52. Chiral oxazolidine and chiral resin-supported oxazolidine ligands in addition of phenylacetylene to aldehydes

Moreover, a soluble chiral polybinaphthol ligand **41** was synthesised by Cheng et al. through polymerisation of (*S*)-5,5'-dibromo-6,6'-di-*n*-butyl-2,2'-binaphthol with (*S*)-2,2'-bis-*n*-hexyloxy-1,1'-binaphthyl-6,6'-boronic acid via palladium-catalysed Suzuki reaction, and further investigated in phenylacetylene addition to both aromatic and aliphatic aldehydes.⁹² Associated to four equivalents of diethylzinc and one equivalent of Ti(Oi-Pr)₄ in toluene, THF, or CH₂Cl₂ as solvent, this novel polymer ligand provided a range of chiral propargylic alcohols in good to high yields and enantioselectivities of up to 90% and 98% ee, respectively, even in the cases of aliphatic aldehydes (Scheme 53). A moderate enantioselectivity of 67% ee was observed only in the case of the reaction of *ortho*-chlorobenzaldehyde. Importantly, it must be noted that this ligand could be easily recovered and reused without loss of activity as well as enantioselectivity.



Scheme 53. Chiral polynaphthol ligand in addition of phenylacetylene to aldehydes

In 2010, the same reactions were carried out by Gou et al. using another type of chiral ligands, such as camphor-derived sulfonylated amino alcohols.⁹³ As shown in Scheme 54, the best results were achieved employing 10 mol% of camphor sulfonylated amino alcohol **42** associated to four equivalents of Ti(Oi-Pr)₄ in the presence of three equivalents of diethylzinc in toluene. A variety of aromatic as well as α,β -unsaturated aldehydes were found to be suitable substrates, leading to the corresponding alcohols in good yields of up to 90% and moderate enantioselectivities of up to 63% ee.

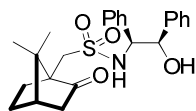
Later, Bauer et al. catalysed the same reactions for the first time with 20 mol% of β -hydroxy sulfonamide **43** derived from D-glucosamine in combination with six equivalents of Ti(Oi-Pr)₄.⁹⁴ It must be noted that these authors previously successfully investigated this family of chiral ligands in enantioselective titanium-promoted dialkylzinc additions to aldehydes (Scheme 15, ref 42b). Performed in CH₂Cl₂ in the presence of 1.2 equivalent of diethylzinc, the process provided various chiral aromatic, aliphatic as well as α,β -unsaturated alcohols in good to high yields of up to 95% and moderate to high enantioselectivities of up to 92% ee (Scheme 54). It was found that the enantioselectivity of the reaction was highly dependent on the nature of the aldehyde used and substitution on its phenyl group. The best result was reached for *ortho*-fluorobenzaldehyde (92% ee) while the reactions of *meta*- and

para-fluorobenzaldehydes gave very low enantioselectivities (12 and 22% ee, respectively). Moreover, a good enantioselectivity (85% ee) was observed for a cycloaliphatic aldehyde (R = Cy), whereas a simple linear aliphatic aldehyde (R = *n*-Pent) gave a lower enantiomeric excess of 49% ee.

Earlier, inspired by their previously reported work dealing with the use of chiral β -sulfonamide alcohol ligands in enantioselective alkynylation processes,⁹⁵ Wang et al. reinvestigated a catalytic system based on the combination of a catalytic amount (20 mol%) of a new readily available and inexpensive chiral β -sulfonamide alcohol **44** with 20 mol% of Ti(Oi-Pr)₄.⁹⁶ The process also needed 3.5 equivalents of diethylzinc and 0.5 equivalent of a terminal base, such as DIPEA, as an additive. As summarised in Scheme 54, the reaction of phenylacetylene with benzaldehyde led to the corresponding propargylic alcohol in good yield (78%) and very high enantioselectivity of 96% ee. The role of DIPEA was to facilitate the formation of the alkynylzinc reagents.



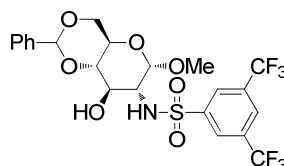
with conditions:



42 (10 mol%)
 Ti(O*i*-Pr)₄ (4 equiv)
 ZnEt₂ (3 equiv)
 toluene, 22 °C

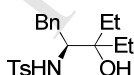
R = Ph: 90% yield, ee = 56%
 R = *p*-MeOC₆H₄: 73% yield, ee = 61%
 R = *p*-Tol: 87% yield, ee = 63%
 R = *p*-FC₆H₄: 77% yield, ee = 56%
 R = *p*-ClC₆H₄: 71% yield, ee = 54%
 R = *p*-BrC₆H₄: 74% yield, ee = 52%
 R = 1-Naph: 83% yield, ee = 51%
 R = 2-furyl: 79% yield, ee = 45%
 R = (*E*)-PhCH=CH: 57% yield, ee = 42%

with conditions:



43 (20 mol%)
 Ti(O*i*-Pr)₄ (6 equiv)
 ZnMe₂ (1.2 equiv)
 CH₂Cl₂, r.t.

R = Ph: 82% yield, ee = 90%
 R = *o*-MeOC₆H₄: 63% yield, ee = 28%
 R = *m*-MeOC₆H₄: 63% yield, ee = 23%
 R = *p*-MeOC₆H₄: 85% yield, ee = 73%
 R = *o*-FC₆H₄: 68% yield, ee = 92%
 R = *p*-FC₆H₄: 59% yield, ee = 22%
 R = *m*-FC₆H₄: 72% yield, ee = 12%
 R = (*E*)-PhCH=CH: 92% yield, ee = 23%
 R = Cy: 75% yield, ee = 85%
 R = *n*-Pent: 95% yield, ee = 49%



44 (20 mol%)
 Ti(O*i*-Pr)₄ (20 mol%)
 ZnEt₂ (3.5 equiv)
 DIPEA (0.5 equiv)
 toluene, 42 °C

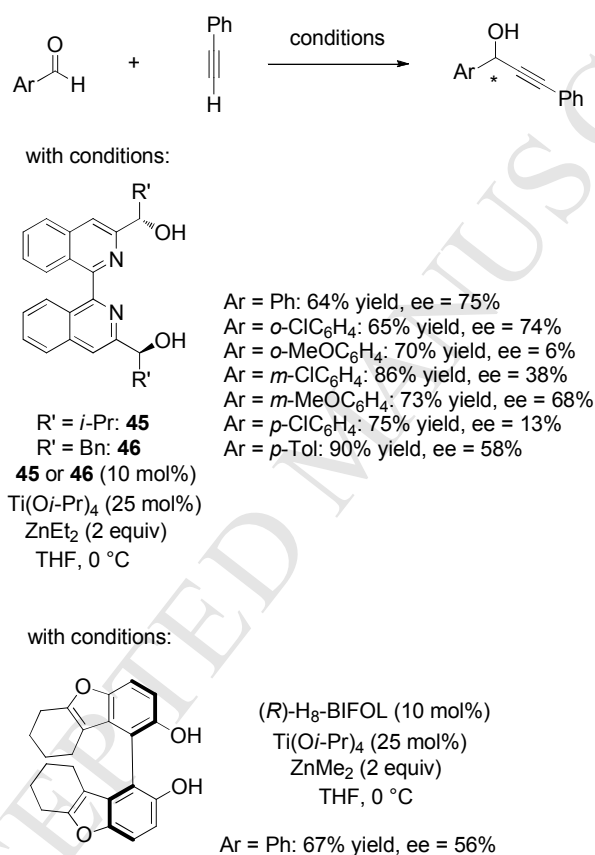
R = Ph: 78% yield, ee = 96%

Scheme 54. Chiral sulfonamide ligands in addition of phenylacetylene to aldehydes (RCHO)

On the other hand, moderate enantioselectivities of up to 75% ee were reported by Kilic et al. for the same reactions of aromatic aldehydes based on the use of novel chiral 4,4'-biquinazoline alcohol ligands synthesised from readily accessible (*S*)-2-acetoxycarboxylic acids.⁹⁷ Even if the enantioselectivities remained moderate, the advantage of this process was the involvement of only 25 mol% of Ti(O*i*-Pr)₄ in combination with 10 mol% of the chiral ligand. These best results were reached by using ligands **45** and **46** in the presence of two equivalents of diethylzinc in THF (Scheme 55). Comparison of the results obtained from

chloro-substituted benzaldehydes and methoxy-substituted benzaldehydes showed that electronic properties had a dramatic effect on the enantioselectivity of the process.

In addition, Dehaen et al. involved the same catalytic amount (25 mol%) of $\text{Ti}(\text{O}i\text{-Pr})_4$ in combination with another type of novel chiral ligands, such as (*R*)- $\text{H}_8\text{-BIFOL}$, to induce the addition of phenylacetylene to benzaldehyde in the presence of two equivalents of dimethylzinc in THF.⁹⁸ The process resulted, however, in a moderate enantioselectivity of 56% ee combined with 67% yield, as shown in Scheme 55.



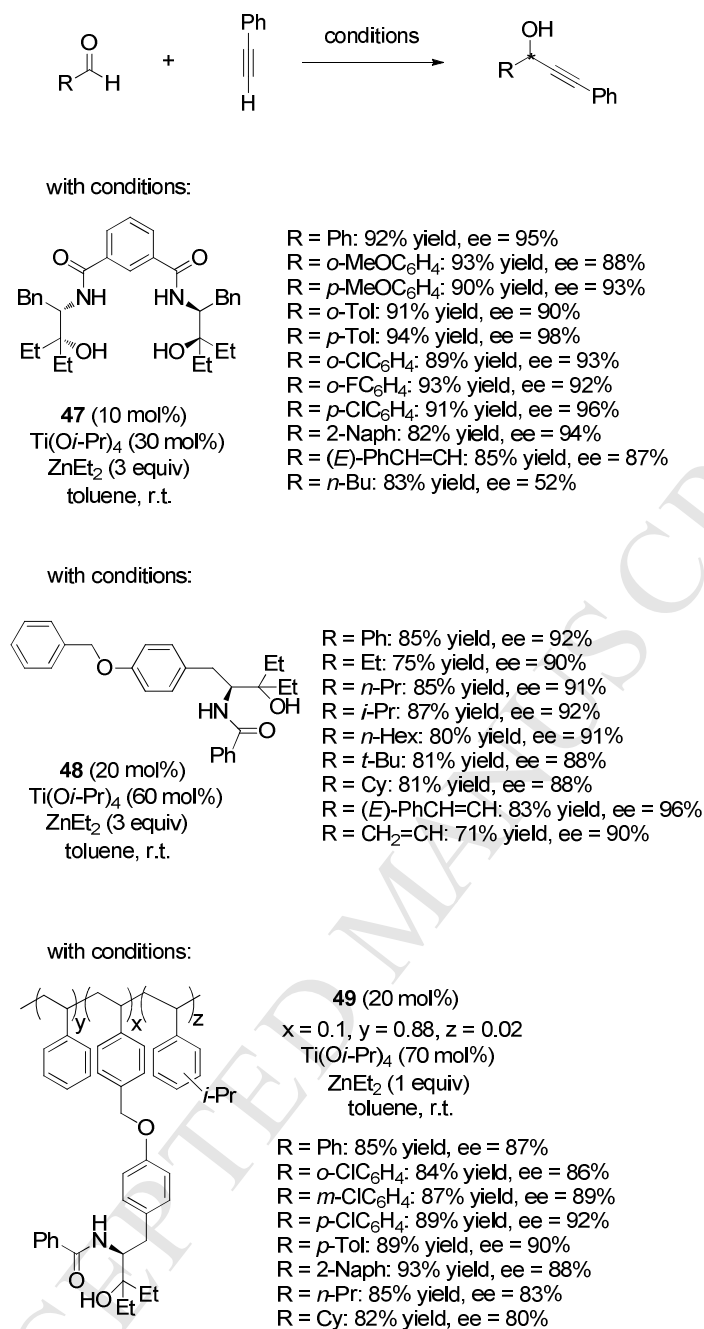
Scheme 55. Chiral 4,4'-biquinazoline alcohol ligands and (*R*)- $\text{H}_8\text{-BIFOL}$ ligand in addition of phenylacetylene to aromatic aldehydes

Several β -hydroxy amide chiral ligands have also been investigated by the group of Hui and Xu. Among them, chiral C_2 -symmetric bis(β -hydroxy amide) **47**, synthesised via the reaction of isophthaloyl dichloride and L-phenylalanine, provided excellent results for the addition of phenylacetylene to benzaldehyde when used at 10 mol% of catalyst loading combined with a catalytic amount (30 mol%) of $\text{Ti}(\text{O}i\text{-Pr})_4$ in the presence of three equivalents of diethylzinc in toluene.⁹⁹ Indeed, general excellent yields and

enantioselectivities of up to 94% and 98% ee, respectively, were obtained with various aromatic aldehydes including benzaldehydes (Scheme 56) while cinnamaldehyde and *n*-pentanal gave lower enantioselectivities of 87 and 52% ee, respectively. The authors have demonstrated that the two β -hydroxy amide moieties in this ligand behave as two independent ligands in the catalytic system.

To further develop efficient catalysts for the asymmetric alkynes addition to aliphatic and vinyl aldehydes, these authors later reported the synthesis of novel β -hydroxy amide ligands from L-tyrosine.¹⁰⁰ Among them, ligand **48** proved to be the most efficient when used at 20 mol% of catalyst loading combined with 60 mol% of Ti(Oi-Pr)₄. As shown in Scheme 56, the reaction of a range of aliphatic and α,β -unsaturated aldehydes performed in toluene in the presence of three equivalents of diethylzinc afforded the corresponding chiral propargylic alcohols in high yields of up to 86% and high enantioselectivities of up to 96% ee. Furthermore, a high enantioselectivity of 92% ee was also obtained in the case of benzaldehyde.

With the aim of developing recyclable ligands of the same family, these authors have synthesised a series of polymer-supported chiral β -hydroxy amides to be investigated in the same additions.¹⁰¹ As shown in Scheme 56, the use of resin **49**, obtained through copolymerisation of the corresponding monomer with styrene and divinyl benzene, at 20 mol% of catalyst loading in combination with 70 mol% of Ti(Oi-Pr)₄ allowed the addition of phenylacetylene to various aromatic aldehydes in the presence of one equivalent of diethylzinc to be achieved in high yields (up to 93%) and enantioselectivities of up to 92% ee. This best enantioselectivity was reached for the alkynylation of *para*-chlorobenzaldehyde. Furthermore, it was worthy to be noticed that this heterogeneous catalytic system was also suitable for aliphatic aldehydes which provided good enantioselectivities of 80-83% ee. Moreover, it must be noted that this resin could be reused four times.

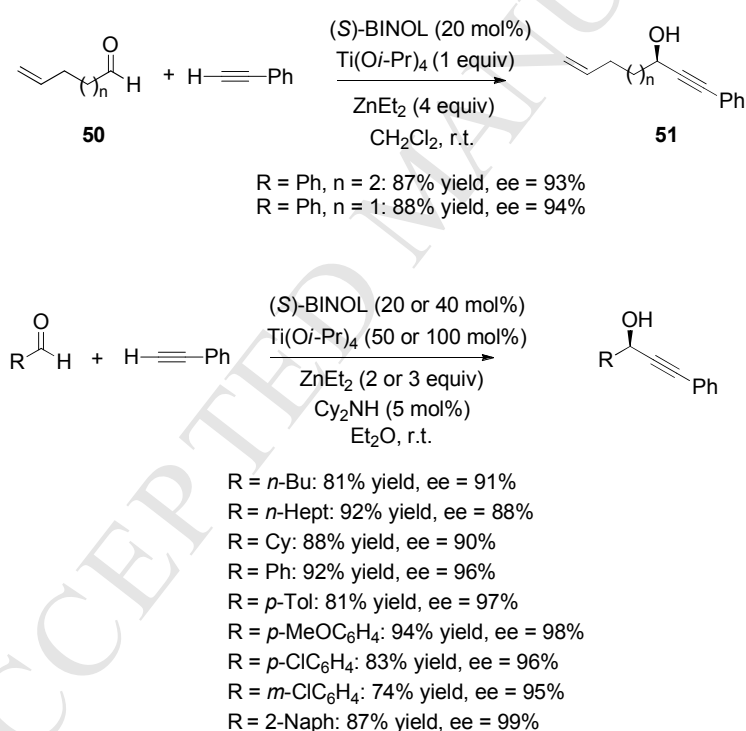


Scheme 56. Chiral β-hydroxy amide ligands in addition of phenylacetylene to aldehydes (RCHO)

Inspired by their pioneering works on enantioselective catalytic alkynylation of aldehydes employing BINOL as ligand,^{89b} Pu et al. later reused (*S*)-BINOL to induce chirality in the addition of phenylacetylene to enals **50**, which provided the corresponding chiral propargylic alcohol-based en-type precursors **51** for Pauson–Khand reactions.¹⁰² As shown in Scheme 57, the combination of 20 mol% of (*S*)-BINOL with one equivalent of *Ti*(*Oi*-Pr)₄ in the presence

of four equivalents of diethylzinc in CH_2Cl_2 provided both high yields (87-88%) and enantioselectivities of up to 94% ee. The formed chiral propargylic alcohols **51** were further successfully submitted to intramolecular Pauson–Khand reaction, allowing the synthesis of optically active 5,5- and 5,6-fused bicyclic products to be achieved with retention of enantiomeric purity.

In 2013, the same authors applied a closely related methodology to other aliphatic aldehydes and also to aromatic ones which provided the corresponding propargylic alcohols with even higher enantioselectivities of up to 99% ee (Scheme 57).¹⁰³ In general, the best enantioselectivities were reached with aromatic aldehydes (up to 99% ee) but aliphatic ones also provided good to high enantioselectivities of 88-91% ee. The process employed dicyclohexylamine as an additive which was supposed to generate a nucleophilic alkynylzinc reagent.

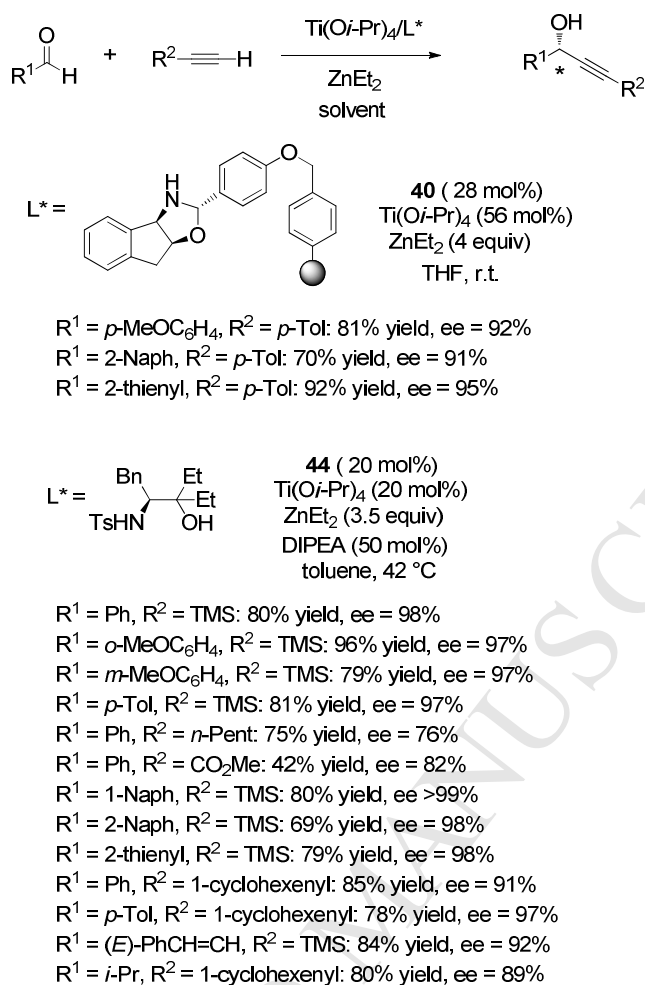


Scheme 57. (S)-BINOL as ligand in additions of phenylacetylene to various aldehydes including enals

3.1.2. Additions of various terminal alkynes. In the last few years, important progress has been particularly made in the variety of terminal alkynes which can be enantioselectively added to aldehydes and also ketones. For example, Mao et al. have extended the scope of the methodology depicted in Scheme 52, using resin-supported oxazolidine ligand **40**, to terminal

alkynes other than phenylacetylene, such as *para*-tolylacetylene, which provided excellent enantioselectivities of up to 95% ee for propargylic alcohols arisen from aromatic and heteroaromatic aldehydes, as shown in Scheme 58.⁹¹ The absolute configuration of the products was not mentioned.

On the other hand, some quite challenging alkynes, such as trimethylsilylacetylene, ethynylcyclohexene, 1-heptyne, and also methyl propiolate were highly enantioselectively added to a range of aromatic, heteroaromatic, α,β -unsaturated, as well as aliphatic aldehydes by Wang et al. on the basis of a novel methodology, in 2009.⁹⁶ These reactions were achieved by using a novel catalytic system based on the combination of a catalytic amount (20 mol%) of a new readily available and inexpensive chiral β -sulfonamide alcohol **44** with 20 mol% of $\text{Ti}(\text{O}i\text{-Pr})_4$. The process also needed 3.5 equivalents of diethylzinc and 0.5 equivalent of a terminal base, such as DIPEA, as an additive. As summarised in Scheme 58, a series of chiral propargylic alcohols were produced under these conditions with good yields of up to 96% and remarkable enantioselectivities of up to >99% ee. The best enantioselectivities were reached by using trimethylsilylacetylene, and ethynylcyclohexene, while 1-heptyne, and methyl propiolate provided lower but acceptable enantioselectivities (76-82% ee). The role of DIPEA was to facilitate the formation of the alkynylzinc reagents. It has to be highlighted that this process is remarkable by its wide scope and homogeneity of its high enantioselectivities. The absolute configuration of some products derived from aromatic aldehydes was assigned as (*R*).

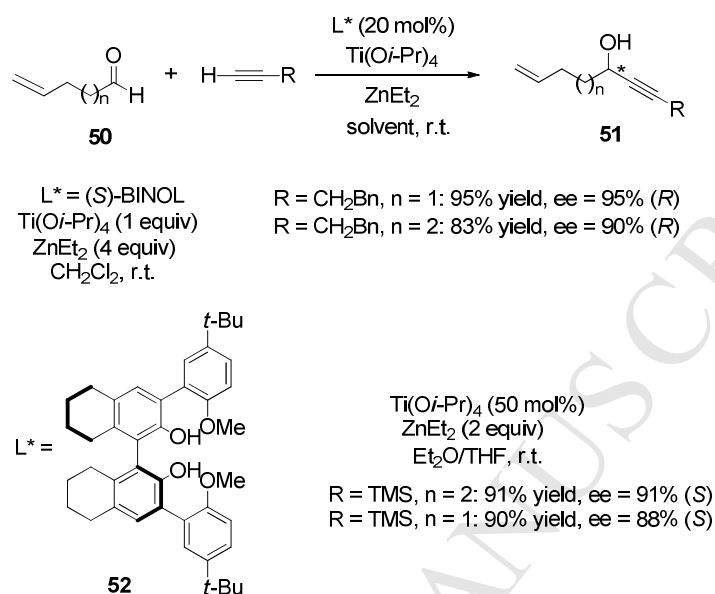


Scheme 58. Chiral resin-supported oxazolidine ligand and chiral β -sulfonamide alcohol ligand in addition of various terminal alkynes to aldehydes

These types of reactions have also been induced by BINOL-derived ligands. For example, Pu et al. have used (*S*)-BINOL to induce chirality in addition of alkynes to enals **50**, which provided the corresponding chiral propargylic alcohol-based en-type precursors **51** for Pauson-Khand reactions.¹⁰² As shown in Scheme 59, the combination of 20 mol% of (*S*)-BINOL with one equivalent of Ti(Oi-Pr)₄ in the presence of four equivalents of diethylzinc in CH₂Cl₂ provided both high yields and enantioselectivities of up to 95% and 95% ee, respectively, in the case of 4-phenyl-1-butyne as substrate.

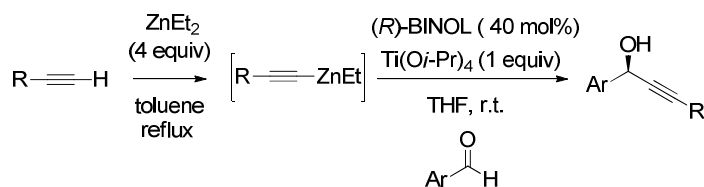
On the other hand, when trimethylsilylacetylene was used as alkyne, the best enantioselectivities obtained in this case of substrate (91% ee) were achieved by using (*S*)-H₈-BINOL-derived ligand **52** at the same catalyst loading (20 mol%) albeit using only 50 mol% of Ti(Oi-Pr)₄ and two equivalents of diethylzinc in a mixed solvent of diethylether/THF (1:5), as shown in Scheme 59. The formed chiral propargylic alcohols **51** were further successfully

submitted to intramolecular Pauson–Khand reactions, allowing the synthesis of optically active 5,5- and 5,6-fused bicyclic products to be achieved with retention of enantiomeric purity.



Scheme 59. (S)-BINOL and (S)-H₈-BINOL-derivative as ligands in addition of alkynes to enals

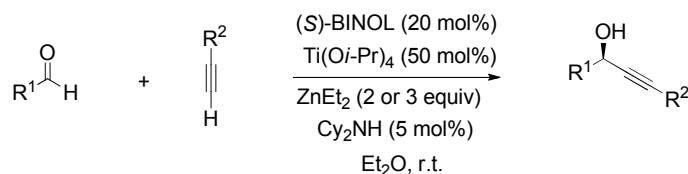
Later, highly enantioselective additions of a range of linear aliphatic alkynes to aromatic aldehydes including those with various substituents on different positions were described by Wang and Yu by using (*R*)-BINOL at 40 mol% of catalyst loading as chiral ligand.¹⁰⁴ As illustrated in Scheme 60, in combination with one equivalent of $\text{Ti}(\text{O}i\text{-Pr})_4$ and four equivalents of diethylzinc in toluene, the use of this ligand allowed useful aromatic chiral propargylic alcohols to be synthesised in good to high yields (up to 92%) and high enantioselectivities of up to 95% ee starting from a range of aromatic aldehydes bearing various substituents on different positions. It was found that reactions with *meta*- and *para*-substituted benzaldehydes gave slightly higher enantioselectivities than those involving *ortho*-substituted benzaldehydes.



Ar = Ph, R = *n*-Hex: 81% yield, ee = 94%
 Ar = *o*-MeOC₆H₄, R = *n*-Hex: 76% yield, ee = 86%
 Ar = *m*-Tol, R = *n*-Hex: 72% yield, ee = 92%
 Ar = *p*-Tol, R = *n*-Hex: 72% yield, ee = 92%
 Ar = *p*-MeOC₆H₄, R = *n*-Hex: 83% yield, ee = 87%
 Ar = *m*-FC₆H₄, R = *n*-Hex: 92% yield, ee = 94%
 Ar = *m*-ClC₆H₄, R = *n*-Hex: 86% yield, ee = 94%
 Ar = *m*-BrC₆H₄, R = *n*-Hex: 89% yield, ee = 95%
 Ar = *o*-ClC₆H₄, R = *n*-Hex: 76% yield, ee = 82%
 Ar = *p*-FC₆H₄, R = *n*-Hex: 84% yield, ee = 92%
 Ar = Ph, R = *n*-Bu: 68% yield, ee = 74%
 Ar = *p*-Tol, R = *n*-Bu: 84% yield, ee = 90%
 Ar = *m*-MeOC₆H₄, R = *n*-Bu: 75% yield, ee = 92%
 Ar = *m*-FC₆H₄, R = *n*-Bu: 73% yield, ee = 86%

Scheme 60. (*R*)-BINOL as ligand in addition of linear terminal alkynes to aromatic aldehydes

Earlier, comparable reactions were also performed by Pu et al. with (*S*)-BINOL as chiral ligand in the presence of a catalytic amount of Ti(Oi-Pr)₄ (50 mol%).¹⁰⁵ In this case, the use of biscyclohexylamine as a Lewis base additive (5 mol%) proved to greatly enhance the enantioselectivity of the reactions of various linear aliphatic alkynes with a variety of aliphatic aldehydes which were found up to 89% ee (Scheme 61). It is interesting to note that 5-chloro-1-pentyne consistently gave better enantioselectivities than the other linear alkynes when reacted with various aldehydes. In this study, the authors have compared the efficiency of (*S*)-BINOL with that of the partially hydrogenated (*S*)-H₈-BINOL, and found that the latter provided lower enantioselectivities. In 2013, the authors extended this methodology to other aldehydes including aromatic and α,β -unsaturated ones, which provided the corresponding propargylic alcohols with even higher enantioselectivities of up to >99% ee (Scheme 61).¹⁰³ It has to be highlighted that this process is remarkable by its wide scope and homogeneity of its high enantioselectivities.



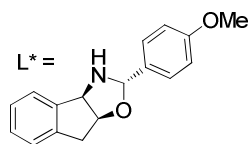
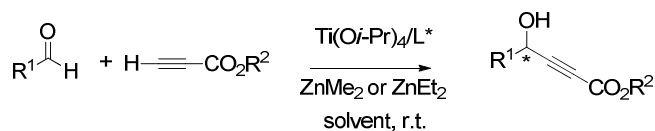
R ¹	R ²	Yield (%)	ee (%)
<i>n</i> -Bu	<i>n</i> -Bu	73	81
<i>n</i> -Bu	<i>n</i> -Hex	61	87
<i>n</i> -Bu	(CH ₂) ₃ Cl	77	88
<i>n</i> -Hept	<i>n</i> -Bu	76	85
<i>n</i> -Hept	<i>n</i> -Hex	67	85
(CH ₂) ₂ CH=CH ₂	(CH ₂) ₃ Cl	70	89
(CH ₂) ₂ CH=CH ₂	<i>n</i> -Bu	63	87
(CH ₂) ₂ CH=CH ₂	<i>n</i> -Hex	59	83
(CH ₂) ₃ Cl	(CH ₂) ₃ Cl	60	88
(CH ₂) ₃ Cl	<i>n</i> -Hex	57	77
BnCH ₂	<i>n</i> -Bu	71	83
BnCH ₂	<i>n</i> -Hex	74	84
BnCH ₂	(CH ₂) ₃ Cl	65	89
Et	CH ₂ Bn	99	84
(<i>E</i>)-PhCH=CH	<i>n</i> -Bu	68	92
Ph	<i>n</i> -Bu	99	99
Ph	TMS	86	98
<i>p</i> -Tol	CH ₂ Bn	80	99
<i>p</i> -ClC ₆ H ₄	CH ₂ Bn	83	>99
<i>p</i> -Tol	<i>n</i> -Bu	93	97
<i>m</i> -Tol	<i>n</i> -Bu	61	98
<i>p</i> -ClC ₆ H ₄	<i>n</i> -Bu	68	97
<i>p</i> -Tol	TMS	81	97
<i>p</i> -MeOC ₆ H ₄	TMS	72	97
2-Naph	TMS	79	97
2-Naph	CH ₂ Bn	82	>99

Scheme 61. (S)-BINOL as ligand in addition of various alkynes to aldehydes with Cy₂NH

γ -Hydroxy- α,β -acetylenic esters containing three different functional groups constitute very important precursors in the synthesis of highly functionalised organic molecules. Although great efforts have been made in asymmetric alkynylations, it must be recognised that few attentions on the enantioselective reactions of alkynoates to aldehydes have been paid until recently. This could be attributed to the higher sensitivity and dissimilar reactivity of alkynoates in comparison with simple aliphatic and aromatic alkynes. The asymmetric reaction of alkynoates with aldehydes was first reported by Pu et al., in 2006.¹⁰⁶ In this work, the reaction between methyl propiolate and aromatic aldehydes was carried out in the presence of diethylzinc, hexamethylphosphoramide (HMPA), and Ti(Oi-Pr)₄, along with (S)-BINOL as chiral ligand, providing the corresponding chiral propargylic alcohols in high

enantioselectivities. More recently, Mao and Guo described the synthesis of chiral γ -hydroxy- α,β -acetylenic esters on the basis of enantioselective titanium-catalysed addition of alkynoates to aromatic aldehydes induced by 20 mol% of easily available chiral oxazolidine **53**.¹⁰⁷ An advantage of this process was the use of a only catalytic amount (40 mol%) of $\text{Ti}(\text{Oi-Pr})_4$ which was combined with two equivalents of HMPA, three equivalents of dimethylzinc, along with 10 mol% of dimethoxy polyethylene glycol (DIMPEG) as an additive in toluene. As shown in Scheme 62, this practical catalytic system allowed a range of chiral γ -hydroxy- α,β -acetylenic esters to be achieved in good yields and enantioselectivities of up to 84% ee. The absolute configuration of the products was not determined.

However, since HMPA is a strong carcinogen, studies of other catalytic systems were quickly reported. In this context, other catalytic systems have been developed avoiding the use of HMPA, such as that reported by Hui et al., in 2009.¹⁰⁸ The latter involved the association of 30 mol% of chiral β -hydroxy amide **48** derived from L-tyrosine with 30 mol% of $\text{Ti}(\text{Oi-Pr})_4$ in the presence of three equivalents of diethylzinc in DME. Under these conditions, aliphatic as well aromatic aldehydes in addition to *trans*-cinnamaldehyde reacted with methyl propiolate to give the corresponding chiral alcohols in good to high yields and enantioselectivities of up to 94% ee, demonstrating the broad generality of this catalytic system for aliphatic and aromatic aldehydes (Scheme 62). It must be noted that the best enantioselectivities were reached in the cases of aromatic aldehydes *para*-substituted by electron-withdrawing as well as electron-donating groups. The highest enantioselectivity of 94% ee was obtained for 2-naphthaldehyde. On the other hand, aliphatic aldehydes provided good to high enantioselectivities of 78-91% ee, with the highest one (91% ee) reached for 2-phenylacetaldehyde. It must be noted that the presence of bulky groups on the aliphatic aldehydes resulted in lower enantioselectivities. A fair result (85% ee) was also obtained in the case of cinnamaldehyde. In some cases of products, the absolute configuration was assigned as (*R*).



53 (20 mol%)
 Ti(Oi-Pr)₄ (40 mol%)
 ZnMe₂ (3 equiv)
 HMPA (2 equiv)
 DIMPEG (10 mol%)
 toluene

with R² = Me:

R¹ = Ph: 78% yield, ee = 84%
 R¹ = *p*-BrC₆H₄: 56% yield, ee = 83%
 R¹ = *p*-ClC₆H₄: 73% yield, ee = 82%
 R¹ = *p*-FC₆H₄: 65% yield, ee = 80%
 R¹ = *p*-Tol: 61% yield, ee = 82%
 R¹ = *p*-MeOC₆H₄: 47% yield, ee = 63%
 R¹ = *o*-ClC₆H₄: 75% yield, ee = 81%
 R¹ = 1-Naph: 73% yield, ee = 83%
 R¹ = 2-Naph: 43% yield, ee = 73%
 with R² = Et:
 R¹ = 1-Naph: 76% yield, ee = 77%
 R¹ = 2-Naph: 70% yield, ee = 74%



48 (30 mol%)
 Ti(Oi-Pr)₄ (30 mol%)
 ZnEt₂ (3 equiv)
 DME

with R² = Me:

R¹ = Et: 89% yield, ee = 87%
 R¹ = *n*-Pr: 87% yield, ee = 87%
 R¹ = *i*-Pr: 90% yield, ee = 86%
 R¹ = *i*-Bu: 88% yield, ee = 88%
 R¹ = *n*-Hex: 76% yield, ee = 86%
 R¹ = Bn: 70% yield, ee = 91%
 R¹ = *t*-Bu: 86% yield, ee = 78%
 R¹ = Cy: 82% yield, ee = 81%

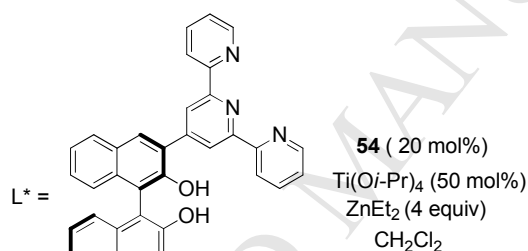
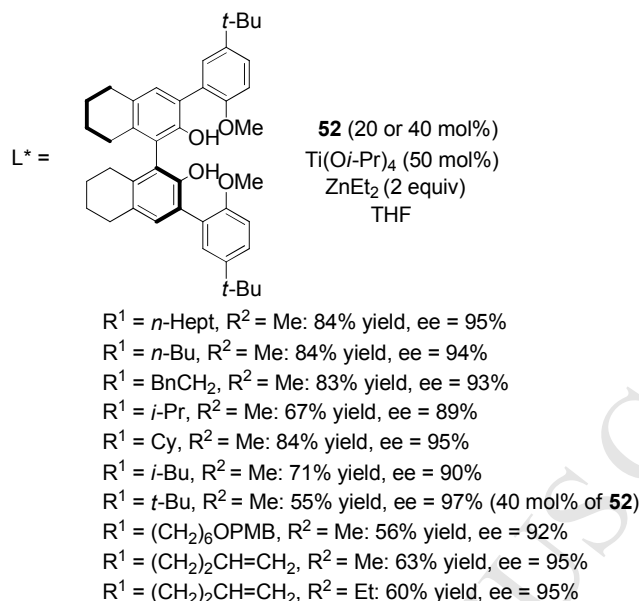
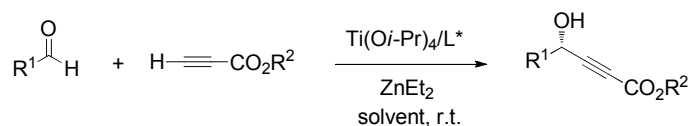
R¹ = (*E*)-PhCH=CH: 68% yield, ee = 85% (*R*)
 R¹ = Ph: 88% yield, ee = 93% (*R*)
 R¹ = *p*-FC₆H₄: 82% yield, ee = 90%
 R¹ = *o*-ClC₆H₄: 85% yield, ee = 87%
 R¹ = *p*-ClC₆H₄: 77% yield, ee = 90%
 R¹ = *p*-BrC₆H₄: 83% yield, ee = 91%
 R¹ = *p*-Tol: 72% yield, ee = 92%
 R¹ = *p*-MeOC₆H₄: 78% yield, ee = 92%
 R¹ = 1-Naph: 79% yield, ee = 90%
 R¹ = 2-Naph: 81% yield, ee = 94%

Scheme 62. Chiral oxazolidine and β-hydroxy amide ligands in addition of alkynoates to aldehydes

On the other hand, Pu et al. developed BINOL-derived ligands to induce this type of reactions. For example, novel H₈-BINOL-based ligand **52** employed at 20 mol% of catalyst loading was found to highly efficiently catalyse the alkyl propiolate addition to aliphatic aldehydes, using only a catalytic amount of Ti(Oi-Pr)₄ with two equivalents of diethylzinc in THF.¹⁰⁹ As shown in Scheme 63, this remarkable novel catalytic system provided general excellent enantioselectivities of up to 97% ee for a wide variety of aliphatic aldehydes under mild conditions. Indeed, good yields (67-71%) and high enantioselectivities (89-95% ee) were obtained for linear, α-branched, and β-branched aliphatic aldehydes. The more bulky trimethylacetaldehyde showed reduced reactivity (55% yield), requiring higher loadings of the

ligand (40 mol% instead of 20 mol%) to give the best enantioselectivity of 97% ee. Functionalised aliphatic aldehydes also gave high enantioselectivities (90-95% ee) as well as 4-pentenal which provided the same high enantioselectivity of 95% ee when reacting with ethyl propiolate and methyl propiolate.

Later, the same authors reported a complementary methodology for the methyl propiolate addition to aromatic aldehydes, employing novel C_1 -symmetric BINOL-terpyridine ligand **54** employed at 20 mol% of catalyst loading, which also used a catalytic amount of $Ti(Oi-Pr)_4$ (50 mol%) in the presence of four equivalents of diethylzinc in CH_2Cl_2 .¹¹⁰ As shown in Scheme 63, general high enantioselectivities of up to 98% ee in combination with high yields of up to 92% were obtained for a range of aromatic aldehydes bearing electron-donating and electron-withdrawing substituents at the *ortho*-, *meta*-, and *para*-positions. It was shown that applying this methodology to aliphatic aldehydes gave poor enantioselectivities (47% ee). The remarkable works reported by the group of Pu summarised in Schemes 61 and 63 in addition to those described by the group of Hui using ligand **48** (Scheme 62) demonstrate that highly efficient alkyl propiolate asymmetric additions to all types of aldehydes are today easily accomplishable under mild reaction conditions.



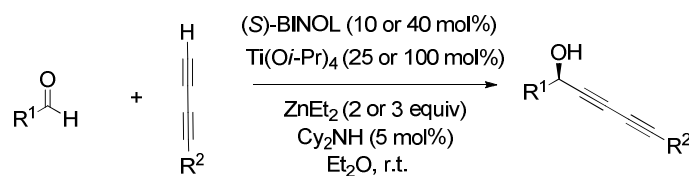
with R² = Me:

R¹ = Ph: 80% yield, ee = 95%
 R¹ = *o*-Tol: 86% yield, ee = 90%
 R¹ = *m*-Tol: 92% yield, ee = 92%
 R¹ = *p*-Tol: 90% yield, ee = 95%
 R¹ = *o*-MeOC₆H₄: 83% yield, ee = 98%
 R¹ = *m*-MeOC₆H₄: 86% yield, ee = 93%
 R¹ = *p*-MeOC₆H₄: 80% yield, ee = 95%
 R¹ = *p*-ClC₆H₄: 89% yield, ee = 98%
 R¹ = *o*-BrC₆H₄: 82% yield, ee = 93%
 R¹ = *p*-FC₆H₄: 88% yield, ee = 87%
 R¹ = Cy: 70% yield, ee = 47%

Scheme 63. Chiral H₈-BINOL-derived ligand and chiral BINOL-terpyridine ligand in addition of alkynoates to aldehydes

3.1.3. Additions of 1,3-diynes and 1,3-enynes. In 2011, Pu et al. demonstrated the challenges associated with the diyne nucleophiles for the asymmetric addition to aldehydes in comparison with simple alkynes.¹¹¹ Indeed, these authors reported a highly enantioselective titanium-promoted addition of various 1,3-diynes to a wide variety of aldehydes. This

remarkable process employed (S)-BINOL as chiral ligand in combination with $\text{Ti}(\text{Oi-Pr})_4$ in the presence of two or three equivalents of diethylzinc along with Cy_2NH as an additive (5 mol%) in diethyl ether as solvent. When reacting aromatic aldehydes containing either electron-donating or -withdrawing groups at the *ortho*-, *meta*-, and *para*-positions, only a catalytic amount (25 mol%) of $\text{Ti}(\text{Oi-Pr})_4$ was sufficient to provide the corresponding chiral diynols in very high enantioselectivities of up to 94% ee and very good yields of up to 98%. On the other hand, the reaction of aliphatic aldehydes required a stoichiometric amount of $\text{Ti}(\text{Oi-Pr})_4$ to provide the corresponding products in comparable enantioselectivities of up to 95% ee (Scheme 64). Furthermore, the addition of diynes to enals gave the corresponding enediynols in general high enantioselectivities (88-95%). It is interesting to note that an α,β -unsaturated enal, such as *trans*-crotonaldehyde, was also found to be well-suited for the catalytic system, since it provided an enantioselectivity of 92% ee. It must be noted that this novel methodology proposed the most generally enantioselective catalyst system for the asymmetric diyne addition to aldehydes. The formed products constituted starting materials to synthesise chiral polycyclic products, such as 5,5,7- and 5,5,8-fused tricyclic products, through Pauson–Khand reaction. This novel and nice methodology has allowed enantioselective additions of 1,3-diynes more challenging than those of simple alkynes to be achieved, opening a novel and efficient synthetic route to the structural framework of many biologically significant molecules.

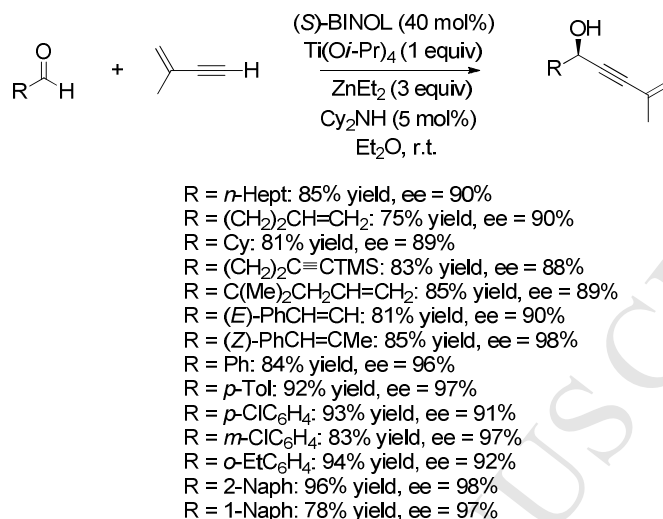


R ¹	R ²	Yield (%)	ee (%)
<i>o</i> -MeOC ₆ H ₄	CH ₂ Bn	81	91
<i>p</i> -MeOC ₆ H ₄	CH ₂ Bn	94	94
<i>p</i> -ClC ₆ H ₄	CH ₂ Bn	94	92
<i>o</i> -Tol	CH ₂ Bn	98	92
2-furyl	CH ₂ Bn	89	85
Ph	CH ₂ Bn	95	94
Ph	TIPS	98	91
Ph	<i>n</i> -Hex	87	94
Ph	CH ₂ OTBS	95	94
Ph	(CH ₂) ₃ OAc	78	86
Ph	(CH ₂) ₃ Cl	95	91
Ph	(CH ₂) ₃ OTBS	97	92
(<i>E</i>)-MeCH=CH	CH ₂ Bn	99	92
<i>n</i> -Bu	CH ₂ Bn	92	92
<i>i</i> -Bu	CH ₂ Bn	90	87
Cy	CH ₂ Bn	91	91
Cy	TIPS	99	89
Cy	Ph	98	92
Cy	CH ₂ OTBS	98	91
Cy	(CH ₂) ₃ OTBS	99	92
(CH ₂) ₂ CH=CH ₂	TIPS	94	95
(CH ₂) ₂ CH=CH ₂	Ph	73	91

Scheme 64. (S)-BINOL as ligand in addition of 1,3-diynes to aldehydes

In addition, Pu et al. recently reported the first highly enantioselective addition of a conjugated enyne to linear aliphatic aldehydes in the presence of a chiral catalyst.¹⁰³ This process involved (S)-BINOL as chiral ligand used at 40 mol% of catalyst loading along with one equivalent of Ti(Oi-Pr)₄, three equivalents of diethylzinc, and 5 mol% of Cy₂NH as an additive in diethyl ether. It afforded the corresponding chiral enynols in remarkable enantioselectivities of up to 98% ee and high yields, as shown in Scheme 65. In addition to linear aliphatic aldehydes, other aliphatic, aromatic, and α,β -unsaturated aldehydes proved compatible to the reaction conditions since the corresponding propargylic alcohols were obtained in comparable results (88-98% ee). The formed chiral enynols were further converted into trienynes, which were submitted to Pauson–Khand and Diels–Alder reactions to achieve important chiral multicyclic products. This remarkable study constituted the first highly enantioselective addition of a conjugated enyne to linear aliphatic aldehydes as well as other aldehydes achieved under very mild reactions conditions in the presence of a chiral

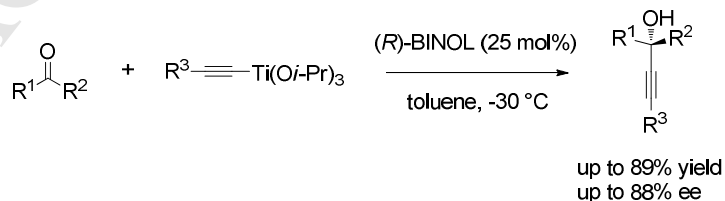
titanium catalyst, and could potentially provide an efficient novel and general method for the asymmetric synthesis of polyquinanes bearing a quaternary carbon centre through subsequent cyclisation of the formed chiral trienynes.



Scheme 65. (S)-BINOL as ligand in addition of 3-methyl-3-buten-1-yne to aldehydes

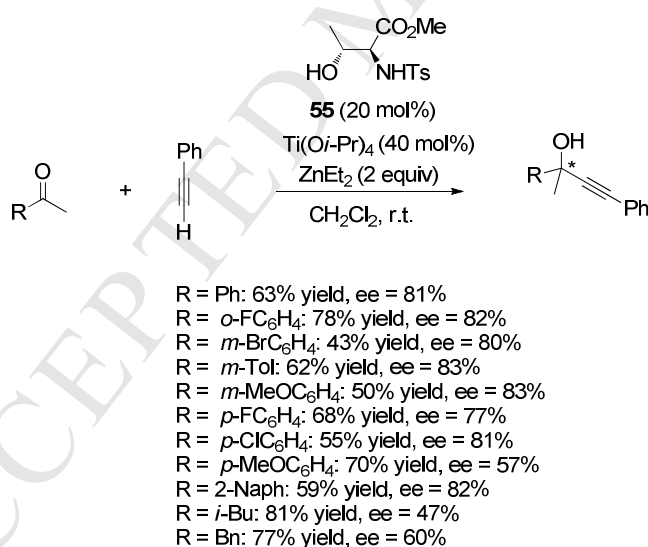
3.2. Ketones as electrophiles

The alkynylation of less reactive ketones is much less developed than that of aldehydes. It was first reported by Tan and co-workers in 1999 and later by Jiang and co-workers in 2002.¹¹² The first catalytic enantioselective alkynylation of ketones using chiral titanium catalysts was reported by Cozzi and Alesi, in 2004.¹¹³ This work involved the direct addition of an alkynyltitanium triisopropoxide to ketones performed in the presence of catalytic amounts of BINOL as chiral ligand to provide the corresponding chiral propargylic alcohols in good to high enantioselectivities of up to 88% ee (Scheme 66).



Scheme 66. First Ti-promoted enantioselective alkynylation of ketones reported by Cozzi and Alesi in 2004

Ever since, several efficient enantioselective titanium-catalysed alkynylations of ketones have been reported using other types of chiral ligands. For example, Wang et al. reported the synthesis of novel chiral hydroxysulfonamide ligands which were further investigated as promoters in the asymmetric addition of phenylacetylene to aromatic ketones.¹¹⁴ Among a range of simple L-amino acids tested, chiral hydroxysulfonamide **55**, employed at 20 mol% of catalyst loading in combination with only 40 mol% of $\text{Ti}(\text{Oi-Pr})_4$ in the presence of two equivalents of diethylzinc in CH_2Cl_2 , was found to be the most efficient ligand, providing the corresponding tertiary chiral propargylic alcohols in enantioselectivities of up to 83% ee along with good yields (43-81%), as shown in Scheme 67. It appeared that electron-donating and electron-withdrawing substituents on acetophenone have little effect on the enantioselectivities of the reactions. For example, 3'-bromoacetophenone and 3'-methylacetophenone gave the corresponding products with 80 and 83% ee, respectively. Moreover, when the scope of the methodology was extended to aliphatic aldehydes, only moderate enantioselectivities (47-60% ee) were obtained along with good yields (77-81%). The absolute configuration of the products was not assigned.



Scheme 67. Chiral hydroxysulfonamide ligand in addition of phenylacetylene to methyl ketones

Trifluoromethyl ketones constitute a class of particularly challenging substrates for the asymmetric titanium-catalysed zinc alkynylide addition because of the presence of the strongly electron-withdrawing fluorine atoms. Indeed, the activating trifluoromethyl group renders the ketone functionality highly reactive and, consequently, has a detrimental effect on

the control of the facial selectivity. In 2011, Ma et al. reported the first effective method for catalysing the asymmetric addition of alkynes to trifluoromethyl ketones.¹¹⁵ This novel methodology involved chiral cinchona alkaloids **56** and **57** as ligands (20 mol%) and had the advantage of using only a catalytic amount of $\text{Ti}(\text{Oi-Pr})_4$. In the presence of two equivalents of diethylzinc and BaF_2 as an additive in CH_2Cl_2 , the reaction of various aromatic alkynes and ketones, including electron-neutral, electron-withdrawing, as well as electron-donating groups afforded the corresponding chiral tertiary alcohols in high enantioselectivities of up to 91% ee, as shown in Scheme 68. It was noteworthy that even aliphatic alkynes also gave the products in good to high yields and enantioselectivities of up to 94% ee. Additionally, the reaction worked with (*E*)-1,1,1-trifluoro-4-phenylbut-en-2-one to afford the corresponding 1,2-adduct in good yield (67%) and enantioselectivity (66% ee). In this work, the authors demonstrated the remarkable effect of the metal fluoride additive, which was proven to be essential for effective asymmetric induction. One advantage of this process was that both enantiomers of trifluoromethylated propargylic tertiary alcohols could be assessed in high yields and enantioselectivities according to the ligands used. Indeed, (*S*)-products were achieved by using ligand **56** while (*R*)-alcohols arose from the employment of ligand **57**. Moreover, it is important to note that this study represented the first effective method for catalysing the asymmetric addition of alkynes to trifluoromethyl ketones.



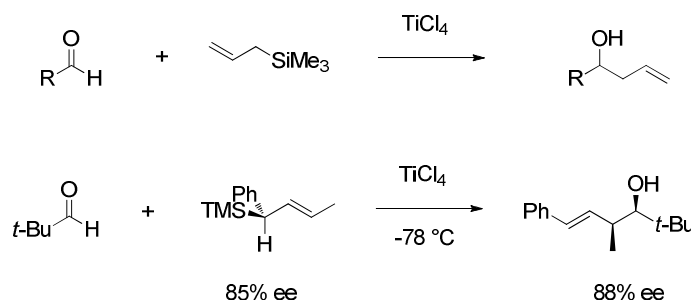
R¹ = *p*-Tol, R² = Ph: 82% yield, ee = 85% (S)



4. Titanium-promoted allylation and vinylation reactions

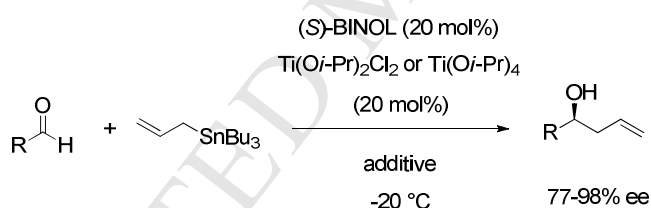
92

to aldehydes the corresponding chiral homoallylic alcohols to be achieved with good enantioselectivities of up to 88% ee (Scheme 69, second equation).



Scheme 69. First Ti-promoted allylations of aldehydes reported by Hosomi and Sakurai in 1976, and by Hayashi and Kumada in 1982

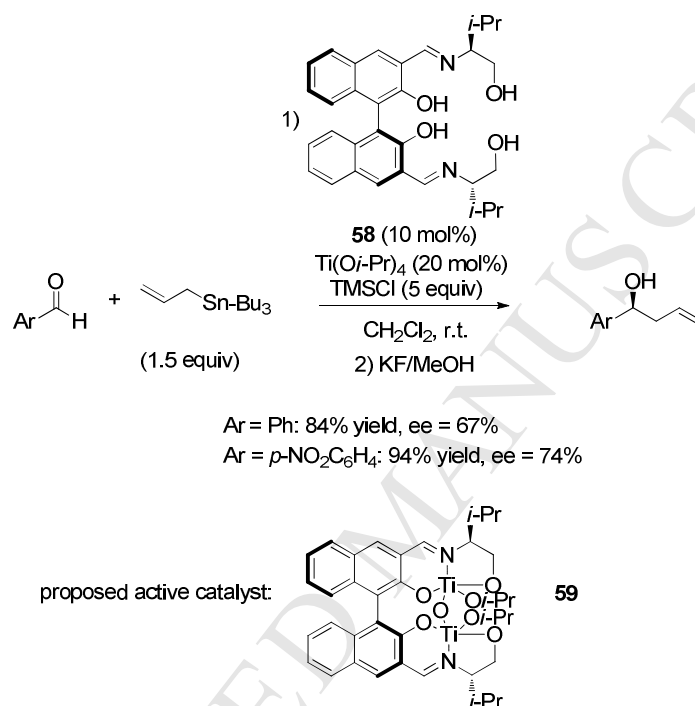
In 1993, the groups of Umani-Ronchi,¹¹⁸ and Keck,¹¹⁹ independently reported the first titanium-catalysed enantioselective reactions of aliphatic and aromatic aldehydes with allyl stannanes using BINOL titanium complex, providing enantioselectivities of up to 98% ee (Scheme 70).



Scheme 70. First Ti-catalysed enantioselective allylation of aldehydes reported by Umani-Ronchi and Keck in 1993

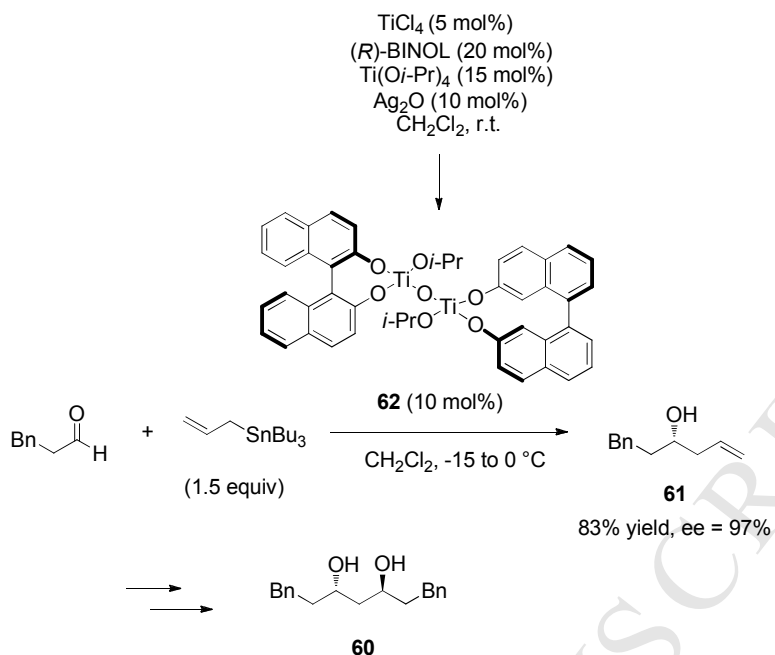
Inspired by these pioneering works, a number of chiral BINOL-derived ligands have been developed by several groups to be used in these enantioselective titanium-catalysed reactions.⁹ As an example, Belokon et al. have reported the use of chiral hexadentate Schiff bases, such as **58**, as titanium ligands in asymmetric allylation of aromatic aldehydes with allyltributyltin (Scheme 71).¹²⁰ The process involved 10 mol% of this chiral ligand in combination with 20 mol% of Ti(Oi-Pr)₄ as catalyst system. The authors demonstrated an unusual effect of TMSCl used as an additive on the effectiveness of the allylation of aldehydes which increased both the yield and the enantioselectivity of the reaction. They proposed that the silylation step, regenerating the initial catalytic species, was the rate limiting

step of the catalytic sequence. As shown in Scheme 71, the best result was obtained for the addition of allyltributyltin to *para*-nitrobenzaldehyde, providing the corresponding chiral allylic alcohol in 94% yield and enantioselectivity of 74% ee. Despite its moderate enantioselectivity, this reaction could be brought to completion after only one hour at room temperature. The authors have proposed dinuclear titanium catalyst **59** as active catalyst of the reaction.



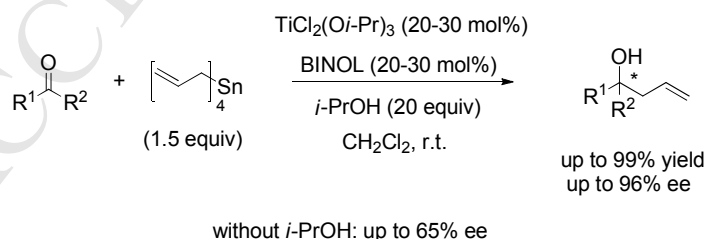
Scheme 71. Chiral hexadentate Schiff base as ligand in allylation of aromatic aldehydes

In 2011, Venkateswarlu et al. reported a concise total synthesis of cytotoxic *anti*-(3*S*,5*S*)-1-(4-hydroxyphenyl)-7-phenylheptane-3,5-diol **60** achieved in six steps with 25% overall yield on the basis of the titanium-catalysed enantioselective allylation of commercially available 3-phenylpropanal.¹²¹ This reaction was induced by (*R*)-BINOL used at 20 mol% of catalyst loading, providing the corresponding chiral allylic alcohol **61** in 83% yield and excellent enantioselectivity of 97% ee, as shown in Scheme 72. The active catalyst **62** was in situ generated from the chiral ligand, a catalytic amount (15 mol%) of $\text{Ti}(\text{O}i\text{-Pr})_4$, and 10 mol% of Ag_2O in CH_2Cl_2 according to Maruoka's method.¹²² The allylic chiral alcohol obtained was further converted into the required 1,3-diol **60**.



Scheme 72. (R) -BINOL as ligand in allylation of 3-phenylpropanal

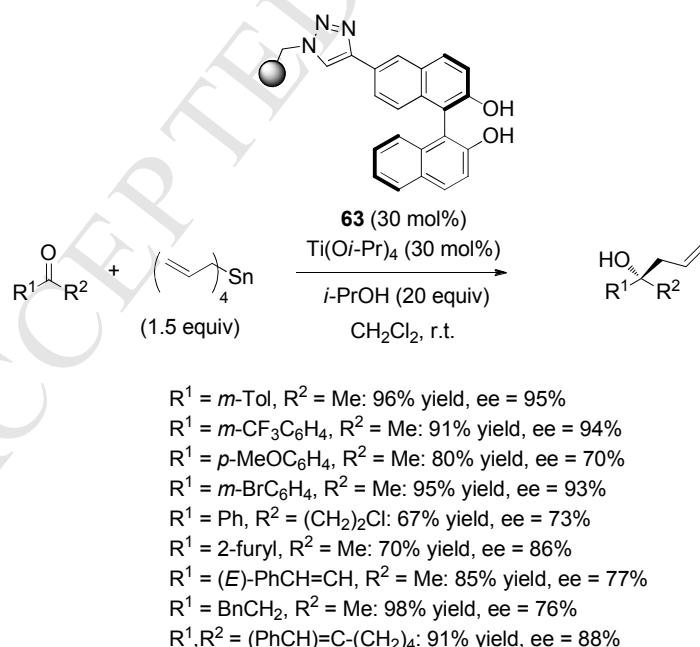
The enantioselective addition of allylmetal derivatives to ketones using chiral titanium complexes has been less developed than the related addition to aldehydes. The first catalytic enantioselective process was published by Tagliavini, in 1999.¹²³ In this work, enantioselectivities of up to 65% ee were obtained for the addition of tetraallyltin to various ketones in the presence of dichlorotitanium diisopropoxide and chiral BINOL. Later, Walsh found that the presence of a large excess of 2-propanol had a favorable impact on the enantioselectivity of the reaction and the enantiomeric excess of the products could be improved up to 96% ee (Scheme 73).¹²⁴



Scheme 73. First asymmetric titanium-catalysed allylation of ketones

Very recently, Pericas and Walsh reported the covalent immobilisation of (R) -BINOL to polystyrene by employing a 1,2,3-triazole linker to heterogenise a titanium-BINOLate catalyst and its application in the enantioselective allylation of ketones.¹²⁵ Therefore, by using a

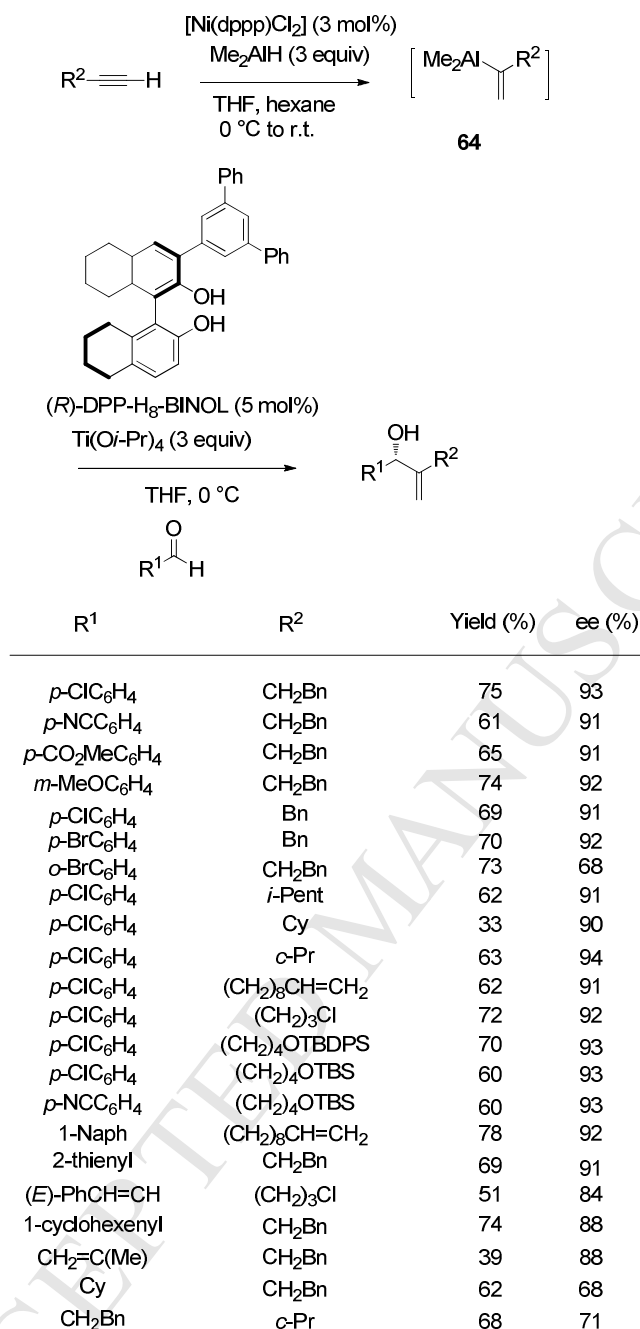
simple synthetic route, enantiopure 6-ethynyl-BINOL was synthesised and anchored to an azidomethylpolystyrene resin through a copper-catalysed alkyne/azide cycloaddition reaction. The polystyrene-supported BINOL ligand **63**, derived from 30 mol% of both (*R*)-BINOL and Ti(Oi-Pr)₄, was converted into its diisopropoxytitanium derivative in situ and used as a heterogeneous catalyst in the asymmetric allylation of a variety of ketones with tetraallyl tin in CH₂Cl₂ at room temperature to provide the corresponding chiral tertiary allylic alcohols in good to high yields of up to 98% and high enantioselectivities of up to 95% ee, as shown in Scheme 74. With the most common substrates, such as substituted acetophenones, the very high enantioselectivities observed were comparable to those recorded with the corresponding homogeneous catalytic system in the case of *ortho*- and *meta*-substituted acetophenones, whereas *para*-substituted acetophenones gave lower enantioselectivities (70% ee).^{124a} It must be noted that cyclic ketones, α,β -unsaturated ketones, and heteroaromatic ketones, such as 2-acetylfuran, also afforded the corresponding alcohols in good enantioselectivities (77-88% ee). Furthermore, the authors demonstrated the reusability of the ligand since both yields and enantioselectivities were preserved after three consecutive reaction cycles. Importantly, it is noteworthy to note that this work constituted one of the very few examples of catalytic creation of quaternary centres involving the use of a heterogenised catalytic species.



Scheme 74. (*R*)-BINOL-derived polystyrene-supported ligand in allylation of ketones

4.2. Vinylations

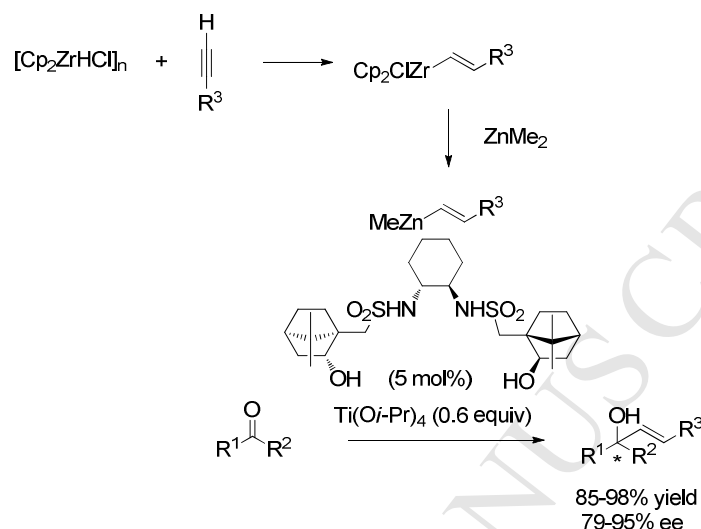
Highly enantioselective catalysts for the asymmetric vinylation of aldehydes have been developed by the groups of Oppolzer,¹²⁶ Wipf,¹²⁷ and others.¹²⁸ Although chiral allylic alcohols with various substitution patterns are in demand, advances in this area have mostly been focused toward the synthesis of β -substituted *E*-allylic alcohols. In contrast, very few methods have been developed for the synthesis of α -substituted allylic alcohols. In this context, Harada et al. recently reported a general one-pot method for the highly enantioselective synthesis of these products starting from alkynes and aldehydes and proceeding through in situ generated vinylaluminum reagents.¹²⁹ This process involved a catalytic amount of chiral ligand (*R*)-DPP-H₈-BINOL (5 mol%) in combination with three equivalents of Ti(Oi-Pr)₄ as catalytic system. The use of Me₂AlH (3 equivalents) was essential in the preliminary nickel-catalysed hydroalumination step to generate vinylaluminum reagents **64** which could not reduce the aldehyde starting material (Scheme 75). Remarkable enantioselectivities of up to 94% ee were achieved at a low catalyst loading of 5 mol% in the subsequent addition reaction to aldehydes. It must be noted that aromatic, heteroaromatic, as well as aliphatic aldehydes were compatible with the reaction albeit aliphatic aldehydes generally provided lower enantioselectivities (68-71% ee). Concerning benzaldehyde derivatives, good yields and high enantioselectivities (90-94% ee) were afforded for *para*- and *meta*-substituted benzaldehydes while only a moderate enantioselectivity (68% ee) was observed for *ortho*-bromo-substituted benzaldehyde. Notably, the reaction tolerated aldehydes that possessed potentially reactive cyano and ester substituents. In addition, good enantioselectivities (84-88% ee) were also obtained with various α,β -unsaturated aldehydes. It has to be highlighted that this one-pot process is remarkable by its wide scope and homogeneity of its high enantioselectivities, allowing a novel entry to chiral α -substituted allylic alcohols to be achieved.



Scheme 75. (R)-DPP-H₈-BINOL as ligand in vinylation of aldehydes

The catalytic asymmetric vinylation of ketones has attracted considerable attention among many research groups due to its more challenging transformations, owing to the significant difference in reactivity between aldehydes and ketones. Despite the success of asymmetric vinylation of aldehydes, vinyl additions to the inert ketones remain challenging for chemists. To counterbalance the reduced reactivity of ketones, a more reactive vinylating agent was needed. An important discovery in the asymmetric vinylation of ketones was reported by Walsh in 2004, using a protocol in which the reaction coupled

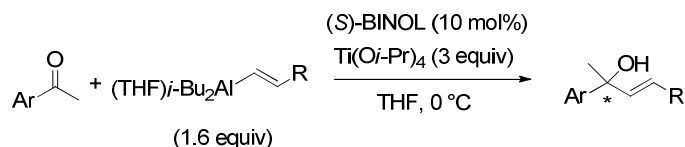
hydrozirconation/transmetalation to zinc with the catalyst to furnish the chiral tertiary alcohols.¹³⁰ In this work, the catalyst was prepared from a chiral bis(sulfonamide) ligand and $\text{Ti}(\text{Oi-Pr})_4$ and employed organozinc reagents as the nucleophiles, to afford chiral tertiary vinylic alcohols in good to excellent enantioselectivities for a range of ketones (Scheme 76).



Scheme 76. Early Ti-catalysed enantioselective vinylation of ketones reported by Walsh in 2004

In 2009, Gau and Biradar decided to prepare vinylaluminum compounds from alkynes and DIBAL-H to be used in the asymmetric addition to methyl ketones because of their high reactivity and the greater Lewis acidity of the aluminum centre.¹³¹ The catalytic system of the reaction was based on a catalytic amount (10 mol%) of (*S*)-BINOL combined with three equivalents of $\text{Ti}(\text{Oi-Pr})_4$. It allowed the synthesis of diversified chiral tertiary allylic alcohols from 1-hexyne and aromatic ketones bearing either electron-donating or electron-withdrawing substituents on the aromatic ring to be achieved in good to excellent enantioselectivities from 86% to 98% ee in combination with general high yields except for 3'-methoxyacetophenone which gave 81% ee (Scheme 77). Differences in enantioselectivities in terms of substituent types on the phenyl group were estimated to be around 10% ee. For example, additions of vinylaluminum compound derived from 1-hexyne to *ortho*-substituted acetophenones, such as 2'-methoxyacetophenone, 2'-methylacetophenone, or 2'-chloroacetophenone, afforded the corresponding allylic alcohols with enantioselectivities of 86, 98, and 90% ee, respectively. Additions to *meta*-substituted acetophenones furnished tertiary allylic alcohols with enantioselectivities ranging from 81 to 93% ee, and additions to *para*-substituted

acetophenones yielded products with enantioselectivities from 85 to 97% ee. Effects of substituent positions were also investigated, and the authors found that differences in enantioselectivities were 6% ee varying from 81 to 87% ee for additions to methoxyacetophenones (86, 81, and 87% ee, respectively, for *ortho*-, *meta*-, and *para*-methoxyacetophenones), 6% ee for additions to methylacetophenones (98, 93, and 92% ee, respectively, for *ortho*-, *meta*-, and *para*-methylacetophenone), and 7% ee for additions to chloroacetophenones (90, 91, and 97% ee, respectively, for *ortho*-, *meta*-, and *para*-chloroacetophenones). For acetonaphthones, the addition of vinylaluminum compound derived from 1-hexyne to 1'-acetonaphthone gave the corresponding product in a lower yield (82%) and with a lower enantioselectivity (87% ee) compared to 92% yield and 92% ee obtained for the product arisen from the addition to 2'-acetonaphthone. The scope of this remarkable process was extended to an α,β -unsaturated ketone, such as *trans*-cinnamaldehyde, which gave the corresponding product in 92% yield and with an enantioselectivity of 82% ee. More importantly, additions of vinylaluminum reagents derived from a variety of alkynes other than 1-hexyne, such as 3-phenyl-1-propyne and 6-chloro-1-hexyne, also produced the corresponding allylic alcohols in excellent enantioselectivities of up to 96% ee (Scheme 77). On the other hand, though the additions of cyclohexylvinyl and cyclohexenylvinyl reagents to 4'-chloroacetophenone produced the corresponding allylic alcohols in excellent yields of 90 and 89%, respectively, lower enantioselectivities of 85 and 88% ee, respectively, were observed.



Ar = Ph, R = *n*-Bu: 91% yield, ee = 92%
 Ar = *m*-MeOC₆H₄, R = *n*-Bu: 80% yield, ee = 81%
 Ar = *p*-MeOC₆H₄, R = *n*-Bu: 93% yield, ee = 87%
 Ar = *o*-MeOC₆H₄, R = *n*-Bu: 87% yield, ee = 86%
 Ar = *o*-Tol, R = *n*-Bu: 87% yield, ee = 98%
 Ar = *p*-Tol, R = *n*-Bu: 92% yield, ee = 92%
 Ar = *m*-Tol, R = *n*-Bu: 87% yield, ee = 93%
 Ar = *p*-ClC₆H₄, R = *n*-Bu: 90% yield, ee = 97%
 Ar = *o*-ClC₆H₄, R = *n*-Bu: 85% yield, ee = 90%
 Ar = *m*-ClC₆H₄, R = *n*-Bu: 88% yield, ee = 91%
 Ar = *p*-CF₃C₆H₄, R = *n*-Bu: 94% yield, ee = 96%
 Ar = *p*-BrC₆H₄, R = *n*-Bu: 89% yield, ee = 91%
 Ar = *p*-NO₂C₆H₄, R = *n*-Bu: 92% yield, ee = 95%
 Ar = *p*-NO₂C₆H₄, R = Cl(CH₂)₄: 93% yield, ee = 96%
 Ar = *p*-ClC₆H₄, R = Bn: 94% yield, ee = 95%
 Ar = *p*-ClC₆H₄, R = Cy: 90% yield, ee = 85%
 Ar = *p*-ClC₆H₄, R = 1-cyclohexenyl: 89% yield, ee = 88%
 Ar = 2-Naph, R = *n*-Bu: 92% yield, ee = 92%
 Ar = 1-Naph, R = *n*-Bu: 82% yield, ee = 87%

with (*E*)-PhCH=CHCOMe and R = *n*-Bu: 92% yield, ee = 82%

Scheme 77. (S)-BINOL as ligand in vinylation of methyl ketones

5. Conclusions

This review demonstrated the important amount of advances in enantioselective titanium-promoted alkylation, arylation, alkynylation, allylation as well as vinylation reactions of carbonyl compounds that have been achieved in the last seven years spanning from classical reactions, such as enantioselective 1,2-nucleophilic additions of organozinc reagents to aldehydes, to those of low reactive organozinc reagents or other organometallic ones to poor electrophilic ketones. For example, good results have been recently reported dealing with enantioselective titanium-promoted dialkylzinc additions to more challenging aliphatic aldehydes than commonly used aromatic ones. Always in the context of organozinc additions, a range of challenging functionalised alkylzinc reagents could be added to aldehydes with high enantioselectivities. Concerning the additions of other organometallic reagents, it must be noted that it is only recently that the first highly efficient enantioselective alkylations of aldehydes with organolithium reagents have been successfully developed. Moreover, the direct additions of highly reactive alkyl and aryl Grignard reagents to all types of aldehydes at room temperature were recently demonstrated to give general remarkable enantioselectivities when induced by chiral titanium catalysts. Importantly, the first direct additions of alkyl- and

aryltitanium reagents to various aldehydes including aliphatic ones performed at room temperature were shown to provide excellent enantioselectivities. Another important advance was the first highly enantioselective direct addition of alkylboranes including functionalised ones to aldehydes including aliphatic ones. Furthermore, in the context of additions to ketones, the first highly efficient enantioselective additions of (2-furyl)- and (2-thienyl)aluminum reagents to ketones have been described. For all these types of nucleophilic reagents, remarkable enantioselectivities were reached for alkylation/arylation reactions. On the other hand, the enantioselective addition of organometallic alkynyl derivatives to carbonyl compounds is today the most expedient route toward chiral propargylic alcohols which constitute strategic building blocks for the enantioselective synthesis of a range of complex important molecules. In the last few years, impressive advances have been made particularly in the variety of alkynes which could be successfully added to aldehydes with remarkable enantioselectivities. Indeed, in addition to the more commonly used phenylacetylene, a range of other terminal (functionalised) alkynes have allowed excellent results to be achieved in enantioselective titanium-promoted alkynylations of aldehydes, such as *para*-tolylacetylene, trimethylsilylacetylene, ethynylcyclohexene, 4-phenyl-1-butyne, 5-chloro-1-pentyne, 1-hexyne, 1-heptyne, 1-octyne, alkynoates, as well as 1,3-diynes and 1,3-enynes. In the context of enantioselective alkynylations of ketones, the first successful use of aryltrifluoromethyl ketones was described. In addition, it is important to note that several supported chiral ligands have been recently successfully applied to the catalysis of almost all types of 1,2-additions, such as enantioselective dialkylzinc additions to ketones, enantioselective alkynylations of aldehydes, and enantioselective allylations of ketones. All these methodologies have a strategically synthetic advantage to form a new C–C bond, a new functionality (alcohol) with concomitant creation of a stereogenic centre in a single transformation. They arose from the extraordinary ability of chiral titanium catalysts to control stereochemistry which can be attributed to their rich coordination chemistry and facile modification of titanium Lewis acid centre by structurally modular ligands. Among the Lewis acidic metal complexes, titanium(IV) is the central metal of choice, because of its high Lewis acidity and relatively short metal-ligand bond lengths, in addition to its high abundance, low cost, and low toxicity. In spite of the important number of publications, however, challenges remain in the context of enantioselective nucleophilic 1,2-additions to carbonyl compounds, such as a better understanding of the role of the active titanium catalysts, and achieving higher turnover numbers of the catalytic cycles constitute an area of interest. Titanium-mediated reactions described in this review are indeed very powerful but their utility remains often hampered by

the need of using superstoichiometric amounts of titanium sources particularly in the cases of alkylation and arylation reactions. Even if titanium is not toxic, abundant, and inexpensive, this remains wasteful and problematic. In this context, the use of substoichiometric amounts of titanium sources or (still rarely used) preformed titanium catalysts have already allowed major advances to be achieved and will have to be more developed in the near future. The ever-growing need for environmentally friendly catalytic processes prompted organic chemists to focus on more abundant first-row transition metals such as titanium to develop new catalytic systems to perform reactions, such as C–C bond formations. Therefore, a bright future is undeniable for more sustainable novel and enantioselective titanium-promoted transformations.

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List of abbreviations

Ad: adamantyl

Ar: aryl

BDMAEE: 2,2'-oxy-bis(*N,N*-dimethylethanamine)

BINOL: 1,1'-bi-2-naphthol

Bn: benzyl

BTME: 1,2-bis(trimethoxysilyl)ethane

Cy: cyclohexyl

DAIB: dimethylamino isoborneol

DIBAL: diisobutylaluminium hydride

DIMPEG: dimethoxy polyethylene glycol

DIPEA: diisopropylethylamine

DMAP: 4-(dimethylamino)pyridine

DME: dimethoxyethane

DPP: 3,5-diphenylphenyl

Dppp: 1,3-bis(diphenylphosphine)propane

ee: enantiomeric excess

FG: functionalised group

HMPA: hexamethylphosphoramide

L: ligand

MCF: mesocellular foam

Mes: mesyl

MOM: methoxymethyl

MTBE: methyl*tert*-butylether

Naph: naphthyl

NOBIN: 2-amino-2-hydroxy-1,1'-binaphthalene

PMB: *para*-methoxybenzyl

r.t.: room temperature

TADDOL: $\alpha,\alpha,\alpha',\alpha'$ -tetraphenyl-2,2-dimethyl-1,3-dioxolane-4,5-dimethanol

TBDPS: *tert*-butyldiphenylsilyl

TBS: *tert*-butyldimethylsilyl

THF: tetrahydrofuran

TIPS: triisopropylsilyl

TMS: trimethylsilyl

Tol: *p*-tolyl

Tr: triphenylmethyl (trityl)

Ts: 4-toluenesulfonyl (tosyl)

Hélène Pellissier carried out her PhD under the supervision of Dr G. Gil in Marseille (France) in 1987. The work was focused on the reactivity of isocyanides. In 1988, she entered the Centre National de la Recherche Scientifique (CNRS) as a researcher. After a postdoctoral period in Professor K. P. C. Vollhardt's group at the University of California, Berkeley, she joined the group of Professor M. Santelli in Marseille in 1992, where she focused on the chemistry of 1,8-bis(trimethylsilyl)-2,6-octadiene and its application to the development of novel very short total syntheses of steroids starting from 1,3-butadiene and benzocyclobutenes. She is currently chargée de recherche (CNRS) at Aix-Marseille Université.

